

# Random-walk SIR models of indoor epidemics: reproduction numbers, discrete individuals and pre-specified tests against outbreaks, contact records and tracer measurements

Xiaoxiao Zhouyi

School of Engineering Mathematics and Technology, University of Bristol, UK

[zhouyixiaoxiao@gmail.com](mailto:zhouyixiaoxiao@gmail.com)

2026

## Abstract

We study SIR epidemics of independent random walkers on a lattice with cell-dependent mobility and infection efficiency. (i) The mean-field  $R_0$  is the spectral radius of a next-generation matrix of the lattice propagator. For same-cell transmission its bounds, limits and monotone decrease in mobility hold, as known for patch models; for other kernels the lower bound and monotonicity can fail. With uniform efficiency,  $R_0$  of a closed room is independent of size, shape and mobility; a leak replaces  $\gamma$  by  $\gamma + \mu_{\text{leak}}$  under stated conditions. (ii) Two-walker theory gives the exact mean number of first-generation secondary cases of a lone index case; it lies below  $R_0$  at low occupancy and mobility (office scene: 2.27 against 5.11) and is not a threshold. (iii) In exact simulations of four closed-room scenes, spatial effects are large only at very low mobility. (iv) Five tests, with protocols fixed before the tested model was run (data-aware, self-attested, not registered), confront the model with 22 of 38 outbreak records, contact records, tracer measurements and a school epidemic. There is no general agreement. Same-site transmission is not supported; walk-generated contacts fail in all six contact datasets. An airborne extension predicts held-out distance gradients better than a well-mixed model (log-score gain +14.35; a constant near/far ratio would also do so with probability 0.65–0.73), but the pre-specified pooled comparison favours the constant ratio (−9.15, nine-tenths from one event whose near zone was chosen on its data); on four held-out flights they tie. Attack rates are fitted; cell-dependent mobility remains untested.

**Keywords:** indoor transmission; lattice random walk; basic reproduction number; next-generation matrix; pair-level theory; stochastic simulation; outbreak data; contact networks; tracer measurements; pre-specified test.

## 1 Introduction

Recorded clusters of SARS-CoV-2 infection arose overwhelmingly indoors [62, 79], and in several well-documented events the cases were not spread evenly over the room. They sat on one side of a call-centre floor [76], close to the index case in an aircraft cabin [56] or on high-speed trains [47], or at neighbouring restaurant tables in the air stream of one air conditioner [63, 65]. The standard quantitative tool for indoor risk is the Wells–Riley equation [81, 92, 103]. It and the models built on it [8, 72] treat the room air as well mixed, so that every susceptible occupant carries the same risk. This article examines what a minimal spatial model adds to that picture: a susceptible–infective–recovered (SIR) epidemic among people who perform independent random walks on a lattice of floor cells, with a mobility  $D_x$  and an infection efficiency  $q_x$  that depend on the cell (workstations, corridors, meeting rooms, barriers).

**Prior work.** Epidemics among moving individuals have been modelled for more than three decades: lattice automata whose individuals move between sites [14, 80], agents moving in continuous space [15, 27, 39], and reaction–diffusion processes on metapopulation networks [10, 21, 100]. Closest to the present model is the random-walk theory of transmission between two walkers, in which infection is a reaction that takes place at a finite rate when an infected and a susceptible walker meet [50, 51, 89]. Giuggioli and Sarvaharman [38] formulate it on a lattice with exact propagators [37, 82] and predict a basic reproduction number analytically for moving susceptible, infected and recovered individuals. The many-walker system itself is a classical object of probability and of statistical physics. For independent continuous-time random walkers on  $\mathbb{Z}^d$ , with infection when an infected and a healthy walker meet, Kesten and Sidoravicius [53, 55] proved that the infected set grows linearly in time with an asymptotic shape, and, when infected walkers recover and can be infected again, that there is a critical recovery rate below which the infection survives with positive probability and above which it dies out [54]. The same process is studied in physics as the diffusive epidemic process, whose absorbing-state transition has been analysed by mean-field theory and simulation [67]; with permanent immunity, as here, simulations on the square lattice show a transition of dynamic-percolation type controlled by the density of walkers [17]. These results concern infinite or large homogeneous lattices. They establish that a threshold exists for walkers; they do not give its value for the small heterogeneous rooms considered here.

For compartmental models the basic reproduction number  $R_0$  is the spectral radius of the next-generation operator [23],  $R_0 = \rho(\mathbf{FV}^{-1})$ , and it is a threshold [96]. Linearised at the disease-free state, the mean-field version of the lattice model is an epidemic patch model with standard incidence, and the relevant theorems are published. For the patch model with symmetric movement, Allen et al. [1] defined  $R_0 = \rho(\mathbf{FV}^{-1})$ , proved the threshold property, showed that the growth bound  $s(\mathbf{F} - \mathbf{V})$  decreases with the dispersal rate between its two limits, and conjectured the same monotonicity for  $R_0$ ; the conjecture was proved by Gao [29] for symmetric movement and by Gao and Dong [30] in general (see also Chen et al. [19]), and it is an instance of the reduction principle of Altenberg [3]. That  $R_0$  lies between the smallest and the largest patch reproduction number, and does not depend on movement when they coincide, is due to Gao and Ruan [31]; the sharper lower bound, which is the fast-dispersal limit, is due to Gao and Dong [30], and the approach to that limit at order one over the dispersal rate was obtained by Tien et al. [94], for a network on which the pathogen rather than the hosts moves, with a next-generation matrix of the same form. Continuum counterparts are given by Allen et al. [2] and Wang and Zhao [101]. (We read the published versions of Gao and Ruan [31] and Tien et al. [94], whose Proposition 2.2 and Section 4.2 contain the results attributed to them, and of Gao and Dong [30]. For Allen et al. [1] and Gao [29] we read only the abstracts; what we attribute to them follows the account of Gao and Dong [30] and agrees with the abstracts as far as they go: the threshold property in the first, the monotone decrease in the dispersal rate for symmetric movement in the second.)

Spatially resolved indoor models also exist: zonal ventilation models [73], transport models for the airborne concentration [61, 99], and pedestrian- or agent-based simulations of supermarkets and of air travel [71, 106]. They resolve the air or the trajectories in more detail than a lattice walk does. What the lattice model offers in return is analytical access to reproduction numbers; what it lacks, as Section 7 shows, is an adequate description of seated people who share air and of who meets whom.

Three bodies of measurement bear on such a model, and each has its own literature. Close-range contacts between people indoors have been recorded with wearable sensors in schools, offices, hospitals and conferences; they are concentrated on few, repeated partners, structured by groups and broadly distributed in duration [18, 33, 34, 88]; random walkers in a plane that slow down near attractive individuals reproduce several of these statistics [86]; and epidemics have been simulated on the recorded networks [32, 87]. The transport of aerosol in vehicles has

been measured with tracers [57, 75, 105], and turbulent diffusion coefficients of rooms have been related to their ventilation [20]. And outbreak investigations report risk by distance: contact tracing on aircraft uses a zone of two rows around a case, a rule whose limits are known [44], and transmission of influenza in schools has been resolved into classes, grades and schools [26]. These are the data against which the model is tested in Section 7; the rules they suggest, a constant ratio of near to far risk and a fixed share of contact within the group, are the generic competitors that the model has to beat.

## Contributions.

- (i) *Mean-field reproduction number* (Sections 3 and 4). We state and prove that  $R_0 = \rho(\beta \mathbf{Q}(\gamma \mathbf{1} - \mathbf{M})^{-1})$ , a next-generation matrix built from the Laplace-transformed lattice propagator. For same-cell transmission we prove that  $\beta \langle q \rangle_\pi / \gamma \leq R_0 \leq \beta q_{\max} / \gamma$ , that  $R_0$  decreases monotonically with the mobility scale between these limits, and that any high-risk zone contributes a lower bound; the lower bound and the monotonicity can fail for other contact kernels. In a closed room with uniform  $q$ ,  $R_0$  does not depend on the size or shape of the room. The genuine boundary effect is a leak (doors, finite dwell time): for uniform  $q$  and the same-cell kernel, or for a dwell time that does not depend on position and any kernel, it replaces  $\gamma$  by  $\gamma + \mu_{\text{leak}}$ , and for heterogeneous  $q$  and the same-cell kernel it gives  $\beta \langle q \rangle_\nu / (\gamma + \mu_{\text{leak}}) \leq R_0^{\text{room}} \leq \beta q_{\max} / (\gamma + \mu_{\text{leak}})$  (Theorem 4.2); for other kernels and a localised door the formula and both bounds can fail.
- (ii) *Discrete individuals* (Section 5). Following the two-walker theory of transmission [38, 51], infection is treated as a partially absorbing defect of the two-walker process. This gives the expected number of secondary cases  $R_1(x)$  of an index case and a closed form for homogeneous periodic rooms and the infinite floor, which approximates rooms with reflecting walls to within 1 % in the cases tested. It explains why counted secondary cases fall below  $R_0$  at low occupancy and what form a finite-size effect takes for discrete people.
- (iii) *Exact stochastic simulation* (Section 6) of four scenes (office, supermarket, classroom, metro carriage) and of interventions, in which secondary cases are counted from the infection tree and not fitted.
- (iv) *Confrontation with data* (Section 7). We assemble 38 records of documented indoor outbreaks and exposure cohorts: a first dataset of 27 records from 31 published sources (23 with counts of cases and of exposed people) and a second dataset of eleven records of ten outbreaks with seat or position data. Two spatial models (M1, M2) are tested against a well-mixed model (M0) on 18 of them in two out-of-sample tests (two records for calibration and five held out; then six and five; four further records enter the level checks and negative controls of the first test) and, in the second test by design, against a constant ratio of near to far risk. Three further analyses address what outbreak counts cannot: contacts generated by the lattice walk are compared with recorded contacts in six indoor settings; the airborne kernel is fixed by published tracer measurements and then confronted with the outbreaks; and the model reduced to zones, with mixing measured in one school, is used to predict an influenza epidemic in 29 others. Each of the five primary tests follows a protocol written before the model under test was run on its data (for the tracer analysis, on outbreaks whose outcome under the other models was already known). The protocols are pre-specified, not pre-registered in the usual sense: they were written with the data already known, they were not lodged with a registry, and their timing is attested by file hashes and file times only. Comparisons added after the results were known are marked as post hoc, and a register of all tests says which were specified in advance (Supplementary Table S39). Everything is reported with its failures.

**What is known and what is added.** Table 1 separates the two. The theorems on the mean-field  $R_0$  are known for patch models. We give complete proofs from standard facts on non-negative matrices (Supplementary Section S2) because the conditions matter: the lower bound  $\beta\langle q \rangle_\pi/\gamma \leq R_0$  holds for the same-cell kernel and fails for general contact kernels, for which we give a three-cell counter-example (Proposition 3.7). The two-walker formalism is prior work, and the saturation of contacts between neighbours that it captures is a classical effect [7, 24, 48]. In particular, Giuggioli and Sarvahaarman [38] already derive the mean number of secondary infections of one infected walker among uniformly placed susceptible walkers in a closed domain from the two-walker formalism and check it against simulation; the one-defect solution is that of Szabo et al. [91]. What is added here is narrower (Section 5): the normalisation of the pair hazard by the occupancy, which makes the pair-level and the mean-field numbers refer to the same model and gives the entrywise comparison  $\mathbf{K}_1 \leq \mathbf{K}$ ; the numbers  $R_1(x)$  and the matrix  $\mathbf{K}_1$  for heterogeneous rooms with contact kernels; the closed form for periodic rooms and the infinite floor with its limits; the comparison with the second generation; and the observation that neither  $\bar{R}_1$  nor the spectral radius of  $\mathbf{K}_1$  is a threshold quantity (Remark 5.7). These statements of what is added rest on a targeted search (Crossref and arXiv queries on infection among random walkers, the diffusive epidemic process and patch-model reproduction numbers, 1 October 2026), not on a systematic review.

**Findings.** In the office scene (234 walkable cells,  $\beta = 0.5 \text{ day}^{-1}$ ,  $\gamma = 0.14 \text{ day}^{-1}$ , mobility scale  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ ) the mean-field number is  $R_0 = 5.11$ , inside the bounds 4.64–5.36 that hold for every layout with the same zone fractions. With 100 discrete people, however, a single index case infects on average  $\bar{R}_1 = 2.27$  others (the exact pair-level expectation; counted  $\hat{R} = 2.265 \pm 0.004$ , mean  $\pm$  standard error over 400,000 runs). Half of the introductions die out: the probability of an outbreak reaching 10 % of the room is 0.50, against 0.78 for the mean-field branching process. The two numbers respond to mobility in opposite ways: at  $D_0 = 10$  and  $100 \text{ cell}^2 \text{ day}^{-1}$ ,  $R_0$  falls to 4.68 and 4.65 while the pair-level number rises to  $\bar{R}_1 = 3.79$  and 4.34. Barriers, separation of zones and capacity limits reduce simulated outbreaks at low mobility and occupancy. The cause is the discreteness of individuals (an infective can infect each neighbour only once and keeps meeting the same ones), not a lower  $R_0$ : under frequency-dependent transmission a capacity limit leaves  $R_0$  unchanged, separation of zones raises it, barrier cells leave it unchanged in a room with uniform  $q$  (Theorem 4.1), and walls between cells cannot lower it (Theorem 3.6). These effects belong to a narrow physical regime. With 1.5 m cells,  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$  is a root-mean-square displacement of about 3 m in a day; people who walk 5 % of the time or all of the time correspond to  $2.4 \times 10^3$  to  $4.9 \times 10^4 \text{ cell}^2 \text{ day}^{-1}$ , at which the  $R_0$  of the closed office exceeds its well-mixed value  $\beta\langle q \rangle/\gamma = 4.64$  by less than 0.003 %. Residence time matters as well: if the office is occupied 8 hours a day instead of 24, an index case infects on average 0.89 to 0.93 people at  $D_0 = 1$  and 1.34 to 1.40 at  $D_0 = 10 \text{ cell}^2 \text{ day}^{-1}$ .

Confronted with data, the model does not show general agreement, and what it gets right is not specific to it (Section 7). The formulation with transmission only between people on the same lattice site is not supported. An extension with a diffusing-aerosol term predicts the decline of risk with distance better than a well-mixed model, but not better than a constant ratio of near to far risk; it fails on several events and admits no common mixing coefficient, and the coefficients fitted to the first dataset lie far below those measured with tracers in vehicles. Mixing shares measured in one school predict an influenza epidemic in 29 others only as well as a generic model with one fitted ratio. The data support a smooth decline of risk with distance and a zone structure dominated by the group, statements that the model shares with simpler ones; attack-rate levels are fitted, not predicted; and because people are seated or at fixed stations in every outbreak that publishes positions, the spatial heterogeneity of mobility remains untested. We present the datasets and the tests as contributions and make no claim of agreement.

**Table 1:** Results of this article, where they are known from, and what is added here. ‘Thm’ numbers refer to the published version of Gao and Dong [30]. No novelty is claimed for rows 1–6 and 8; in rows 9 and 10 the data are those of the cited sources; what is added is their compilation and digitisation and the tests.

| Result                                                                                                     | Known from                                                                                                                                   | Added here                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. $R_0 = \rho(\mathbf{F}\mathbf{V}^{-1})$ and its threshold property                                      | [23, 96]; for the patch model [1]                                                                                                            | proof in lattice notation; the modal formula, of which the largest diagonal term is a lower bound                                                                                                                                                                             |
| 2. $\beta\langle q \rangle_\pi / \gamma \leq R_0 \leq \beta q_{\max} / \gamma$                             | between the smallest and largest patch numbers [31]; lower bound [30] Thm 3.3; continuum case [2]; heterogeneity raises $R_0$ [42]           | the condition (same-cell kernel) and a counter-example for other contact kernels                                                                                                                                                                                              |
| 3. $R_0$ decreases with mobility, between the limits of row 2                                              | conjectured in [1]; proved in [29] (symmetric movement) and [30] Thm 2.3 (general); [3, 19]                                                  | size of the effect in four indoor scenes, at low and at realistic mobility                                                                                                                                                                                                    |
| 4. Uniform $q$ : $R_0 = \beta q / \gamma$ whatever the movement                                            | [31]; loss through absorbing boundaries [16, 84]                                                                                             | consequence for rooms (no size effect when closed); leaky-room formula $R_0^{\text{room}} = \beta q / (\gamma + \mu_{\text{leak}})$ (same-cell kernel)                                                                                                                        |
| 5. Zone lower bound; expansions for fast and slow mixing; edge-wise monotonicity                           | fast-dispersal expansion [94]; the others are elementary consequences of the same matrix theory, for which a targeted search found no source | statements and proofs for indoor lattices                                                                                                                                                                                                                                     |
| 6. Hotspot maps (Perron vectors, elasticities)                                                             | standard eigenvector sensitivity                                                                                                             | application to the scenes; a caveat on transients                                                                                                                                                                                                                             |
| 7. Infection as a reaction between two walkers; mean number of secondary infections of one infected walker | [38, 50, 51, 89]; one defect [91]; local saturation [7, 24, 48]; thresholds for walkers on infinite lattices [17, 54, 67]                    | occupancy-normalised hazard and $\mathbf{K}_1 \leq \mathbf{K}$ ; $R_1(x)$ and $\mathbf{K}_1$ for heterogeneous rooms with kernels; closed form for periodic rooms and the infinite floor; second generation; not a threshold                                                  |
| 8. Diffusing-aerosol exposure term; decline of risk with distance                                          | [20, 61, 98]; two-row rule on aircraft [44]                                                                                                  | used only as the extension tested in Section 7; no claim that it predicts better than a constant ratio                                                                                                                                                                        |
| 9. Outbreak investigations                                                                                 | the published sources of Supplementary Section S6; an earlier compilation of outbreaks evaluated with a well-mixed model [78]                | 38 records: near/far strata for eight records of the first dataset, the seat map of the call centre, the seat-offset matrix of the train cohort, and eleven records of the second dataset with seat or position data; two pre-specified out-of-sample tests and their outcome |
| 10. Recorded contacts, tracer measurements, class structure of school epidemics                            | [18, 26, 33, 57, 88, 105]; walkers that slow down near others [86]                                                                           | pre-specified tests of the walk’s contacts, of the kernel with a measured coefficient and of the zone-level reduction against these data, with negative or weak outcomes                                                                                                      |

**How the results were checked.** Each strand of this work (the theory, the simulations, the outbreak dataset, the two outbreak tests and the analyses of contact records, tracer measurements and zone structure) was re-checked with separately written code, within the same project and on the same machine: the second pass re-derived the proofs, re-implemented the computations, including a second exact simulator, retyped the outbreak counts from the sources and searched for counter-examples. It was not carried out by anyone independent of the project, and it is not an external review; where it re-ran analyses, compared kernels or added comparisons, it reused data loaders, event geometries, the transcribed seat maps and, for timing benchmarks and schedules, the simulator of the analyses (Supplementary Section S11.1). Below, ‘the re-check’ refers to that second pass and ‘second simulator’ to its simulator. The reported results use these checked implementations. Analyses outside the scope of the second pass and comparisons introduced after their outcomes were known are identified explicitly. What each re-check covered is described in Supplementary Section S11.

**Outline.** Section 2 defines the model, the four scenes and the parameter regime. Section 3 treats the basic reproduction number of the mean-field model and Section 4 room size, leaks and hotspot maps. Section 5 develops the pair-level theory for discrete individuals, and Section 6 reports the simulations of the scenes and of interventions. Section 7 confronts the model with documented outbreaks, recorded contacts, tracer measurements and a school epidemic, and Section 8 discusses what stands and the limits. The Supplementary Material holds the proofs, the numerical verification, the outbreak datasets, the protocols, extended tables and the register of all tests; its sections, equations, figures and tables are numbered with the prefix S.

## 2 Model, scenes and parameter regime

This section fixes the model: one movement rule, a mean-field model with a contact kernel, an individual-based model of which it is the mean field, and three reproduction numbers that must be kept apart. It then describes the four scenes and states the physical regime of the parameters. In Sections 2–6 lengths are measured in lattice cells and time in days.

### 2.1 Movement on the lattice

The floor of a room is covered by a square lattice of spacing  $a$ . Cells occupied by walls, furniture or other barriers are removed; the remaining  $n$  walkable cells form the set  $\Omega$ , and two walkable cells are neighbours if they share a side. Cell  $x$  has a mobility (diffusion coefficient)  $D_x = D_0 d_x > 0$ , where  $D_0$  is the mobility scale and  $d_x$  a dimensionless zone multiplier. A person in cell  $x$  hops to a neighbouring cell  $y$  at rate

$$\omega_{x \rightarrow y} = \omega_{y \rightarrow x} = \frac{2D_x D_y}{D_x + D_y} \quad \text{for neighbouring cells } x, y \in \Omega, \quad (1)$$

the harmonic mean of the two mobilities. This is the interface coefficient of finite-volume discretisations of diffusion in composite media [77]: the net flow between two cells is  $\omega_{x \rightarrow y}$  times the difference of their occupation numbers, so that (1) is a discretisation of Fickian diffusion,  $\partial_t \rho = \nabla \cdot (D \nabla \rho)$ , with reflecting walls. The movement generator is the  $n \times n$  matrix

$$M_{yx} = \omega_{x \rightarrow y} \quad (y \neq x), \quad M_{xx} = - \sum_{y \neq x} \omega_{x \rightarrow y}, \quad \mathbf{1}^\top \mathbf{M} = 0, \quad (2)$$

with  $M_{yx} = 0$  if  $x \neq y$  are not neighbours. The column index is the source cell: the vector of expected occupation numbers obeys  $\dot{N} = \mathbf{M}N$ , and  $\mathbf{1}^\top \mathbf{M} = 0$  says that nobody enters or leaves through the walls. Because  $D_x = D_0 d_x$ , we have  $\mathbf{M} = D_0 \mathbf{M}_0$ , where  $\mathbf{M}_0$  is the generator at  $D_0 = 1$ . The propagator  $P_t(y|x) = (e^{\mathbf{M}t})_{yx}$  is the probability that a walker that started in cell

$x$  is in cell  $y$  at time  $t$ . Closed forms of the propagator are known for homogeneous rectangular lattices [37]; for the heterogeneous scenes of this article every quantity is computed numerically from  $\mathbf{M}$ .

**Assumption A.** The walkable region is connected:  $\mathbf{M}$  is irreducible. The population is at movement equilibrium,  $N_x(t) \equiv N_x^* = N \pi_x$ , where  $\boldsymbol{\pi} > 0$  is the stationary distribution of  $\mathbf{M}$  ( $\mathbf{M}\boldsymbol{\pi} = 0$ ,  $\mathbf{1}^\top \boldsymbol{\pi} = 1$ ;  $\pi_x = 1/n$  for the symmetric rule (1)).

Under rule (1) the matrix  $\mathbf{M}$  is symmetric, so the uniform distribution is the stationary one and people do not accumulate in slow zones. The assumption is phrased with a general  $\boldsymbol{\pi}$  because several results of Section 3 hold for any irreducible generator. The harmonic mean is a modelling choice. A rule in which the rate depends on the departure cell only,  $\omega_{x \rightarrow y} = D_x$ , is a different model: it satisfies detailed balance with  $\pi_x \propto 1/D_x$ , so that people accumulate where they move slowly, and its continuum limit is  $\partial_t \rho = \nabla^2(D\rho)$ . With lengths in cells,  $D_x$  is in  $\text{cell}^2 \text{day}^{-1}$ ; in physical units the rate (1) is  $2D_x D_y / ((D_x + D_y)a^2)$ .

## 2.2 Mean-field model

An infective in cell  $x$  makes infectious contacts at rate  $\beta q_x$ , where  $\beta$  is the transmission rate and  $q_x \geq 0$  the dimensionless infection efficiency of the cell ( $\mathbf{Q} = \text{diag } q$ ,  $q \neq 0$ ). A contact lands in cell  $y$  with probability  $P_{xy} = P(y|x)$ , where the contact kernel  $\mathbf{P}$  is row-stochastic, and reaches a person chosen uniformly among those present in  $y$  (frequency-dependent incidence). Infectives recover at rate  $\gamma$ . With  $S_y, I_y, R_y$  the expected numbers of susceptible, infective and recovered people in cell  $y$ ,

$$\begin{aligned} \dot{S}_y &= (\mathbf{M}\mathbf{S})_y - \frac{S_y}{N_y^*} \beta \sum_x q_x P_{xy} I_x, & \dot{I}_y &= (\mathbf{M}\mathbf{I})_y + \frac{S_y}{N_y^*} \beta \sum_x q_x P_{xy} I_x - \gamma I_y, \\ \dot{R}_y &= (\mathbf{M}\mathbf{R})_y + \gamma I_y. \end{aligned} \quad (3)$$

The *same-cell kernel*  $\mathbf{P} = \mathbf{1}$  gives the local incidence  $\beta q_y S_y I_y / N_y^*$ ; the system is then a susceptible–infective–recovered patch model with one patch per cell, and the results known for such models apply (Section 3). The simulations use the *1 m kernel*:  $P(y|x)$  is uniform over the walkable cells whose centres lie within 1 m of the centre of  $x$  (line of sight through barriers is not checked). It coincides with the same-cell kernel when  $a > 1$  m; without it the lattice spacing would silently set the contact range, with cell areas from  $0.25 \text{ m}^2$  to  $4 \text{ m}^2$  in the scenes below. The range of 1 m is a modelling choice.

Summing the three equations gives  $\dot{N} = \mathbf{M}\mathbf{N}$  for  $N = S + I + R$ , so that  $N_y(t) = N_y^*$  for all  $t$  under Assumption A. Writing  $S_y/N_y^* = 1 - (I_y + R_y)/N_y^*$ ,

$$\dot{I} = \mathbf{J}\mathbf{I} + O(|I|(|I| + |R|)), \quad \mathbf{J} = \mathbf{M} + \mathbf{F} - \gamma \mathbf{1}, \quad \mathbf{F} = \beta \mathbf{P}^\top \mathbf{Q}, \quad (4)$$

with  $\mathbf{F} = \beta \mathbf{Q}$  for the same-cell kernel. Two features of (4) are used repeatedly. First,  $\mathbf{J}$  contains neither the number of people nor their density: under frequency-dependent incidence, crowding does not enter the linearised dynamics. Second, the neglected term is  $-\text{diag}((I + R)/N^*) \mathbf{F}\mathbf{I} \leq 0$ , so that every solution of (3) satisfies  $\dot{I} \leq \mathbf{J}\mathbf{I}$  componentwise for all  $t \geq 0$ .

## 2.3 Individual-based model

The individual-based model has  $N$  people who perform independent continuous-time random walks on  $\Omega$  with generator  $\mathbf{M}$  (rule (1), uniform  $\boldsymbol{\pi}$ ) and who are placed independently and uniformly at time 0. Each person is susceptible, infective or recovered. While an infective is located at  $x$  and a susceptible at  $y$ , the infective infects that susceptible at rate

$$h(x, y) = \frac{\beta q_x P_{xy}}{\bar{\rho}}, \quad \bar{\rho} = \frac{N-1}{n} \quad (\text{frequency-dependent calibration}), \quad (5)$$

where  $\bar{\rho}$  is the mean number of *other* people per cell. The other  $N - 1$  walkers are independent and uniformly distributed, so the expected rate at which an infective at  $x$  infects in a fully susceptible room at movement equilibrium is  $\sum_y h(x, y) \bar{\rho} = \beta q_x$ , exactly the rate assumed in (3), for every  $N$ . In the density-dependent (DD) calibration, used only in Section 6.4 and always labelled DD,  $\bar{\rho}$  in (5) is frozen at its value  $\bar{\rho}_{\text{ref}}$  for a reference occupancy; the hazard of a pair is then fixed and the contact rate becomes  $\beta q_x \bar{\rho} / \bar{\rho}_{\text{ref}}$ , proportional to occupancy. Infectives recover at rate  $\gamma$ : the model is of SIR type with an exponential infectious period of mean  $1/\gamma$  and no latent class.

The process is a continuous-time Markov chain with three kinds of event — a hop, a recovery, and the infection of a susceptible located at  $y$ , at rate  $\sum h(x_i, y)$  summed over the current positions  $x_i$  of the infectives — and is simulated exactly by the direct method of Gillespie [35, 36]; there is no time step and hence no discretisation error. The infection tree (infectior, generation, time and cell of every infection) is recorded, so that secondary cases are counted, not fitted. The algorithm and the verification of the simulator are in Supplementary Section S11.

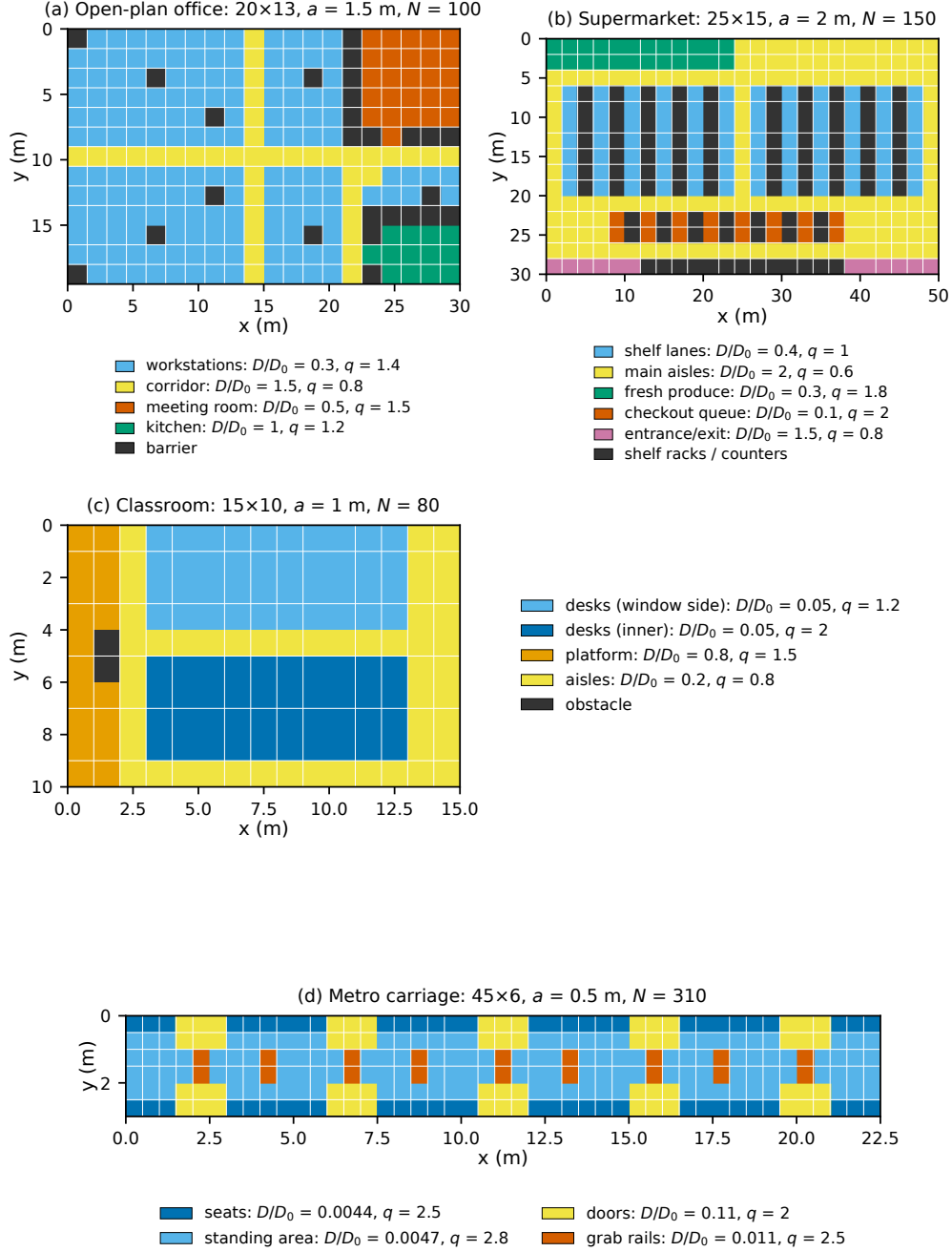
Model (3) is the mean-field approximation of this process: it replaces the numbers of susceptibles and infectives in each cell by their expectations and neglects their correlations. The two models agree on the infection pressure of a case: a case born at  $x$  exerts the expected pressure  $\beta \mathbb{E}_x \int_0^\tau q(X_t) dt$  over its infectious period  $\tau$ , which is the column sum  $n_{\text{off}}(x)$  of the next-generation matrix of Section 3. What the mean field ignores is that a discrete person can be infected only once and that an infective keeps meeting the same few neighbours (Section 5). The approximation becomes accurate when many people share a contact neighbourhood and mobility is high. In the office scene with 1% of the people initially infective, the largest difference between the mean simulated prevalence curve and the solution of (3), both as fractions of  $N$ , is 0.42 for  $N = 100$  and  $D_0 = 1$  (200 runs), 0.058 for  $N = 1000$  and  $D_0 = 10$  (200 runs), and 0.004 for  $N = 5000$  and  $D_0 = 100$  (40 runs; checked only by re-running the same code). The normalisation (5) by the mean number of *other* people matters. A rule that divides instead by the instantaneous number of all people in the cell, the infective included, defines a different process in which an infective alone in its cell transmits to nobody; at the occupancies of the scenes it is sub-critical (Supplementary Remark S4.1).

## 2.4 Three reproduction numbers

Three quantities are called reproduction numbers in this article.

1. The *basic reproduction number*  $R_0 = \rho(\mathbf{K})$  of the mean-field model, the spectral radius of the next-generation matrix  $\mathbf{K} = \mathbf{F}(\gamma \mathbf{1} - \mathbf{M})^{-1}$  (Theorem 3.1). It is the threshold of (4):  $R_0 > 1$  if and only if the growth rate  $\lambda_1 = s(\mathbf{J})$  is positive.
2. The *pair-level* numbers of the individual-based model (Definition 5.2):  $R_1(x)$ , the expected number of secondary cases of a lone index case born at  $x$  when the other  $N - 1$  people are susceptible and only the index transmits; its average  $\bar{R}_1$  over a uniformly placed index; and the spectral radius  $\rho(\mathbf{K}_1)$  of a pair-level next-generation matrix.  $R_1(x)$  is exact for the first generation. None of the three is a threshold quantity (Remark 5.7).
3. The *counted* number  $\hat{R}$ : the mean number of people infected directly by the index case in simulation, read from the infection tree. It depends on the law of the index cell (uniform, or weighted by the Perron vector of  $\mathbf{K}$ ) and on whether later cases also transmit and so compete for susceptibles; both are stated wherever  $\hat{R}$  is quoted.

In the office scene at  $D_0 = 1$  they are  $R_0 = 5.107$ ,  $\bar{R}_1 = 2.266$  and  $\hat{R} = 2.265 \pm 0.004$  (standard error; 400,000 runs, uniform index, only the index transmits).



**Figure 1:** Floor plans of the four scenes. Each small square is one lattice cell of side  $a$ ; colours mark the zones, whose mobility multiplier  $D/D_0$  and infection efficiency  $q$  are given in the legends; dark cells are barriers and are not part of  $\Omega$ . The panel titles give the lattice size in cells, the spacing  $a$  and the number of people  $N$ . The arrangement of the zones and the zone values are illustrative assumptions (see text).

**Table 2:** The four scenes: lattice size in cells, spacing  $a$ , number  $n$  of walkable cells, number  $N$  of people, mean number  $\bar{\rho} = (N - 1)/n$  of other people per cell, means over the walkable cells of the mobility multiplier  $d = D/D_0$  and of the infection efficiency  $q$ , the well-mixed value  $\beta\langle q \rangle/\gamma$  and the upper bound  $\beta q_{\max}/\gamma$  of  $R_0$  for  $\beta = 0.5 \text{ day}^{-1}$  and  $\gamma = 0.14 \text{ day}^{-1}$ , and the mean number of cells in the 1 m contact kernel (1.0 means the same-cell kernel). The zone values of  $d$  and  $q$  are in Supplementary Table S1; all of them are illustrative assumptions, and none is a measurement.

| Scene          | lattice        | $a$ (m) | $n$ | $N$ | $\bar{\rho}$ | $\langle d \rangle$ | $\langle q \rangle$ | $\beta\langle q \rangle/\gamma$ | $\beta q_{\max}/\gamma$ | kernel cells |
|----------------|----------------|---------|-----|-----|--------------|---------------------|---------------------|---------------------------------|-------------------------|--------------|
| Office         | $20 \times 13$ | 1.5     | 234 | 100 | 0.423        | 0.561               | 1.300               | 4.643                           | 5.357                   | 1.0          |
| Supermarket    | $25 \times 15$ | 2       | 278 | 150 | 0.536        | 1.239               | 0.914               | 3.263                           | 7.143                   | 1.0          |
| Classroom      | $15 \times 10$ | 1       | 148 | 80  | 0.534        | 0.192               | 1.318               | 4.706                           | 7.143                   | 4.6          |
| Metro carriage | $45 \times 6$  | 0.5     | 270 | 310 | 1.144        | 0.028               | 2.536               | 9.056                           | 10.000                  | 11.1         |

## 2.5 The four scenes

The model is evaluated on four scenes: an open-plan office, a supermarket, a classroom and a metro carriage (Table 2, Fig. 1). The same scene files are used for the theory and for the simulations. The lattice sizes and spacings, the zones with their values of  $d$  and  $q$ , the numbers of people and the arrangement of the zones on the floor are illustrative assumptions; they are not measurements and are not derived from any standard. In the office the zones cover 60, 15, 10, 5 and 10 % of the 260 cells of the floor (workstations, corridor, meeting room, kitchen and barriers; the 234 walkable cells exclude the barriers); in the supermarket shelf racks and counters act as barriers; the classroom desks are split into a window block ( $q = 1.2$ ) and an inner block ( $q = 2.0$ ); and the carriage has doors, seats and grab rails. The metro mobilities follow a crowding rule  $d = c n_d^{-\eta}$  with  $\eta = 2$  in the standing area and  $\eta = 1$  elsewhere, evaluated at the crowd density  $n_d = 310/67.5 \text{ m}^2 = 4.59 \text{ people per m}^2$ ; their area mean is 0.028. The quantity that enters the bounds of Section 3 is the mean of  $q$  over walkable cells, which is 1.300, 0.914, 1.318 and 2.536. The 1 m kernel is the same-cell kernel in the office and the supermarket; it contains the cell and up to four neighbours in the classroom (4.6 cells on average) and up to 13 cells in the carriage (11.1 on average). Results for the same-cell kernel in these two scenes are reported alongside as a sensitivity case.

## 2.6 Parameters and physical regime

The default epidemiological parameters are  $\beta = 0.5 \text{ day}^{-1}$  and  $\gamma = 0.14 \text{ day}^{-1}$ , a mean infectious period of 7.1 days and  $\beta/\gamma = 3.571$ . They are illustrative values and are not calibrated to any disease or dataset. The same holds for the default mobility scale  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ , and this value matters for every result below. It is used as a reference low-mobility value at which the spatial effects of the model are visible; most results are also given at  $D_0 = 10$  and  $100 \text{ cell}^2 \text{ day}^{-1}$ , and  $R_0$  of every scene at walking mobility (Table 3).

**What  $D_0 = 1$  means.** In the office ( $a = 1.5 \text{ m}$ ),  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$  is  $2.25 \text{ m}^2$  per day. The root-mean-square displacement  $\sqrt{4Dt}$  of a walker with  $d = 1$  is then 3 m in a day and 8 m in a mean infectious period, in a room of  $30 \text{ m} \times 19.5 \text{ m}$ . A walker in the meeting room of this scene (26 cells) leaves it at the principal rate  $\mu_Z = 0.00877 \text{ day}^{-1}$ : its residence time  $1/\mu_Z$  is 114 days, sixteen infectious periods. In the metro carriage at  $D_0 = 1$  a passenger makes on average 0.075 hops of 0.5 m per day: the carriage is frozen on the time scale of an infectious period.

**Walking mobility.** For a walker whose velocity autocorrelation decays as  $\langle \mathbf{v}(0) \cdot \mathbf{v}(t) \rangle = v^2 e^{-t/\tau}$ , the diffusion coefficient in two dimensions is  $D = \frac{1}{2} \int_0^\infty \langle \mathbf{v}(0) \cdot \mathbf{v}(t) \rangle dt = v^2 \tau / 2$  [59, 93]. With an assumed walking speed  $v = 1.3 \text{ m s}^{-1}$ , a typical value for pedestrians [102], and an assumed persistence time  $\tau = 1.5 \text{ s}$ , this gives  $D = 1.27 \text{ m}^2 \text{ s}^{-1}$ , that is,  $4.9 \times 10^4 \text{ cell}^2 \text{ day}^{-1}$  for  $a = 1.5 \text{ m}$ ,

**Table 3:** Basic reproduction number  $R_0$  of the mean-field model ( $\beta = 0.5 \text{ day}^{-1}$ ,  $\gamma = 0.14 \text{ day}^{-1}$ ) for the mobility scales  $D_0$  (in  $\text{cell}^2 \text{ day}^{-1}$ ) used in this article and for walking mobility; in brackets and in the columns headed ‘excess’, the excess of  $R_0$  over the well-mixed value  $\beta\langle q\rangle/\gamma$  in per cent. For the office and the supermarket the 1 m kernel is the same-cell kernel; for the classroom and the metro carriage both kernels are given. The walking values of  $D_0$  are  $v^2\tau/2$  with  $v = 1.3 \text{ m s}^{-1}$  and  $\tau = 1.5 \text{ s}$ , converted with the lattice spacing of each scene, for people who walk 5 % of the time and all the time.

| Scene          | kernel    | $\beta\langle q\rangle/\gamma$ | $R_0$ (excess, %) at $D_0 =$ |              |              | walking 5 %       |        | walking           |         |
|----------------|-----------|--------------------------------|------------------------------|--------------|--------------|-------------------|--------|-------------------|---------|
|                |           |                                | 1                            | 10           | 100          | $D_0$             | excess | $D_0$             | excess  |
| Office         | same cell | 4.643                          | 5.107 (10.0)                 | 4.676 (0.72) | 4.646 (0.06) | $2.4 \times 10^3$ | 0.0026 | $4.9 \times 10^4$ | 0.00013 |
| Supermarket    | same cell | 3.263                          | 4.394 (34.6)                 | 3.391 (3.93) | 3.275 (0.37) | $1.4 \times 10^3$ | 0.0268 | $2.7 \times 10^4$ | 0.00134 |
| Classroom      | 1 m       | 4.706                          | 5.866 (24.7)                 | 4.931 (4.79) | 4.731 (0.53) | $5.5 \times 10^3$ | 0.0098 | $1.1 \times 10^5$ | 0.00049 |
|                | same cell |                                | 6.182 (31.4)                 | 4.960 (5.42) | 4.733 (0.57) | $5.5 \times 10^3$ | 0.0105 | $1.1 \times 10^5$ | 0.00053 |
| Metro carriage | 1 m       | 9.056                          | 9.241 (2.1)                  | 9.164 (1.20) | 9.078 (0.25) | $2.2 \times 10^4$ | 0.0013 | $4.4 \times 10^5$ | 0.00006 |
|                | same cell |                                | 9.836 (8.6)                  | 9.308 (2.79) | 9.086 (0.33) | $2.2 \times 10^4$ | 0.0016 | $4.4 \times 10^5$ | 0.00008 |

and  $2.4 \times 10^3 \text{ cell}^2 \text{ day}^{-1}$  if people walk 5 % of the time. Over the four scenes these walking values lie between  $1.4 \times 10^3$  and  $4.4 \times 10^5 \text{ cell}^2 \text{ day}^{-1}$  (Table 3):  $D_0 = 1$  is three to almost six orders of magnitude below the mobility of people who walk.

**Consequence.** At  $D_0 = 1$  the heterogeneity of  $q$  raises  $R_0$  above the well-mixed value  $\beta\langle q\rangle/\gamma$  by 2 to 35 % (1 m kernel), depending on the scene. At walking mobility the excess is below 0.003 % in the office and below 0.03 % in every scene: in the mean-field model a closed room in which people walk is well mixed, and  $R_0 = \beta\langle q\rangle/\gamma$  for all practical purposes. All spatial effects in closed rooms reported in Sections 3–6 are large only at low mobility, and this holds for the pair-level correction of Section 5 as well. Leaks through a localised door are the exception: their effect grows with mobility (Section 4.2). Results are therefore given for the low default  $D_0 = 1$  and for  $D_0 = 10$  and 100, where the room approaches the well-mixed limit; the later sections refer back to this paragraph and do not repeat it. A small  $D_0$  could be read as a stand-in for people who stay at a desk or a seat, but a random walk without home positions is not an adequate model of such people: a slow walker drifts away and does not return, and the stationary occupancy is uniform over all walkable cells, corridors and aisles included. Section 7.7 shows the consequence against recorded contacts, at the mobility that matches recorded contact durations: a free walk calibrated to the amount and duration of contact in six indoor settings gives each person too many partners in five of the six settings and no group structure in any.

**Closed room.** Unless stated otherwise, each scene is a closed room occupied 24 hours a day by the same  $N$  people for the whole epidemic. This is an idealisation for the office and the classroom, and it is not a physical scenario for the supermarket and the metro carriage, for which the scene files assume visits of 27 and 20 minutes; Section 6.4 gives the corresponding numbers per visit and for a room that is occupied during part of the day.

### 3 The basic reproduction number of the mean-field model

This section determines the basic reproduction number  $R_0$  of the mean-field model (3), its bounds and its dependence on mobility. Throughout, Assumption A holds,  $\mathbf{V} = \gamma\mathbf{1} - \mathbf{M}$ , the column index of a matrix is the source cell, and inequalities between vectors or matrices are entrywise. Each statement names the contact kernel it needs: most require the same-cell kernel,  $\mathbf{F} = \beta\mathbf{Q}$ , with  $q \geq 0$ ,  $q \neq 0$ . The main results are known for epidemic patch models (Section 1, Table 1), and no novelty is claimed for them. Complete proofs from standard facts on non-negative matrices are given in Supplementary Section S2, because they fix the conditions under which each statement holds (which kernel, which symmetry of the hop rates). The zone bound, the

slow-mixing coefficient and the edge-wise monotonicity are elementary consequences of the same matrix theory; a targeted search (Crossref queries on patch-model reproduction numbers and dispersal, 1 October 2026) found no source for them, which does not establish that there is none.

**Theorem 3.1** (Next-generation matrix). *Let Assumption A hold, let  $\mathbf{F} \geq 0$ ,  $\mathbf{F} \neq 0$ ,  $\mathbf{V} = \gamma \mathbf{1} - \mathbf{M}$ , and define*

$$\mathbf{K} = \mathbf{FV}^{-1}, \quad R_0 = \rho(\mathbf{K}), \quad \lambda_1 = s(\mathbf{J}), \quad \mathbf{J} = \mathbf{M} + \mathbf{F} - \gamma \mathbf{1}. \quad (6)$$

- (a)  $\mathbf{V}^{-1} = \int_0^\infty e^{-\gamma t} e^{\mathbf{M}t} dt > 0$  entrywise. For  $\mathbf{F} = \beta \mathbf{P}^\top \mathbf{Q}$ ,  $K_{yx} = \beta \sum_z P_{zy} q_z \int_0^\infty e^{-\gamma t} P_t(z|x) dt$  is the expected number of infections generated in cell  $y$  by one infective introduced in cell  $x$  of a fully susceptible room; for the same-cell kernel,  $K_{yx} = \beta q_y \int_0^\infty e^{-\gamma t} P_t(y|x) dt$ .
- (b)  $R_0 > 0$ , and  $R_0$  is the unique  $R > 0$  with  $s(\mathbf{M} + \mathbf{F}/R - \gamma \mathbf{1}) = 0$ .
- (c)  $\lambda_1 < 0 \iff R_0 < 1$ ,  $\lambda_1 = 0 \iff R_0 = 1$ ,  $\lambda_1 > 0 \iff R_0 > 1$ .
- (d) Every solution of (3) satisfies  $I(t) \leq e^{\mathbf{J}t} I(0)$  componentwise for all  $t \geq 0$ . Hence if  $R_0 < 1$  the infective population is bounded by a decaying exponential from  $t = 0$  on,  $\mathbf{1}^\top I(t) \leq C e^{\lambda_1 t} \mathbf{1}^\top I(0)$  with  $\lambda_1 < 0$  and  $C$  independent of the initial condition. If  $R_0 > 1$  the disease-free state is unstable and the linearised solution grows like  $e^{\lambda_1 t}$ .

In a closed SIR model the infectives vanish for every value of  $R_0$ ; the content of (d) is that for  $R_0 < 1$  there is no growth at any time, whereas for  $R_0 > 1$  a small seed grows until the depletion of susceptibles stops it. The column sums of  $\mathbf{K}$  are the expected numbers of secondary cases of an index case introduced in cell  $x$ , the offspring number  $n_{\text{off}}(x)$ ; for every row-stochastic kernel

$$n_{\text{off}}(x) = \sum_y K_{yx} = \beta \mathbb{E}_x \int_0^\tau q(X_t) dt, \quad \frac{\beta q_{\min}}{\gamma} \leq n_{\text{off}}(x) \leq \frac{\beta q_{\max}}{\gamma}, \quad \sum_x \pi_x n_{\text{off}}(x) = \frac{\beta \langle q \rangle_\pi}{\gamma}, \quad (7)$$

where  $X_t$  is the walk started at  $x$  and  $\tau$  its exponential infectious period. In the office scene at  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ ,  $n_{\text{off}}(x)$  ranges from 3.995 to 5.335 over the cells, with uniform mean 4.643. The next-generation matrix is  $\mathbf{K} = \mathbf{F} \hat{G}(\gamma)$ , where  $\hat{G}(\gamma) = (\gamma \mathbf{1} - \mathbf{M})^{-1}$  is the Laplace transform of the lattice propagator at the recovery rate; for reversible movement  $R_0$  also has a variational (Rayleigh) form and a modal matrix form in the eigenmodes of the movement generator, whose largest diagonal entry is only a lower bound (Supplementary Theorem S2.4 and Corollary S2.5).

**Theorem 3.2** (Bounds; same-cell kernel). *Let Assumption A hold and  $\mathbf{F} = \beta \mathbf{Q}$  with  $q \geq 0$ ,  $q \not\equiv 0$ . Then, for every irreducible generator (reversible or not),*

$$\frac{\beta \langle q \rangle_\pi}{\gamma} \leq R_0 \leq \frac{\beta q_{\max}}{\gamma}, \quad \beta \langle q \rangle_\pi - \gamma \leq \lambda_1 \leq \beta q_{\max} - \gamma. \quad (8)$$

*If  $q$  is constant all four are equalities; if  $q$  is not constant all four inequalities are strict.*

By (7),  $\beta \langle q \rangle_\pi / \gamma$  is the mean number of secondary cases of an index case placed according to the stationary distribution.  $R_0$  is larger because secondary cases are not born at stationarity: they are born preferentially in cells of high  $q$  and begin their infectious period there. Mobility erases this memory of the birthplace (Theorem 3.3). With the same-cell kernel there is therefore no mechanism by which spatial separation could lower the mean-field  $R_0$  below the well-mixed value  $\beta \langle q \rangle_\pi / \gamma$ ; what separation does to discrete individuals is a different question (Sections 5 and 6).

**Theorem 3.3** (Mobility; same-cell kernel). *Let  $\mathbf{M} = D_0 \mathbf{M}_0$  with  $\mathbf{M}_0$  an irreducible generator with stationary distribution  $\pi$ , and  $\mathbf{F} = \beta \mathbf{Q}$ . Then*

- (a)  $\lim_{D_0 \rightarrow \infty} R_0(D_0) = \beta \langle q \rangle_\pi / \gamma$ ;

- (b)  $\lim_{D_0 \rightarrow 0^+} R_0(D_0) = \beta q_{\max}/\gamma$ ;
- (c)  $\lambda_1(D_0)$  is convex and non-increasing, and  $R_0(D_0)$  is non-increasing; if  $q$  is not constant,  $R_0(D_0)$  is strictly decreasing on  $(0, \infty)$ .

No reversibility is needed.

**Proposition 3.4** (Expansions). *In the setting of Theorem 3.3:*

- (i) as  $D_0 \rightarrow \infty$ ,  $R_0(D_0) = \frac{\beta \langle q \rangle_\pi}{\gamma} + \frac{\beta C}{\langle q \rangle_\pi D_0} + O(D_0^{-2})$ , where  $C = \mathbf{q}^\top \mathbf{y} \geq 0$  and  $\mathbf{y}$  solves  $-\mathbf{M}_0 \mathbf{y} = \mathbf{Q} \boldsymbol{\pi} - \langle q \rangle_\pi \boldsymbol{\pi}$ ,  $\mathbf{1}^\top \mathbf{y} = 0$  (for symmetric rates,  $C = n^{-1} \delta q^\top (-\mathbf{M}_0)^+ \delta q$  with  $\delta q = \mathbf{q} - \langle q \rangle \mathbf{1}$  and  $(\cdot)^+$  the pseudo-inverse);
- (ii) as  $D_0 \rightarrow 0$ ,  $R_0(D_0) = \frac{\beta q_{\max}}{\gamma} \left(1 - \frac{D_0 \mu_Z^0}{\gamma}\right) + o(D_0)$ , where  $Z = \{x : q_x = q_{\max}\}$  and  $\mu_Z^0 = -s(\mathbf{M}_0[Z, Z])$ .

**Proposition 3.5** (Zone bound). *Let Assumption A hold and  $\mathbf{F} = \beta \mathbf{Q}$ . For every non-empty  $Z \subset \Omega$ ,*

$$R_0 \geq \frac{\beta q_Z}{\gamma + \mu_Z}, \quad q_Z = \min_{x \in Z} q_x, \quad \mu_Z = -s(\mathbf{M}[Z, Z]) \geq 0. \quad (9)$$

Here  $\mu_Z$  is the principal decay rate of a walker that is killed when it steps out of  $Z$ . By Proposition 3.4(ii) the zone bound with  $Z = \{x : q_x = q_{\max}\}$  is sharp to first order as  $D_0 \rightarrow 0$ . An expression of the form  $\beta q/(\gamma + \mu)$  thus does occur in the closed room, but as a lower bound contributed by a high-risk zone from which infectives escape into lower-risk space, with  $q$  and  $\mu$  those of the zone and not of the room.

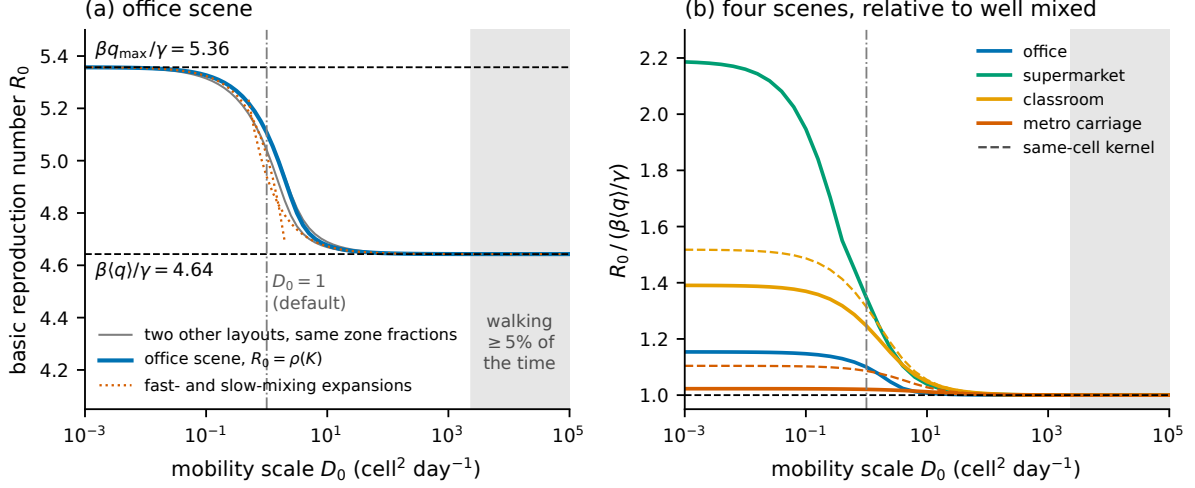
**Theorem 3.6** (Edge-wise monotonicity; symmetric rates, same-cell kernel). *If  $\omega_{x \rightarrow y} = \omega_{y \rightarrow x}$  for all edges and  $\mathbf{F} = \beta \mathbf{Q}$ , then  $R_0$  and  $\lambda_1$  are non-increasing functions of each individual edge rate. Consequently, under rule (1) and for a fixed set of cells  $\Omega$ , lowering  $D$  in any set of cells or closing edges (walls between cells) cannot lower  $R_0$ .*

The theorem concerns operations that keep the set of cells. Turning cells into barrier cells removes them, and their  $q$ , from  $\Omega$ ; this changes  $\langle q \rangle_\pi$  and both bounds of Theorem 3.2, and  $R_0$  can then fall (Section 4.4). The symmetry is essential: under a departure-site rule a change of  $D$  also changes the stationary occupancy, and in 300 random instances multiplying  $D$  by 3 in one cell raised  $R_0$  in 201, lowered it in 98 and left it unchanged (to  $10^{-10}$ ) in one.

**Proposition 3.7** (Contact kernels). *Let  $\mathbf{F} = \beta \mathbf{P}^\top \mathbf{Q}$  with  $\mathbf{P}$  row-stochastic. Then*

- (i) *Theorem 3.1 holds;*
- (ii)  $\beta q_{\min}/\gamma \leq R_0 \leq \beta q_{\max}/\gamma$ , and  $R_0 = \beta q_0/\gamma$  if  $q \equiv q_0$ ;
- (iii)  $R_0(D_0) \rightarrow \beta \langle q \rangle_\pi/\gamma$  as  $D_0 \rightarrow \infty$  and  $R_0(D_0) \rightarrow \rho(\beta \mathbf{P}^\top \mathbf{Q})/\gamma$  as  $D_0 \rightarrow 0$ ;
- (iv) *the lower bound  $\beta \langle q \rangle_\pi/\gamma \leq R_0$  of Theorem 3.2 and the monotonicity of Theorem 3.3(c) can fail. Example: three cells in a row, hop rate  $D_0$  between neighbours,  $\mathbf{P}$  uniform over a cell and its neighbours,  $q = (1, 0, 1)$ : then  $R_0(D_0) = (\beta/\gamma)(\gamma + 4D_0)/(2\gamma + 6D_0)$ , which is  $0.5 \beta/\gamma$  at  $D_0 = 0$ , below  $\beta \langle q \rangle/\gamma = 0.667 \beta/\gamma$ , and increases with  $D_0$ .*

With a contact kernel, an infective in a high- $q$  cell infects people in neighbouring cells; if those cells have low  $q$  and nobody moves, the new cases transmit little, and the size bias that raises  $R_0$  for the same-cell kernel is reversed. In the two scenes in which the 1 m kernel differs from the same-cell kernel the lower bound happens to hold ( $5.866 \geq 4.706$  in the classroom and  $9.241 \geq 9.056$  in the metro carriage at  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ ), but it is not guaranteed.



**Figure 2:** Mean-field basic reproduction number  $R_0 = \rho(K)$  against the mobility scale  $D_0$  ( $\text{cell}^2 \text{ day}^{-1}$ ), for  $\beta = 0.5 \text{ day}^{-1}$  and  $\gamma = 0.14 \text{ day}^{-1}$ ;  $R_0$  does not depend on the number of people  $N$  (frequency-dependent incidence), and no simulation is involved. (a) Office scene (234 walkable cells, same-cell kernel):  $R_0$  (thick line), the bounds  $\beta\langle q\rangle/\gamma$  and  $\beta q_{\max}/\gamma$  of Theorem 3.2 (dashed), the two expansions of Proposition 3.4 (dotted), and two other floor plans with the same zone fractions (thin grey). (b) The four scenes, as the ratio of  $R_0$  to the well-mixed value  $\beta\langle q\rangle/\gamma$ , with the 1 m contact kernel (solid) and the same-cell kernel (dashed; the two coincide for the office and the supermarket). The dash-dotted vertical line marks the default  $D_0 = 1$ ; the shaded band is  $D_0 \geq 2.4 \times 10^3$ , which for the lattice spacing of the office ( $a = 1.5 \text{ m}$ ) is the mobility of people who walk at least 5% of the time (Section 2.6; the values for the other scenes are in Table 3).

### 3.1 The scenes

For every floor plan with the zone fractions of the office scene ( $\langle q \rangle = 1.300$ ,  $q_{\max} = 1.5$ ), Theorem 3.2 gives  $4.643 \leq R_0 \leq 5.357$ . In the scene itself  $R_0 = 5.107$  at  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ , with growth rate  $\lambda_1 = 0.602 \text{ day}^{-1}$  (doubling time 1.15 days); two other floor plans with the same zone fractions give 5.109 and 5.041. The curve  $R_0(D_0)$  decreases from the upper to the lower bound (Fig. 2a), and the two expansions of Proposition 3.4 are  $R_0 \simeq 4.643 + 0.296/D_0$  as  $D_0 \rightarrow \infty$  and  $R_0 \simeq 5.357(1 - 0.0626 D_0)$  as  $D_0 \rightarrow 0$  ( $D_0$  in  $\text{cell}^2 \text{ day}^{-1}$ ). The first is accurate to 0.004 at  $D_0 = 10$  and to  $3 \times 10^{-5}$  at  $D_0 = 100$ , the second to 0.003 at  $D_0 = 0.1$ . The set  $Z$  of the slow-mixing expansion is the meeting room (26 cells,  $q = 1.5$ ), whose quasi-stationary residence time at  $D_0 = 1$  is  $1/\mu_Z = 114$  days; its zone bound (9) is 5.041. These are properties of this floor plan, in which the meeting room is connected to the corridor through a single cell. Over the four scenes the excess of  $R_0$  over the well-mixed value at  $D_0 = 1$  is largest in the supermarket (+34.6%), where  $q_{\max}$  is more than twice  $\langle q \rangle$  (Table 3, Fig. 2b).

For a two-zone room, a workspace ( $q_W = 1.5$ ,  $D = 0.3 D_0$ ) and a central corridor band ( $q_C = 0.5$ ,  $D = 2 D_0$ ) on a  $20 \times 13$  lattice with  $\langle q \rangle = 1.2$ , Theorem 3.2 gives  $R_0 \geq 1.2 \beta/\gamma$ : separating a low-risk corridor from a high-risk workspace cannot lower the mean-field  $R_0$ . Numerically  $R_0 = 4.972 = 1.39 \beta/\gamma$  at  $D_0 = 1$ ,  $1.24 \beta/\gamma$  at  $D_0 = 10$  and  $1.20 \beta/\gamma$  for  $D_0 \geq 100$ . Four other arrangements of the same two zones, with 69–70% workspace, give  $1.43 \beta/\gamma$  (corridor band along one wall) and  $1.22$ – $1.23 \beta/\gamma$  (aisles, a grid of aisles, scattered corridor cells) at  $D_0 = 1$ : the finer the two zones are interleaved, the closer  $R_0$  is to the well-mixed value.

Every statement of this section was tested on random instances of four families of generators, and the proofs were re-derived and the computations re-implemented in the re-check (Supplementary Section S11.2); over 2,000 random instances, for example, the largest violation of the bounds (8) was  $4 \times 10^{-14}$ , at the level of rounding error.

## 4 Room size, leaks and hotspot maps

Section 3 showed that mobility and the heterogeneity of  $q$  together place  $R_0$  between  $\beta\langle q\rangle_\pi/\gamma$  and  $\beta q_{\max}/\gamma$ . This section asks what the geometry of the room adds. None of it is claimed as new: the invariance of Theorem 4.1 is known for patch models whose patches have equal reproduction numbers [31]; loss through an absorbing boundary is the critical-patch-size mechanism of Skellam [16, 84]; the quasi-stationary objects of Theorem 4.2 go back to Darroch and Seneta [22]; and Theorem 4.4 is Perron–Frobenius theory together with the sensitivity of a simple eigenvalue [11, 46]. For the leaky-room formulation as stated here we know of no source; we searched only for the patch-model results of Section 1. Proofs are in Supplementary Section S3.

### 4.1 Closed rooms: no finite-size effect

**Theorem 4.1** (Closed room, uniform  $q$ ). *Let Assumption A hold,  $q \equiv q_0$  and  $\mathbf{F} = \beta\mathbf{P}^\top\mathbf{Q}$  with any row-stochastic kernel. Then  $R_0 = \beta q_0/\gamma$  and  $\lambda_1 = \beta q_0 - \gamma$  for every room size, shape, barrier layout, mobility field and mobility scale, and every index case causes  $n_{\text{off}}(x) = \beta q_0/\gamma$  secondary cases in expectation in the mean-field model, wherever it starts.*

Reflecting walls lose nobody: all column sums of  $\mathbf{K}$  equal  $\beta q_0/\gamma$ . Equivalently, for uniform  $q$  the modal matrix of Supplementary Corollary S2.5 is diagonal with entries  $\beta q_0/(\gamma + \mu_j)$ , and the largest is the entry of the uniform mode,  $\mu_0 = 0$ ; the first non-zero eigenvalue  $\mu_1 = 2D[1 - \cos(\pi/L)] \approx \pi^2 D/L^2$  of a square room of side  $L$  (cells) plays no role. Square rooms with  $L = 3, \dots, 40$  and  $D = 0.5 \text{ cell}^2 \text{ day}^{-1}$  give  $\rho(\mathbf{K}) = \beta q/\gamma$  to within  $6 \times 10^{-15}$  in relative terms, and forty random rooms with barrier cells, mobilities spread over four decades, symmetric and non-symmetric generators and random contact kernels reproduce  $R_0 = \beta q_0/\gamma$ ,  $\lambda_1 = \beta q_0 - \gamma$  and  $n_{\text{off}}(x) = \beta q_0/\gamma$  to within  $10^{-13}$ . In a closed room, geometry therefore enters  $R_0$  only through the interplay of mobility with the heterogeneity of  $q$ .

### 4.2 Leaky rooms

A genuine boundary effect requires that people leave. Suppose that a person in cell  $x$  leaves the room at rate  $\kappa_x \geq 0$  (through a door cell, an open side, or after a dwell time that does not depend on position) and is replaced at once by a susceptible person entering through the same cell, so that the occupancy stays at  $N^*$ ; only infections that take place inside the room are counted. An infective then follows the killed walk with generator  $\mathbf{M}_\kappa = \mathbf{M} - \text{diag } \kappa$ .

**Theorem 4.2** (Leaky room; same-cell kernel). *Let Assumption A hold,  $\mathbf{F} = \beta\mathbf{Q}$ ,  $\kappa \geq 0$ ,  $\kappa \not\equiv 0$ , and put  $\mathbf{M}_\kappa = \mathbf{M} - \text{diag } \kappa$ ,  $\mathbf{V}_\kappa = \gamma\mathbf{1} - \mathbf{M}_\kappa$ ,  $R_0^{\text{room}} = \rho(\beta\mathbf{Q}\mathbf{V}_\kappa^{-1})$  and  $\mu_{\text{leak}} = -s(\mathbf{M}_\kappa)$ . Let  $\mathbf{u}_\kappa, \ell_\kappa > 0$  be the right and left Perron vectors of  $\mathbf{M}_\kappa$  and  $\nu = (\ell_\kappa \circ \mathbf{u}_\kappa)/(\ell_\kappa^\top \mathbf{u}_\kappa)$ . Then*

- (1) *Theorem 3.1(a)–(c) holds with  $\mathbf{M}$  replaced by  $\mathbf{M}_\kappa$ ;*
- (2)  *$0 < \mu_{\text{leak}} \leq \langle \kappa \rangle_\pi := \sum_x \pi_x \kappa_x$ ;  $\mu_{\text{leak}}$  is non-decreasing in every  $\kappa_x$ ; and for  $\kappa = k \mathbf{1}_\Delta$  with a door set  $\Delta \subsetneq \Omega$ ,  $\mu_{\text{leak}} \uparrow \mu_\Delta^{\text{abs}} := -s(\mathbf{M}[\Omega \setminus \Delta, \Omega \setminus \Delta])$  as  $k \rightarrow \infty$ ;*
- (3) *if  $q \equiv q_0$ , then  $R_0^{\text{room}} = \beta q_0/(\gamma + \mu_{\text{leak}})$ ;*
- (4) *in general,  $\beta\langle q \rangle_\nu/(\gamma + \mu_{\text{leak}}) \leq R_0^{\text{room}} \leq \beta q_{\max}/(\gamma + \mu_{\text{leak}})$ , where  $\langle q \rangle_\nu = \sum_x \nu_x q_x$ .*

$\mathbf{u}_\kappa$  is the quasi-stationary distribution of a walker conditioned on still being in the room [22],  $1/\mu_{\text{leak}}$  its asymptotic mean residence time, and  $\nu$  the stationary law of the walk conditioned never to leave. As  $\kappa \rightarrow 0$ ,  $\nu \rightarrow \pi$  and (4) reduces to Theorem 3.2. Two cases are explicit.

**Corollary 4.3** (Explicit cases). (a) Uniform leak (dwell time). If  $\kappa_x \equiv 1/T$ , then  $\mu_{\text{leak}} = 1/T$ ,  $\nu = \pi$ , and every result of Section 3 for the same-cell kernel holds for  $R_0^{\text{room}}$  with  $\gamma$  replaced by  $\gamma + 1/T$ ; in particular

$$\frac{\beta \langle q \rangle_\pi T}{1 + \gamma T} \leq R_0^{\text{room}} \leq \frac{\beta q_{\text{max}} T}{1 + \gamma T}, \quad (10)$$

and an infective who enters at a  $\pi$ -distributed position causes  $\beta \langle q \rangle_\pi T / (1 + \gamma T)$  infections in the room in expectation. For any row-stochastic kernel, with  $R_0^{\text{room}} := \rho(\beta \mathbf{P}^\top \mathbf{Q} \mathbf{V}_\kappa^{-1})$ , every result of Section 3 that holds for that kernel holds for  $R_0^{\text{room}}$  with  $\gamma$  replaced by  $\gamma + 1/T$ ; in particular  $R_0^{\text{room}} = \beta q_0 / (\gamma + 1/T)$  when  $q \equiv q_0$ .

(b) Corridor with absorbing ends. For  $L$  cells in a row with hop rate  $w$  between neighbours, uniform  $q$ , and  $\kappa = w$  at the two end cells (a hop beyond either end leaves the room),

$$\mu_{\text{leak}} = 2w \left[ 1 - \cos \frac{\pi}{L+1} \right] = \frac{\pi^2 w}{(L+1)^2} + O(wL^{-4}), \quad R_0^{\text{room}} = \frac{\beta q}{\gamma + 2w \left[ 1 - \cos \frac{\pi}{L+1} \right]}. \quad (11)$$

With lattice spacing  $a$ ,  $w = D/a^2$  and physical length  $\ell = La$ , (11) tends to  $\beta q / (\gamma + \pi^2 D / \ell^2)$ , which is thus the reproduction number of a corridor with *absorbing* ends, not of a closed room. The mechanism is the critical patch size of spatial ecology [16, 84]: growth at rate  $\beta q - \gamma$  competes with loss through the boundary at rate  $\mu_{\text{leak}}$ , and  $R_0^{\text{room}} > 1$  only if the corridor is longer than about  $\pi \sqrt{D / (\beta q - \gamma)}$ . With  $w = 1 \text{ day}^{-1}$  and  $\beta q = 0.5 \text{ day}^{-1}$  this continuum estimate is 5.2 cells, and on the lattice  $R_0^{\text{room}}$  first exceeds one at  $L = 5$  (0.958 at  $L = 4$ , 1.226 at  $L = 5$ ).

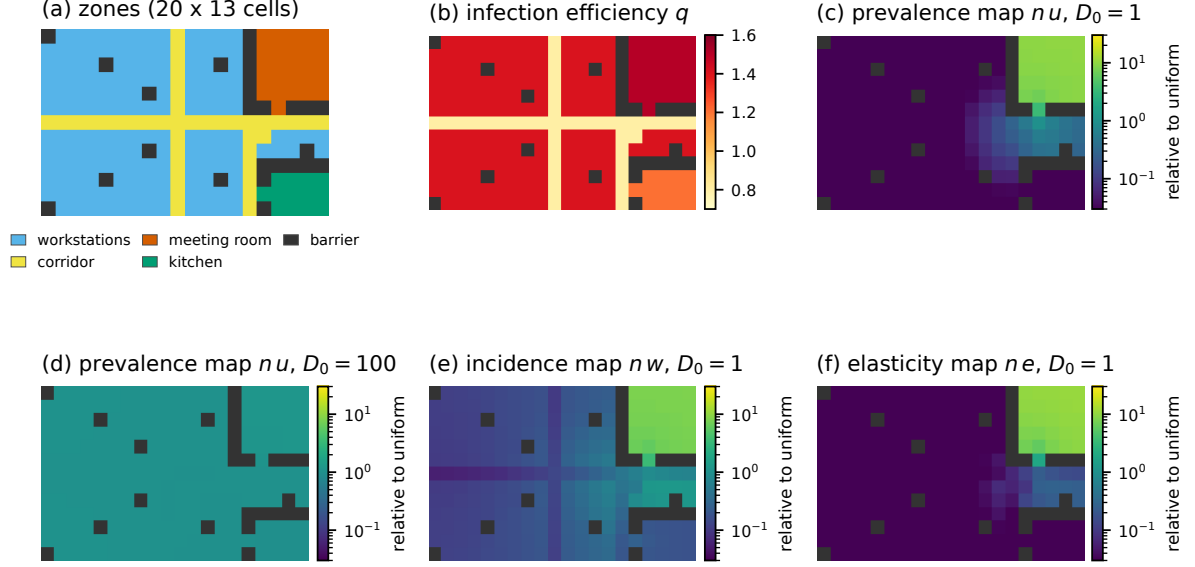
Parts (3) and (4) do not extend to other contact kernels with a localised door: in a three-cell example with a door at one end the formula of part (3) is off by  $-3.0\%$  for a nearest-neighbour kernel, and a skewed kernel exceeds the upper bound of part (4) by  $10.9\%$  for one of the two door positions (Supplementary Remark S3.1). For heterogeneous  $q$  and a localised door the substitution  $\gamma \rightarrow \gamma + \mu_{\text{leak}}$  is not exact either: in the office scene with a door with  $\kappa = 100 \text{ day}^{-1}$  at the end of a corridor arm and  $D_0 = 10$ ,  $R_0^{\text{room}} = 4.220$  where the substitution gives 4.133, below the lower bound 4.139 of part (4), because a door in a low- $q$  zone weights the walk conditioned to stay towards higher  $q$ .

Leaks lower the reproduction number, the more so the smaller the room (Supplementary Fig. S1a). Two cautions apply. First, with  $D = 0.5 \text{ cell}^2 \text{ day}^{-1}$  these leak rates are small ( $\mu_{\text{leak}} \leq 0.1 \text{ day}^{-1}$ , residence times from 10 days to 19 years). For a fixed geometry and a door whose exit rate is itself a hop rate ( $\kappa = D$ )  $\mu_{\text{leak}}$  is proportional to the mobility scale, so at  $D = 2.4 \times 10^3 \text{ cell}^2 \text{ day}^{-1}$  (Section 2.6) the same residence times would lie between 3 minutes and 1.5 days. A door with a fixed exit rate  $\kappa$  behaves differently: its leak rate grows with mobility but stays below  $\langle \kappa \rangle_\pi$ ; in the office with one door cell and  $\kappa = 1 \text{ day}^{-1}$ ,  $R_0^{\text{room}} / R_0$  is 0.99998, 0.977, 0.971 and 0.970 at  $D_0 = 1, 10, 100$  and 2,400. For applications the dwell time  $T$  should therefore be an input, as in Corollary 4.3(a), not the output of a random walk towards a door. Second,  $R_0^{\text{room}}$  counts the infections of one stay. An infective who returns to the same room every day makes many stays during one infectious period, so a per-visit number is not the reproduction number of a setting that is visited repeatedly; Section 6.4 treats daily schedules.

### 4.3 Hotspot maps

**Theorem 4.4** (Hotspot maps; same-cell kernel). Let Assumption A hold,  $\mathbf{F} = \beta \mathbf{Q}$  and  $q_x > 0$  for all  $x$ . Let  $\mathbf{u}, \ell > 0$  be the right and left Perron vectors of  $\mathbf{J}$  ( $\mathbf{1}^\top \mathbf{u} = 1$ ),  $\mathbf{w}, \mathbf{z} > 0$  those of  $\mathbf{K}$  ( $\mathbf{K} \mathbf{w} = R_0 \mathbf{w}$ ,  $\mathbf{z}^\top \mathbf{K} = R_0 \mathbf{z}^\top$ ,  $\mathbf{1}^\top \mathbf{w} = 1$ ), and let  $\xi > 0$  solve  $\beta \mathbf{Q} \xi = R_0 (\gamma \mathbf{1} - \mathbf{M}) \xi$ .

- (i) (prevalence)  $e^{\mathbf{J}t} I(0) = e^{\lambda_1 t} [(\ell^\top I(0) / \ell^\top \mathbf{u}) \mathbf{u} + O(e^{-\delta' t})]$  for every  $\delta' < \delta$ , where  $\delta = \lambda_1 - \max\{\text{Re } \lambda : \lambda \in \sigma(\mathbf{J}) \setminus \{\lambda_1\}\} > 0$ , and with  $\delta' = \delta$  if  $\mathbf{J}$  is diagonalisable, in particular for reversible movement.  $\mathbf{u} \propto \pi$  if and only if  $q$  is constant.



**Figure 3:** Hotspot maps of the office scene in the mean-field model ( $\beta = 0.5 \text{ day}^{-1}$ ,  $\gamma = 0.14 \text{ day}^{-1}$ , same-cell kernel;  $n = 234$  walkable cells of the  $20 \times 13$  lattice). (a) Zones. (b) Infection efficiency  $q$ . (c), (d) Prevalence map  $\mathbf{u}$  (right Perron vector of  $\mathbf{J}$ ) at  $D_0 = 1$  and  $D_0 = 100 \text{ cell}^2 \text{ day}^{-1}$ . (e) Incidence map  $\mathbf{w}$  (right Perron vector of  $\mathbf{K}$ ) and (f) elasticity map  $e_x = \partial \ln R_0 / \partial \ln q_x$  at  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ . Maps are multiplied by  $n$ , so that 1 is the uniform value; the colour scale is logarithmic.

(ii) (incidence)  $\mathbf{K}^g I_0 = R_0^g [(\mathbf{z}^\top I_0 / \mathbf{z}^\top \mathbf{w}) \mathbf{w} + o(1)]$  as  $g \rightarrow \infty$ ;  $\mathbf{w} \propto \mathbf{Q}\xi$ ; and the first generation of an index case placed according to  $\boldsymbol{\pi}$  is  $(\mathbf{K}\boldsymbol{\pi})_x = \beta q_x \pi_x / \gamma$ .

(iii) (elasticity)  $e_x := \partial \ln R_0 / \partial \ln q_x = z_x w_x / (\mathbf{z}^\top \mathbf{w}) \geq 0$  and  $\sum_x e_x = 1$ . For reversible movement  $\mathbf{z} \propto \Pi^{-1} \xi$ ,  $\Pi = \text{diag } \boldsymbol{\pi}$ , and  $e_x \propto q_x \xi_x^2 / \pi_x$ .

(iv) (limits) Let  $\mathbf{M} = D_0 \mathbf{M}_0$ . As  $D_0 \rightarrow \infty$ :  $\mathbf{u} \rightarrow \boldsymbol{\pi}$ ,  $\mathbf{w} \rightarrow \mathbf{Q}\boldsymbol{\pi} / \langle q \rangle_\pi$  and  $e_x \rightarrow q_x \pi_x / \langle q \rangle_\pi$ . As  $D_0 \rightarrow 0$ :  $\mathbf{u}$ ,  $\mathbf{w}$  and  $\mathbf{e}$  concentrate on  $Z = \{x : q_x = q_{\max}\}$ .

The three maps answer different questions:  $\mathbf{u}$  says where infectives are in the exponential phase,  $\mathbf{w}$  where infections are generated, and  $\mathbf{e}$  where a reduction of  $q$  pays, since lowering  $q$  by a small fraction  $\varepsilon$  on a set  $A$  of cells lowers  $R_0$  by the fraction  $\varepsilon \sum_{x \in A} e_x$  to first order. The lowest mode of the diffusion operator, which is constant under reflecting walls, carries no spatial information; the squared amplitude that matters in the reversible case,  $e_x \propto q_x \xi_x^2 / \pi_x$ , is that of the principal vector of a generalised eigenproblem that involves  $q$ .

Figure 3 shows the maps for the office scene, in which the meeting room (26 cells, 11.1 % of the walkable floor of 234 cells,  $q = 1.5$ ) is joined to the rest of the floor by a single lattice edge. At  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$  the meeting room holds 95.9 % of  $\mathbf{u}$ , 79.2 % of  $\mathbf{w}$  and 98.1 % of the elasticity:  $R_0 = 5.107$  is in effect the reproduction number of the meeting room, whose zone bound is 5.041. The elasticity share of the meeting room falls to 20.8 % at  $D_0 = 10$  and 13.4 % at  $D_0 = 100 \text{ cell}^2 \text{ day}^{-1}$ , close to its fast-mixing limit of 12.8 %. The prevalence map is an asymptotic object. It is reached after a time of order  $1/\delta$ , whereas the exponential phase started by one index case among  $N$  people lasts about  $\ln N / \lambda_1$ . At  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ ,  $1/\delta = 16$  days and  $\ln 100 / \lambda_1 = 7.6$  days: in the linearised dynamics a point seed at a workstation far from the meeting room is still at total-variation distance 0.996 from  $\mathbf{u}$  when the total has risen from  $10^2$  to  $10^4$  times the seed, so an outbreak seeded far from the hotspot does not reach the map before saturation (Supplementary Remark S3.3 in Section S3.2). For discrete individuals at this mobility the mean-field maps do not describe where infections occur either (Section 6.3).

#### 4.4 Interventions in the mean-field model

Supplementary Section S3.3 works out interventions on the office scene. Slowing the corridors raises  $R_0$ , as Theorem 3.6 requires (corridor mobility  $\times 0.1$ :  $5.107 \rightarrow 5.222$ ). Walls closing between 10 % of the workstation–workstation edges and as many as connectivity allows raise  $R_0$  by 0.009 % to 0.080 % at  $D_0 = 1$ , and none of 80 random draws lies below the baseline at any of three mobilities. Turning 39 workstation cells into barrier cells removes floor with  $q$  above the mean: the well-mixed value falls by 1.5 %, and  $R_0$  changes by +0.065 % at  $D_0 = 1$ , where it is pinned by the meeting room, and by  $-1.07$  % and  $-1.49$  % at  $D_0 = 10$  and  $100$ . Halving  $q$  in a targeted set of cells chosen by the elasticity map, recomputed as cells are treated, lowers  $R_0$  more than random placement at  $D_0 = 1$  (by 0.35 to 0.48 between 5 % and 75 % treated area), but makes little difference at  $D_0 = 100$ , where targeting reduces to treating the cells with the largest  $q$ . Uniform measures scale  $R_0$  exactly: multiplying  $\beta$  by  $s_\beta$  and every  $q_x$  by  $s_q$  multiplies  $R_0$  by  $s_\beta s_q$ .

All of this belongs to the low-mobility regime. In the mean-field model the closed office is well mixed for  $D_0 \gtrsim 100 \text{ cell}^2 \text{ day}^{-1}$ :  $R_0 = \beta \langle q \rangle_\pi / \gamma$  up to the correction of Proposition 3.4, the maps of Theorem 4.4 reduce to  $\pi$  and  $\mathbf{Q}\pi / \langle q \rangle_\pi$ , and the floor plan enters only through the area-weighted mean of  $q$ . What survives at realistic mobility is the absence of a size effect in a closed room, the proportional scaling in  $\beta$  and  $q$ , and the leak in its dwell-time form (10). A size dependence does exist for discrete individuals, for a different reason; it is the subject of Section 5.

### 5 Discrete individuals: a pair-level theory

The next-generation matrix of Section 3 counts infectious *contacts*: its column sum  $n_{\text{off}}(x)$ , eq. (7), is the expected time-integrated transmission rate of a case born in cell  $x$ . In the mean-field model every contact meets a fresh susceptible. In the individual-based model of Section 2.3 this is not so: each person can be infected only once, and an infective who moves slowly keeps meeting the same few neighbours, so that later contacts with a person already infected are wasted. This local saturation of contacts is classical. It is the reason why epidemics with household structure need reproduction numbers defined for groups and not for individuals [7], why pair approximations on lattices and networks predict slower invasion than mass action [48], and more generally why discrete spatial models differ from their mean-field equations [24]. At the default parameters (0.43 people per cell,  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ ) the effect is a factor of two: a lone index case in the office scene infects on average 2.27 people, against  $R_0 = 5.11$ .

Treating infection as a reaction between two random walkers is not new. Kenkre and Sugaya [51] and Sugaya and Kenkre [89] computed the infection probability of a pair of walkers with and without confinement, and the approach is developed in the monograph of Kenkre and Giuggioli [50]. Closest to what follows, Giuggioli and Sarvaharman [38] map transmission at co-location onto a partially absorbing site of the two-walker lattice walk in discrete time. For one infected walker among  $W$  susceptible walkers with uniformly random initial positions and a geometric infectious period they obtain the mean number of secondary infections as  $W$  times the position-averaged generating function of the transmission probability (their Section 6), call it the basic reproduction number, and check it against simulations in a square domain with reflecting walls. The number  $\bar{R}_1$  defined below is the continuous-time counterpart of that quantity. The solution technique for one partially absorbing site goes back to Szabo et al. [91]; lattice formulations with several reactive defects are given by Giuggioli [37] and Giuggioli and Sarvaharman [38], and the analogous formalism for inert heterogeneities by Sarvaharman and Giuggioli [82]. What is added here is listed in Section 1.

## 5.1 The pair problem

Throughout this section movement follows the symmetric rule (1), so that  $\mathbf{M}$  is symmetric and  $\pi_x = 1/n$ , and the  $N$  people start independently from the uniform distribution. One of them, the index case, becomes infectious at time 0 in cell  $x$  and stays infectious for a time  $\tau$  that is exponential with rate  $\gamma$  and independent of the movement. Consider one other person, who is in cell  $y$  at time 0. The pair  $Z_t = (X_t, Y_t)$  is a Markov chain on  $\Omega \times \Omega$  with generator  $\mathcal{M} = \mathbf{M} \otimes \mathbf{1} + \mathbf{1} \otimes \mathbf{M}$  (column index = source state). While the pair is in state  $(x', y')$  the index infects the other person at the rate  $h(x', y') = \beta q_{x'} P_{x'y'}/\bar{\rho}$  of eq. (5), where  $\bar{\rho} = (N - 1)/n$ .

**Proposition 5.1** (Pair infection probability). *Let  $p(x, y)$  be the probability that the index case infects the given other person before recovering, when no one else transmits. With  $\mathcal{M} = \mathbf{M} \otimes \mathbf{1} + \mathbf{1} \otimes \mathbf{M}$  and  $\mathbf{H} = \text{diag } h$  on  $\Omega \times \Omega$ ,*

$$\mathbf{1} - p = \gamma(\gamma \mathbf{1} + \mathbf{H} - \mathcal{M}^\top)^{-1} \mathbf{1}. \quad (12)$$

Equation (12) is the Feynman–Kac formula  $1 - p(z) = \mathbb{E}_z \exp(-\int_0^\tau h(Z_t) dt)$  for the pair chain killed at rate  $\gamma + h$ : infection is a partially absorbing defect of the two-walker process. Since  $\mathbf{M}$  is symmetric, the matrix in (12) is symmetric positive definite, and the system is solved by conjugate gradients on the  $n^2$  pair states (54,756 for the office scene).

**Definition 5.2** (Pair-level reproduction numbers).  $R_1(x) = (N - 1)n^{-1} \sum_y p(x, y)$  is the expected number of secondary cases of a lone index case born at  $x$  when the other  $N - 1$  people are susceptible, independent and uniformly distributed and only the index transmits;  $\bar{R}_1 = n^{-1} \sum_x R_1(x)$ . The pair-level next-generation matrix  $\mathbf{K}_1$  has entries  $(K_1)_{zx} =$  expected number of people infected by that index case *while they are located in cell  $z$* ; its column sums are  $R_1(x)$ .

$R_1(x)$  is exact for the individual-based model: the  $N - 1$  pairs formed by the index and each other person are identically distributed, and although they are dependent (they share the index's path and infectious period) expectations add. When later cases also transmit, the index infects at most as many people, since some are infected by others first.  $\mathbf{K}_1$  is obtained from  $n$  further linear solves; for the scenes of this article all of them take a few seconds.

**Proposition 5.3** (Saturation). *For every scene, every row-stochastic contact kernel and every  $N \geq 2$ ,  $0 \leq (K_1)_{zx} \leq K_{zx}$  for all  $z, x$ , with strict inequality wherever  $K_{zx} > 0$ . Consequently  $R_1(x) < n_{\text{off}}(x)$  for every  $x$ ,  $\bar{R}_1 < \beta \langle q \rangle / \gamma$  and  $\rho(\mathbf{K}_1) \leq R_0$ .*

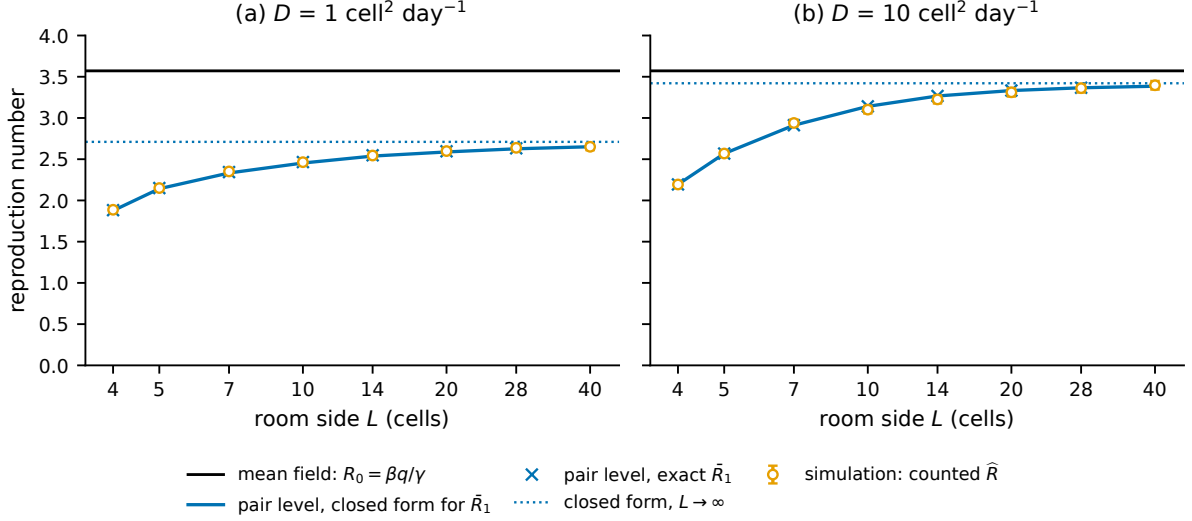
The mean-field matrix thus counts exposure, and the pair-level matrix discounts the exposure of people who are already infected. For the same-cell kernel Theorem 3.2 gives  $\bar{R}_1 < \beta \langle q \rangle / \gamma \leq R_0$ : a uniformly placed lone index case infects fewer people than even the well-mixed lower bound of  $R_0$ .

## 5.2 Homogeneous periodic rooms: a closed form

**Proposition 5.4** (Homogeneous periodic room). *On an  $L_1 \times L_2$  periodic lattice with hop rate  $D$  to each of the four neighbours, uniform  $q$  and the same-cell kernel, let  $h_0 = \beta q / \bar{\rho}$ . Then  $p(x, y) = h_0 g(y - x) / (1 + h_0 g_0)$ , where  $g(r) = \int_0^\infty e^{-\gamma t} \mathbb{P}(\text{relative walk at 0 at time } t \mid \text{start at } r) dt$  and  $g_0 = g(0)$ , and*

$$R_1 = \frac{\beta q / \gamma}{1 + (\beta q / \bar{\rho}) g_0}, \quad g_0 = \frac{1}{n} \sum_k \frac{1}{\gamma + 2\mu_k}, \quad \mu_k = 2D(2 - \cos k_1 - \cos k_2), \quad (13)$$

where  $k = (k_1, k_2)$  runs over  $\frac{2\pi}{L_1} \{0, \dots, L_1 - 1\} \times \frac{2\pi}{L_2} \{0, \dots, L_2 - 1\}$ .  $R_1$  is increasing in  $D$  and in  $\bar{\rho}$ . Limits:  $D \rightarrow \infty$  at fixed  $n$ :  $R_1 \rightarrow (\beta q / \gamma) / (1 + \beta q / (\gamma(N - 1)))$ ;  $D \rightarrow 0$ :  $R_1 \rightarrow \bar{\rho} h_0 / (h_0 + \gamma)$ .



**Figure 4:** Finite-size effect for discrete people. Homogeneous  $L \times L$  rooms with reflecting walls,  $q = 1$ , same-cell kernel,  $\bar{\rho} \approx 0.381$  others per cell ( $N = 7$  at  $L = 4$  to  $610$  at  $L = 40$ ), mobility (a)  $D = 1$  and (b)  $D = 10 \text{ cell}^2 \text{ day}^{-1}$ . Black line: mean-field  $R_0 = \beta q / \gamma = 3.571$ . Blue line: closed form (13) with the reflecting-wall spectrum; crosses: exact pair solve (12) ( $L \leq 28$ ); dotted line: infinite lattice (Corollary 5.5). Circles: counted secondary cases  $\hat{R}$  of a uniformly placed index case when only the index transmits, 20,000 runs per point, bars are 95 % confidence intervals.

Formula (13) has the structure of a renewal argument. If infectious contacts with the given person continued after infection, their expected number would be  $h_0 g(r)$ , the mean-field exposure. Given that there is at least one contact, the pair is co-located at that moment and the expected number of contacts is  $1 + h_0 g_0$ . The ratio of the two is the probability of at least one contact. The two limits are the Markovian epidemic in a well-mixed population of  $N$  people, where each pair interacts at rate  $\beta q / (N - 1)$ , and a frozen room, where the index infects only the  $\bar{\rho}$  people who share its cell on average, each with probability  $h_0 / (h_0 + \gamma)$ . A direct solve of (12) on four periodic lattices reproduces (13) to 13 digits.

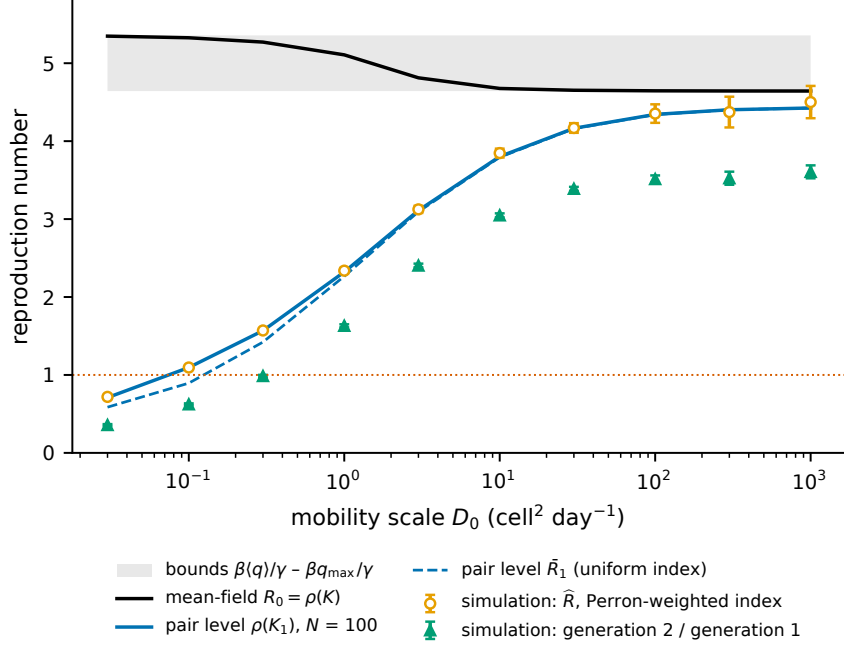
**Corollary 5.5** (Infinite floor). *For  $L_1, L_2 \rightarrow \infty$  at fixed  $\bar{\rho}$ ,  $g_0 = \frac{2}{\pi} \frac{\mathcal{K}(k_*)}{\gamma + 8D}$  with modulus  $k_* = \frac{8D}{\gamma + 8D}$ , where  $\mathcal{K}$  is the complete elliptic integral of the first kind. In particular  $R_1 < \beta q / \gamma$  for every finite  $D$ : the correction does not vanish for large rooms.*

For  $8D \gg \gamma$  the logarithmic singularity of  $\mathcal{K}$  gives  $g_0 \approx \ln(64D/\gamma)/(8\pi D)$ : the correction decays with mobility only as  $\ln D/D$ , which reflects the recurrence of the planar walk.

**Numerical observation 5.6** (Reflecting walls). For a rectangular room with reflecting walls, formula (13) with  $\mu_k = 2D(2 - \cos(k_1\pi/L_1) - \cos(k_2\pi/L_2))$ ,  $k_i = 0, \dots, L_i - 1$  (cell-averaged  $g_0$ ) reproduces  $\bar{R}_1$  from the exact pair solve (12) to within 0.9 % for  $L = 4, \dots, 28$  and  $D = 0.51, 1, 10$  (largest deviation at  $D = 0.51$ ,  $L = 7$ ).

### 5.3 The form of a finite-size effect

For the mean-field model a closed room has no size effect (Theorem 4.1). Discrete people do show a size effect, and (13) is its form. Fig. 4 (numbers in Supplementary Table S6) compares the mean-field number, the closed form, the exact pair solve and simulation in homogeneous  $L \times L$  rooms with 0.381 other people per cell (99 people on the 260 cells of a  $20 \times 13$  lattice without barriers; the office scene, with 234 walkable cells, has 0.423).



**Figure 5:** Reproduction numbers of the office scene against the mobility scale  $D_0$  ( $N = 100$ , same-cell kernel,  $\beta = 0.5 \text{ day}^{-1}$ ,  $\gamma = 0.14 \text{ day}^{-1}$ ). Black line: mean-field  $R_0 = \rho(\mathbf{K})$ ; grey band: its bounds  $\beta\langle q \rangle/\gamma = 4.643$  and  $\beta q_{\max}/\gamma = 5.357$ . Solid blue line: the spectral radius  $\rho(\mathbf{K}_1)$  of the pair-level next-generation matrix; dashed blue line: the average  $\bar{R}_1$  over a uniformly placed index case, which is lower at small  $D_0$ . Circles: counted secondary cases  $\bar{R}$  of an index case born according to the Perron vector of  $\mathbf{K}_1$  when only the index transmits. Triangles: number of second-generation cases divided by the number of first-generation cases when the index and its direct cases transmit. Bars are 95 % confidence intervals; numbers of runs and values in Supplementary Table S7. The dotted line marks the value one.

At  $D = 1$  a lone index case infects on average 1.88 people in a  $4 \times 4$  room with 7 people, 2.60 in a  $20 \times 20$  room with 153 people and 2.71 on an infinite floor at the same occupancy, against 3.57 in the mean-field model at every size. The correction is a lattice Green’s function. It decreases with mobility: at  $L = 10$  the counted number rises from 2.46 at  $D = 1$  to 3.10 at  $D = 10$ . And it does not vanish as  $L \rightarrow \infty$ : the infinite-floor value is 76 % of  $\beta q/\gamma$  at  $D = 1$  and 96 % at  $D = 10$ . The cause is the repeated meeting of the same people, in a small room because there are few of them and on a large floor because a slow planar walker returns to its neighbours; nothing leaks through the walls. The closed form lies inside the 95 % interval of the counted value in 23 of the 24 rooms scanned, and the exact solve in 19 of the 21 rooms for which it was computed; the second simulator agrees with the exact solve within 1.7 standard errors in the seven rooms re-run, including the two exceptions.

#### 5.4 Mobility: opposite trends at the two levels

The mean-field  $R_0$  of the office scene decreases with mobility (Theorem 3.3), from 5.35 at  $D_0 = 0.03$  to 4.64 at  $D_0 = 1000$ ; the pair-level numbers increase,  $\bar{R}_1$  from 0.59 to 4.42: a faster index case meets more distinct people (Fig. 5). At  $D_0 = 1$ ,  $\bar{R}_1 = 2.27$  is 44 % of  $R_0 = 5.11$ ; at  $D_0 = 10$  it is 81 % and at  $D_0 = 100$  it is 93 %. At  $D_0 = 1000$ ,  $\bar{R}_1 = 4.425$  is close to  $(\beta\langle q \rangle/\gamma)/(1 + \beta\langle q \rangle/(\gamma(N - 1))) = 4.435$ , the value for a well-mixed population of 100 people (compare the limit  $D \rightarrow \infty$  in Proposition 5.4). Counted secondary cases agree: samples of 400,000, 400,000 and 100,000 runs give  $2.265 \pm 0.004$ ,  $3.793 \pm 0.006$  and  $4.340 \pm 0.015$  at  $D_0 = 1$ , 10 and 100, against  $\bar{R}_1 = 2.266$ , 3.793 and 4.341. On the infinite homogeneous lattice at  $\bar{\rho} = 0.381$  and  $\beta q = 0.5 \text{ day}^{-1}$  the ratio  $R_1/R_0$  is 0.5 at  $D = 0.24$ , 0.9 at  $D = 3.5$  and 0.99 at  $D = 52 \text{ cell}^2 \text{ day}^{-1}$  (Supplementary Fig. S3). The mean-field number is therefore adequate

when many people share a contact neighbourhood or when people move far enough during an infectious period to meet new ones; the scenes at  $D_0 = 1$  satisfy neither condition. At  $D_0 \approx 2.4 \times 10^3 \text{ cell}^2 \text{ day}^{-1}$  (Section 2.6) the ratio on the infinite lattice is 0.9997, and what remains in a finite room is the well-mixed finite-population factor.

## 5.5 Later generations: not a threshold

The pair level is exact for the first generation and stops there. A secondary case is not a lone index case: it starts in or next to the cell of its infector, who cannot be infected again, and near the other cases its infector has caused, so part of its neighbourhood is already used up. In the office at  $D_0 = 1$  the second generation is 1.63 (1.61–1.65) times the first, against  $\rho(\mathbf{K}_1) = 2.32$ . The ratio of these two numbers is 0.50 at  $D_0 = 0.03$ , 0.70 at  $D_0 = 1$  and 0.80–0.81 for  $D_0 \geq 10$ ; the last value is that of a well-mixed population of 100, in which the index and its direct cases compete for the same susceptibles (0.81 in 200,000 runs of the well-mixed model with the same design).

*Remark 5.7* ( $R_1$  and  $\rho(\mathbf{K}_1)$  are not threshold quantities). Neither  $\bar{R}_1 > 1$  nor  $\rho(\mathbf{K}_1) > 1$  implies that large outbreaks are possible. In the metro carriage with the same-cell kernel at  $D_0 = 1$  (310 people on 270 cells; in the median cell a passenger waits 48 days between hops)  $\rho(\mathbf{K}_1) = 2.24$  and  $\bar{R}_1 = 1.51$ , and the counted number of secondary cases of a lone index case is 1.52 (1.49–1.54; 20,000 runs). Yet the second generation is 0.61 times the first, and no outbreak reached 10 % of the passengers in 2,000 runs of one simulator and 3,000 of another (mean attack rate 1.1 %): the index infects the people in its own cell, and they have no one left to infect. The mean-field value for the same case is  $R_0 = 9.84$ . An intervention must therefore not be judged by ‘ $R_1 < 1$ ’, and  $\rho(\mathbf{K}_1)$  should not be read as a basic reproduction number of the discrete system. That a threshold exists for infection among lattice random walkers is known: rigorously for independent walkers on  $\mathbb{Z}^d$  with recovery and reinfection [54], and from simulations of the diffusive epidemic process, including its variant with permanent immunity [17, 67]. Those results concern infinite or large homogeneous lattices and do not give a computable threshold for the small heterogeneous rooms considered here. No quantity in this article predicts the threshold or the final size of the individual-based model at low mobility; these are obtained by simulation in Section 6.

In summary, three numbers describe the first steps of an outbreak among discrete people at low mobility, and they must be kept apart: the mean-field  $R_0$ , which counts exposure and whose matrix bounds the pair-level one from above (Proposition 5.3); the pair-level  $R_1(x)$ , which is the exact expected number of people a lone index case infects; and the growth from one generation to the next, which is lower again and for which we have no formula.

## 6 Exact stochastic simulation of four indoor scenes

Sections 3–5 gave two reproduction numbers that are computed without simulation: the mean-field  $R_0$  and the pair-level  $\bar{R}_1$ . This section reports what the individual-based model of Section 2.3 does in the four scenes of Table 2, how far each number describes it, and what the model says about interventions. All results are given for  $D_0 = 1, 10$  and  $100 \text{ cell}^2 \text{ day}^{-1}$ , which all lie far below walking mobility (Section 2.6); the departures from mean-field theory reported below are large at  $D_0 = 1$  and small at  $D_0 = 100$ .

### 6.1 Simulator and protocol

The individual-based model is simulated exactly, event by event (Supplementary Fig. S25); secondary cases are counted on the recorded infection tree. Each replicate has its own stream of the xoshiro256\*\* generator [13]. The simulator agrees in distribution with a separately written pure-Python implementation, with the exact final-size law of the one-cell model, with

the next-generation matrix through an exposure identity, and with the mean-field equations at high occupancy and mobility (Supplementary Table S41). The scene results at  $D_0 = 1$  and most of those at  $D_0 = 10$  were in addition re-run with the second simulator; the two agree within Monte Carlo error, the two largest differences being 1.6 and 2.1 standard errors (Supplementary Section S5.1).

People start from the stationary (uniform) distribution and one of them, hence at a uniformly distributed cell, is infectious at time 0. For each scene, mobility and contact kernel, 2,000 epidemics are run until no infective is left. The *attack rate* is the fraction of the  $N$  people ever infected, the index case included. An outbreak is called *major* if its attack rate is at least 10 %, and  $P_{\text{maj}}$  is the fraction of major outbreaks, with a Wilson 95 % interval; the threshold is a convention whose influence is quantified in Supplementary Section S5.2. The counted number of secondary cases  $\hat{R}$  is the mean over 2,000 further runs in which only the index case transmits. The mean-field counterparts are  $R_0$ , the solution of (3), and the outbreak probability  $P_{\text{br}}$  of the branching process in which a case walks with generator  $\mathbf{M}$ , recovers at rate  $\gamma$  and, while in cell  $x$ , produces new cases at rate  $\beta q_x$  placed in cell  $y$  with probability  $P_{xy}$ ; the extinction probabilities  $s_x$  of its lines of descent are the minimal non-negative solution [5] of

$$\gamma(1 - s_x) + \beta q_x s_x \left( \sum_y P_{xy} s_y - 1 \right) + \sum_y \omega_{x \rightarrow y} (s_y - s_x) = 0, \quad P_{\text{br}} = 1 - \frac{1}{n} \sum_{x \in \Omega} s_x \quad (14)$$

## 6.2 Outcomes in the four scenes

**Secondary cases.** The pair-level  $\bar{R}_1$  reproduces the counted  $\hat{R}$  in all 18 rows of Table 4: the largest deviation is 2.3 standard errors (office,  $D_0 = 100$ ) and no other row exceeds 2. The mean-field  $R_0$  does not: at  $D_0 = 1$  the ratio  $\hat{R}/R_0$  is 0.44, 0.45, 0.45 and 0.60 in the office, supermarket, classroom and metro carriage, rising to 0.84–0.93 at  $D_0 = 100$ . Later generations grow more slowly still: over the full epidemics at  $D_0 = 1$  the second generation is 1.29, 1.38, 1.32 and 1.84 times as numerous as the first (second simulator: 1.39, 1.40, 1.29, 1.87), where  $\rho(\mathbf{K}_1) = 2.32, 2.20, 2.83$  and 5.72.

**Low mobility,  $D_0 = 1$ .** The mean-field description fails for every whole-epidemic outcome in the office, the supermarket and the classroom. In the office  $P_{\text{maj}} = 0.496$  (0.474–0.518) against  $P_{\text{br}} = 0.781$ ; major outbreaks infect 44.3 % (95 % interval of the mean 42.9–45.6 %) of the people where the mean-field equations give 98.9 %, and their prevalence peaks at 12.7 % on day 23.2 where the mean-field equations give 45.1 % on day 11.5 from a uniform seed and 32.8 % on day 20.4 from a single cell. The attack-rate distributions are not bimodal: 27 %, 28 % and 43 % of all runs end with an attack rate between 10 % and 50 %, against at most 0.6 % at  $D_0 = 10$ . Spread is local and slow, and outbreaks stall at every size. The metro carriage is closer to its mean field ( $P_{\text{maj}} = 0.879$  against 0.888) although its passengers hardly move, because it holds 1.15 people per cell and the 1 m kernel covers 11.1 cells on average, so that an infective has about 13 people within reach.

**Higher mobility,  $D_0 \geq 10$ .** With the 1 m kernel the mean field is adequate for the outbreak probability and the attack rate:  $P_{\text{br}}$  exceeds  $P_{\text{maj}}$  by 0.01–0.04 at  $D_0 = 10$  and by at most 0.014 at  $D_0 = 100$ , and the attack rate of major outbreaks is 1.5–4.9 percentage points below the mean-field value at  $D_0 = 10$  and within 0.5 point at  $D_0 = 100$ . Peak height and peak day are a different matter: at  $D_0 = 10$  the simulated peaks are lower and about four to five days later than those of the mean-field equations even when these are started from a single cell.

**Outbreak probability.** Early extinction of a chain started by one case occurs in every scene. The number of secondary cases of an index case is over-dispersed, with a variance 2.6 to 5.0 times the mean, and 10 % to 31 % of index cases infect no one; over-dispersion lowers the outbreak

**Table 4:** The four scenes in exact simulation and in mean-field theory, for mobility scale  $D_0$  (cell<sup>2</sup> per day),  $\beta = 0.5 \text{ day}^{-1}$ ,  $\gamma = 0.14 \text{ day}^{-1}$ , frequency-dependent incidence; 2,000 epidemics per row, one uniformly placed index case, closed room.  $R_0$ : mean-field basic reproduction number.  $\bar{R}_1$ : pair-level expected number of secondary cases of a uniformly placed index case (Definition 5.2).  $\hat{R} \pm \text{SE}$ : counted mean number of secondary cases, index case alone transmitting (2,000 further runs).  $P_{\text{maj}}$ : fraction of runs with attack rate at least 10 % (Wilson interval).  $P_{\text{br}}$ : mean-field branching value (14). Attack rate: mean over major outbreaks and over all runs. Peak: mean over major outbreaks of the maximal prevalence (0.5-day grid) and of its day. ODE: attack rate (%) / peak prevalence (%) / peak day of the mean-field equations started from one infective spread uniformly, and peak prevalence (%) / peak day for one infective in a single cell (mean over seed cells; 1 m kernel only). A dash means that there was no major outbreak.

| $D_0$                                                                                                                                                                      | $R_0$ | $\bar{R}_1$ | $\hat{R} \pm \text{SE}$ | $P_{\text{maj}}$ (95 % CI) | $P_{\text{br}}$ | attack rate (%) |      | peak |      | mean-field ODE     |             |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|-------------------------|----------------------------|-----------------|-----------------|------|------|------|--------------------|-------------|--|
|                                                                                                                                                                            |       |             |                         |                            |                 | major           | all  | %    | day  | uniform seed       | point seed  |  |
| <i>Office</i> , $N = 100$ , $n = 234$ , 1 m kernel (identical to the same-cell kernel); $\beta\langle q \rangle/\gamma = 4.64$ , $\beta q_{\text{max}}/\gamma = 5.36$      |       |             |                         |                            |                 |                 |      |      |      |                    |             |  |
| 1                                                                                                                                                                          | 5.11  | 2.27        | $2.26 \pm 0.06$         | 0.496 (0.474–0.518)        | 0.781           | 44.3            | 23.2 | 12.7 | 23.2 | 98.9 / 45.1 / 11.5 | 32.8 / 20.4 |  |
| 10                                                                                                                                                                         | 4.68  | 3.79        | $3.90 \pm 0.09$         | 0.762 (0.743–0.781)        | 0.783           | 96.3            | 73.8 | 39.0 | 16.5 | 99.0 / 45.7 / 11.5 | 44.1 / 12.8 |  |
| 100                                                                                                                                                                        | 4.65  | 4.34        | $4.11 \pm 0.10$         | 0.770 (0.751–0.788)        | 0.784           | 98.8            | 76.4 | 48.4 | 12.6 | 99.0 / 45.9 / 11.5 | 45.9 / 11.6 |  |
| <i>Supermarket</i> , $N = 150$ , $n = 278$ , 1 m kernel (identical to the same-cell kernel); $\beta\langle q \rangle/\gamma = 3.26$ , $\beta q_{\text{max}}/\gamma = 7.14$ |       |             |                         |                            |                 |                 |      |      |      |                    |             |  |
| 1                                                                                                                                                                          | 4.39  | 1.99        | $1.97 \pm 0.05$         | 0.436 (0.414–0.458)        | 0.667           | 42.0            | 19.4 | 10.1 | 31.5 | 93.6 / 28.9 / 17.0 | 23.7 / 25.7 |  |
| 10                                                                                                                                                                         | 3.39  | 2.86        | $2.86 \pm 0.07$         | 0.648 (0.627–0.669)        | 0.685           | 92.1            | 60.0 | 30.6 | 22.6 | 95.4 / 33.4 / 17.5 | 33.1 / 18.0 |  |
| 100                                                                                                                                                                        | 3.28  | 3.15        | $3.04 \pm 0.08$         | 0.686 (0.665–0.706)        | 0.692           | 95.3            | 65.7 | 36.1 | 19.3 | 95.7 / 33.7 / 18.5 | 33.7 / 18.3 |  |
| <i>Classroom</i> , $N = 80$ , $n = 148$ , 1 m kernel (4.6 cells on average); $\beta\langle q \rangle/\gamma = 4.71$ , $\beta q_{\text{max}}/\gamma = 7.14$                 |       |             |                         |                            |                 |                 |      |      |      |                    |             |  |
| 1                                                                                                                                                                          | 5.87  | 2.58        | $2.62 \pm 0.06$         | 0.614 (0.592–0.635)        | 0.767           | 39.2            | 25.2 | 14.1 | 16.3 | 98.4 / 42.6 / 11.0 | 31.7 / 18.1 |  |
| 10                                                                                                                                                                         | 4.93  | 3.72        | $3.72 \pm 0.09$         | 0.750 (0.731–0.768)        | 0.777           | 94.0            | 70.9 | 35.0 | 17.3 | 98.8 / 45.0 / 10.5 | 41.3 / 12.8 |  |
| 100                                                                                                                                                                        | 4.73  | 4.32        | $4.30 \pm 0.10$         | 0.779 (0.760–0.797)        | 0.785           | 98.6            | 77.2 | 48.8 | 12.2 | 99.1 / 46.5 / 11.0 | 46.5 / 11.0 |  |
| <i>Classroom</i> , $N = 80$ , $n = 148$ , same-cell kernel; $\beta\langle q \rangle/\gamma = 4.71$ , $\beta q_{\text{max}}/\gamma = 7.14$                                  |       |             |                         |                            |                 |                 |      |      |      |                    |             |  |
| 1                                                                                                                                                                          | 6.18  | 1.56        | $1.53 \pm 0.04$         | 0.276 (0.256–0.295)        | 0.766           | 19.7            | 7.6  | 8.5  | 15.2 | 98.0 / 40.0 / 11.0 |             |  |
| 10                                                                                                                                                                         | 4.96  | 3.20        | $3.15 \pm 0.07$         | 0.695 (0.674–0.715)        | 0.777           | 89.1            | 62.5 | 29.7 | 20.7 | 98.8 / 44.5 / 10.5 |             |  |
| 100                                                                                                                                                                        | 4.73  | 4.23        | $4.39 \pm 0.10$         | 0.771 (0.753–0.789)        | 0.785           | 98.6            | 76.4 | 48.2 | 12.3 | 99.1 / 46.5 / 11.0 |             |  |
| <i>Metro carriage</i> , $N = 310$ , $n = 270$ , 1 m kernel (11.1 cells on average); $\beta\langle q \rangle/\gamma = 9.06$ , $\beta q_{\text{max}}/\gamma = 10.00$         |       |             |                         |                            |                 |                 |      |      |      |                    |             |  |
| 1                                                                                                                                                                          | 9.24  | 5.58        | $5.59 \pm 0.10$         | 0.879 (0.864–0.893)        | 0.888           | 93.7            | 82.4 | 24.9 | 21.9 | 100.0 / 64.1 / 7.0 | 35.0 / 16.3 |  |
| 10                                                                                                                                                                         | 9.16  | 6.32        | $6.44 \pm 0.13$         | 0.878 (0.863–0.892)        | 0.888           | 98.4            | 86.5 | 27.5 | 21.0 | 100.0 / 64.3 / 7.0 | 36.3 / 15.7 |  |
| 100                                                                                                                                                                        | 9.08  | 7.54        | $7.58 \pm 0.16$         | 0.891 (0.877–0.904)        | 0.889           | 99.9            | 89.0 | 35.3 | 16.5 | 100.0 / 64.6 / 7.0 | 42.2 / 13.8 |  |
| <i>Metro carriage</i> , $N = 310$ , $n = 270$ , same-cell kernel; $\beta\langle q \rangle/\gamma = 9.06$ , $\beta q_{\text{max}}/\gamma = 10.00$                           |       |             |                         |                            |                 |                 |      |      |      |                    |             |  |
| 1                                                                                                                                                                          | 9.84  | 1.51        | $1.56 \pm 0.04$         | 0.000 (0.000–0.002)        | 0.888           | –               | 1.1  | –    | –    | 100.0 / 61.4 / 7.0 |             |  |
| 10                                                                                                                                                                         | 9.31  | 3.00        | $3.05 \pm 0.07$         | 0.349 (0.329–0.371)        | 0.888           | 17.3            | 7.7  | 5.0  | 23.4 | 100.0 / 62.9 / 7.0 |             |  |
| 100                                                                                                                                                                        | 9.09  | 6.10        | $6.24 \pm 0.14$         | 0.858 (0.842–0.873)        | 0.889           | 98.6            | 84.7 | 24.0 | 24.4 | 100.0 / 64.4 / 7.0 |             |  |

probability [64], and a Galton–Watson process fed with the simulated offspring law reproduces  $P_{\text{maj}}$  to within 0.011 wherever the attack-rate distribution is bimodal (Supplementary Table S8).

**The contact kernel** is a modelling choice with large consequences when cells are smaller than the contact range. With the same-cell kernel the metro carriage (0.5 m cells) produces no outbreak at  $D_0 = 1$ : none of 2,000 runs reaches an attack rate of 10 % (none of 3,000 in the second simulator), although  $R_0 = 9.84$ ,  $\bar{R}_1 = 1.51$  and  $\rho(\mathbf{K}_1) = 2.24$  (Remark 5.7). With the 1 m kernel the same carriage is the most explosive scene ( $P_{\text{maj}} = 0.879$ ). Nothing computed in this article predicts the threshold of the discrete system at low mobility; it has to be simulated.

### 6.3 Where infections happen

For a uniformly placed index case the cell in which a case of generation  $g$  is infected has a distribution proportional to  $\mathbf{K}^g \mathbf{1}$  in the mean-field model and to  $\mathbf{K}_1^g \mathbf{1}$  in the pair-level description. In the office at  $D_0 = 1$  the mean-field map is uncorrelated with the simulated one in the first

generation and anti-correlated in the next two ( $r = 0.09, -0.23, -0.37$ ; 40,000 outbreaks), whereas the pair-level map gives  $r = 0.88, 0.82, 0.71$  (Supplementary Table S9 and Fig. S6). The hottest cells are workstation cells that border a corridor, not the meeting room that dominates the mean-field maps of Section 4.3: the meeting room falls from the average rate to 0.68 by the third generation, where the mean field predicts 1.44, consistent with local depletion. A statement such as ‘infection concentrates in the meeting room’ is thus a mean-field statement; for discrete people at low mobility the map has to be taken from  $\mathbf{K}_1$  or from simulation.

## 6.4 Interventions

Two calibrations of the pair hazard (5) are used. Under the *frequency-dependent* (FD) calibration  $\bar{\rho} = (N - 1)/n$  is evaluated for the room at hand, so that an infective in cell  $x$  makes infectious contacts at rate  $\beta q_x$  however crowded the room is. Under the *density-dependent* (DD) calibration  $\bar{\rho}$  is frozen at the value  $\bar{\rho}_{\text{ref}}$  of a baseline room, so that the hazard per pair is fixed and the contact rate grows with crowding. In the mean field

$$R_0^{\text{DD}} = \frac{\bar{\rho}}{\bar{\rho}_{\text{ref}}} R_0^{\text{FD}}, \quad \bar{\rho} = \frac{N - 1}{n}. \quad (15)$$

Which law describes people indoors is an empirical matter [9, 69] that the model cannot decide; every result below says which one is used. Details are in Supplementary Section S5.

**Separating zones.** In a two-zone room ( $20 \times 13$  cells, a workspace on 70 % of the floor with mobility  $0.3 D_0$  and a central corridor band on 30 % with mobility  $2 D_0$ ) a contrast  $c$  redistributes  $q$  at fixed area mean  $\langle q \rangle = 1.2$ ,

$$q_C = 1.2(1 - c), \quad q_W = 1.2(1 + \frac{3}{7}c). \quad (16)$$

By Theorem 3.2,  $R_0 > \beta \langle q \rangle / \gamma = 4.286$  as soon as  $c \neq 0$ ; at  $c = 0.583$  ( $q_W = 1.5$ ,  $q_C = 0.5$ )  $R_0 = 4.972$  at  $D_0 = 1$ , 16 % above the uniform room. For discrete individuals at  $D_0 = 1$  and  $N = 100$  everything that is counted *falls* when the contrast is switched on (Supplementary Fig. S7):  $\hat{R}$  of an index case placed with the Perron vector of  $\mathbf{K}_1$  goes from 2.75 to 2.10 (−24 %), the probability of a major outbreak from 0.541 to 0.460 and the mean attack rate from 0.347 to 0.231 (−33 %). The cause is the coincidence of high efficiency with low mobility: an infective in the workspace keeps meeting the same neighbours, and its contacts fall on people who are already infected. With the same contrast and uniform mobility  $\hat{R}$  rises slightly ( $2.76 \rightarrow 2.82$ ). At  $D_0 = 10$  the contrast changes the outbreak probability and the attack rate by less than 2 %.

**Barriers.** In a uniform room ( $20 \times 13$  cells,  $N = 100$ ,  $q = 1.2$ , same-cell kernel) with randomly placed, nested barrier cells and identical starts in all arms,  $R_0$  stays at 4.29 under FD (Theorem 4.1). At  $D = 1$  the mean attack rate falls by  $0.040 \pm 0.003$  at 10 % barriers and by  $0.223 \pm 0.011$  at 30 % (second simulator:  $0.047 \pm 0.004$  and  $0.235 \pm 0.011$ ), and the peak of major outbreaks is delayed by  $0.81 \pm 0.12$  day at 10 % (second simulator:  $0.82 \pm 0.10$ ), while the outbreak probability falls much less ( $0.70 \rightarrow 0.69 \rightarrow 0.62$ ): barriers act by stopping outbreaks part-way on a fragmented floor, not by preventing take-off. At  $D = 10$ , 10 % barriers leave the attack rate unchanged ( $+0.005 \pm 0.005$ ). Under DD barriers also crowd people together, and at  $D = 10$  the attack rate rises (Supplementary Table S11).

**Occupancy.** Under FD the mean-field  $R_0$  of the office does not depend on the number of people. Discrete people feel occupancy through saturation: at  $D_0 = 1$  the counted  $\hat{R}$  is  $1.47 \pm 0.04$  for  $N = 50$ ,  $2.29 \pm 0.05$  for 100 and  $3.70 \pm 0.09$  for 400, following  $\bar{R}_1 = 1.52, 2.27, 3.65$ ; the elasticity of  $\bar{R}_1$  with respect to  $N$  near  $N = 100$  is  $+0.48$  at  $D_0 = 1$  and  $+0.17$  at  $D_0 = 10$ , against 0 for  $R_0$ . Under DD,  $R_0$  is proportional to  $N - 1$ . Whether a capacity limit works is therefore an assumption about contact behaviour, not an output of the model.

**Uniform measures** (‘ventilation’,  $q \rightarrow q/2$ , and ‘masks’,  $\beta \rightarrow 0.3\beta$ ; both are inputs, not results) scale  $R_0$  exactly ( $5.11 \rightarrow 2.55$  and  $1.53$  at  $D_0 = 1$ ), but the pair-level number less than proportionally ( $\bar{R}_1 = 2.27 \rightarrow 1.51$  and  $1.05$ ), because a lower hazard also wastes fewer contacts. Halving  $q$  on the quarter of the cells with the largest elasticity lowers  $R_0$  more than treating the quarter with the smallest, but for discrete people at  $D_0 = 1$  it makes no measurable difference which quarter is treated (Supplementary Table S12). A pair-level number below one does not mean control: with ventilation, half occupancy and partitions together under DD,  $\bar{R}_1 = 0.82$  and still 16.5 % of the runs reach an attack rate of 10 %.

**Dwell time.** Every number so far assumes that the same closed group stays in the room for the whole infectious period. For a room occupied during a fraction  $f$  of every day, with movement and transmission switched on only during the occupied intervals and recovery proceeding at all times, the linearised model is solved exactly and its threshold parameter is

$$R_0^{(f)} = \rho\left(\mathbf{F}\left(\frac{\gamma}{f} \mathbf{1} - \mathbf{M}\right)^{-1}\right) = f R_0(f D_0) \quad (17)$$

(Supplementary Proposition S5.1 in Section S5.10): a duty cycle scales  $R_0$  by  $f$  and also lowers the effective mobility to  $f D_0$ . The exact pair-level expectation with a schedule follows from a contraction fixed-point equation for the pair chain (Supplementary Remark S5.2, same section). Table 5 gives the numbers. In an office occupied eight hours a day ( $f = 1/3$ ) the mean-field number drops from 5.11 to 1.75 at  $D_0 = 1$ , and the index case infects 0.889 to 0.930 people, depending on when in the working day it becomes infectious; simulation with the schedule imposed explicitly agrees within 1.6 standard errors (2,000,000 runs). With a class that meets one hour a day no run in 2,000 reaches a 10 % attack rate. Counted per visit, an index case causes 0.175 infections in an office day of 8 hours at  $D_0 = 1$ , 0.027 in a one-hour lecture, 0.018 in a 20-minute metro ride and 0.008 in a supermarket visit of 27 minutes, so that the closed-room ranking of the scenes (metro carriage first, Table 4) does not survive realistic dwell times; per-visit numbers are, however, not reproduction numbers of settings that are visited repeatedly, for which the residence-time formulation of multi-setting models applies [12].

**At higher mobility.** The pair-level number is exact for the first generation at any mobility and costs one linear solve, so each effect can be followed to  $D_0 = 2.4 \times 10^3 \text{ cell}^2 \text{ day}^{-1}$ . The effects of the contrast in  $q$  and of barriers under FD shrink below 1 % at  $D_0 = 100$  and below 0.05 % at  $D_0 = 2.4 \times 10^3$  (Supplementary Table S14); simulation at  $D_0 = 100$  agrees. Three things survive: uniform reductions of  $q$  or  $\beta$ , which act on  $\bar{R}_1$  almost in proportion; occupancy under DD (under FD a residue of about 5 % remains for halving the office, a finite-population and not a spatial effect); and the time spent in the room. Epidemics at  $D_0 = 2.4 \times 10^3$  were not simulated.

**Table 5:** Dwell time (1 m kernel, FD, default occupancies). (a) Room occupied by the same closed group for  $24f$  hours of every day:  $R_0$  for continuous occupancy, the duty-cycle value  $R_0^{(f)}$  of (17), the pair-level  $\bar{R}_1$  with  $\gamma \rightarrow \gamma/f$  (‘rule’), the exact pair-level expectation for an index that becomes infectious at the start of an occupied interval or at a uniformly distributed time within it, and the probability of a major outbreak (2,000 epidemics with the schedule; 0.0005 is one run). For the supermarket and the metro carriage the fellow visitors change at every visit; the entry is the well-mixed value  $f\beta\langle q\rangle/\gamma$ , the expected number of people one infective infects there over the infectious period (one 27-minute visit or two 20-minute rides a day). (b) One visit of length  $T$ : expected number of infections caused by one infective; mean field and counted (200,000 runs,  $\pm$  standard error).

| (a) Duty cycle       |       |       |             |                    |              |                     |                   |
|----------------------|-------|-------|-------------|--------------------|--------------|---------------------|-------------------|
| scene, hours per day | $D_0$ | $R_0$ | $R_0^{(f)}$ | $\bar{R}_1$ , rule | exact, start | exact, uniform time | $P(\text{major})$ |
| office, 8            | 1     | 5.11  | 1.754       | 0.889              | 0.930        | 0.889               | 0.061             |
| office, 8            | 10    | 4.68  | 1.596       | 1.335              | 1.397        | 1.336               | 0.302             |
| classroom, 1         | 1     | 5.87  | 0.271       | 0.182              | 0.194        | 0.182               | 0.000             |
| classroom, 1         | 10    | 4.93  | 0.258       | 0.187              | 0.200        | 0.187               | 0.0005            |
| supermarket, 0.45    | 1     | 4.39  |             | 0.061              | –            | –                   | –                 |
| metro carriage, 0.67 | 1     | 9.24  |             | 0.252              | –            | –                   | –                 |

| (b) One visit     |             |            |                     |                     |
|-------------------|-------------|------------|---------------------|---------------------|
| visit             | $T$ (hours) | mean field | counted, $D_0 = 1$  | counted, $D_0 = 10$ |
| office day        | 8.00        | 0.2117     | $0.1753 \pm 0.0009$ | $0.1983 \pm 0.0010$ |
| lecture           | 1.00        | 0.0274     | $0.0270 \pm 0.0004$ | $0.0270 \pm 0.0004$ |
| metro ride        | 0.33        | 0.0176     | $0.0178 \pm 0.0003$ | $0.0171 \pm 0.0003$ |
| supermarket visit | 0.45        | 0.0086     | $0.0083 \pm 0.0002$ | $0.0086 \pm 0.0002$ |

## 7 Confrontation with data

Sections 3–6 are statements about a model. This section asks what data say about it. The data are of four kinds: documented outbreaks (38 records in two datasets; 18 are calibrated on or held out in the two distance-gradient tests, and four more enter the level checks and negative controls of the first test), RFID contact records from six indoor settings, tracer measurements in vehicles and rooms, and an influenza epidemic in 29 schools. Five primary tests were run: two out-of-sample tests against outbreaks, and one analysis each of the contact records, of the tracer measurements and of the zone-level structure. Each follows a protocol fixed before the model under test was run on its data. We call the protocols *pre-specified* and not pre-registered, for three reasons that apply to all of them. They are *data-aware*: the outbreak counts, contact records and school data had been read before the protocols were written, and choices such as strata, priors and exclusions were made with that knowledge. They are *model-blind*: no output of the model under test existed when they were frozen (for the tracer analysis, the outcome of the outbreaks under the other models was already known). And their timing is *self-attested*: they were frozen by SHA-256 hashes of files on the author’s machine, without an external time stamp, version-control history or registry; the protocol files released with the code are redacted copies (every changed line is listed in the repository), and the hashes refer to the unredacted originals, so this timing cannot be checked from the repository alone (Supplementary Section S8). Comparisons added after the results were known are called *post hoc* and are marked as such; Supplementary Table S39 lists every test with its status. Times are in hours and lengths in metres throughout this section.

### 7.1 Outbreak records

The first dataset has 27 records from 31 published sources, each with a registered DOI, 23 of them with counts of cases and of exposed people: five offices and other workplaces, four retail premises, five classrooms and class-like rooms, seven vehicle-cabin records, two restaurants, two

choir or church events, and two records of large settings outside the scope of the model (a cruise ship and worker dormitories), recorded for reference as examples of shared settings that a model of one closed room does not cover; they take part in no comparison (Supplementary Section S6). We found no documented outbreak in a metro carriage with a known number of exposed riders (the search was not exhaustive); bus, coach, train and aircraft cabins stand in for it, and the supermarket has one record with denominators and no geometry. The second dataset, collected after the first test, is restricted to events that publish the position of every exposed person or counts by distance: eight flights [6, 45, 52, 74, 85, 90, 95, 107], a hospital ward observed twice with the same index patient [104, 108], and a meat-processing line [40], which is record O4 of the first dataset with its counts by distance added. The 38 records of the two datasets therefore describe 35 distinct outbreaks: besides the processing line, the two records of the ward and those of the coach and the minibuss (records T2 and T3) are two exposure cohorts of one outbreak each. The pathogens are SARS-CoV-2 (five events), SARS-CoV-1 (three), influenza A(H1N1)pdm09 (two) and *Mycobacterium tuberculosis* (one).

The records themselves show a distance gradient. With ‘near’ defined as within two rows of a source on aircraft (the zone of the contact-tracing rule discussed by Hertzberg and Weiss [44]), the index patient’s own bay or cubicle in the ward, and 8 m on the processing line, the near/far risk ratio pooled over the eleven records of the second dataset is 3.98 (95 % interval 2.32–6.82; random effects, DerSimonian–Laird), with moderate heterogeneity (Cochran’s  $Q = 18.9$  on 10 degrees of freedom,  $p = 0.041$ ;  $I^2 = 0.47$ ; Supplementary Fig. S15); with fixed effects it is 3.16. The two ward records share the index patient and are treated here as independent studies; without the second of them the pooled ratio is 4.64 (2.57–8.40). The 8 m zone is the radius that the source of the processing line singled out as the most significant on these same data; without the line the pooled ratio is 3.30 (1.99–5.45; post hoc), and over the eight flights alone 3.75 (2.01–7.00). By pathogen it is 6.66 (3.38–13.1) for SARS-CoV-2, 2.34 (1.53–3.59) for SARS-CoV-1, and 1.15 (0.35–3.72) on the two influenza flights, which show no detectable gradient. Over the four held-out events of the first test with distance strata the pooled ratio with the same estimator is 2.05 (1.15–3.65; 2.00 with fixed effects), and these four are mutually compatible (Cochran’s  $Q = 3.77$  on 3 degrees of freedom,  $p = 0.29$ ;  $I^2 = 0.20$ ). The two pooled values rest on different selections (all eleven records of the second dataset, calibration events included, against four held-out events of the first) and are not a comparison of settings. No adjustment was made for companions or households seated together, and seat maps are published selectively. This is a statement about data, not a test of any model.

## 7.2 Three models and two generic competitors

In a single event of a few hours there is one infectious person and no second generation. All models give the infection probability of susceptible  $j$  in event  $k$  as

$$p_j = 1 - \exp(-\varepsilon_k X_j), \quad (18)$$

with one infectiousness parameter  $\varepsilon_k$  per event and an exposure  $X_j$  computed on a rectangular lattice with reflecting walls, one site per seat position and physical pitches  $(a_x, a_y)$ ;  $P_t(y|x)$  is the propagator of the symmetric walk with hop rates  $D/a_x^2$  and  $D/a_y^2$ . With the index case at  $x_0$ , susceptible  $j$  at  $x_j$  for a time  $T_j$ , and  $n$  sites,

$$\text{M0 : } X_j = \frac{1}{n} \left[ \frac{T_j}{\Lambda} - \frac{1 - e^{-\Lambda T_j}}{\Lambda^2} \right], \quad (19)$$

$$\text{M1 : } X_j = \int_0^{T_j} \sum_x P_t(x|x_0) P_t(x|x_j) dt = \int_0^{T_j} P_{2t}(x_j|x_0) dt \quad (D = D_p), \quad (20)$$

$$\text{M2 : } X_j = \int_0^{T_j} (T_j - s) e^{-\Lambda s} P_s(x_j|x_0) ds \quad (D = D_{\text{air}}). \quad (21)$$

**Table 6:** The three models compared. Each has one infectiousness parameter  $\varepsilon_k$  per event. The mixing coefficient is shared by all events of a setting class (first test: vehicle cabins, seated rooms; second test: aircraft cabins, rooms); the removal rate  $\Lambda$  is measured or integrated over a prior and is never fitted.

| Model | Transmission                                             | What walks on the lattice | Shared parameter |
|-------|----------------------------------------------------------|---------------------------|------------------|
| M0    | well-mixed air, eq. (19)                                 | nothing                   | none             |
| M1    | between two people on the same site, eq. (20)            | people                    | $D_p$            |
| M2    | inhalation of quanta emitted by the index case, eq. (21) | quanta                    | $D_{\text{air}}$ |

M0 is the transient Wells–Riley model of a well-mixed room [28, 81, 103]. M1 is the model of Section 2: people walk, and transmission occurs only between two people on the same site (the same-cell kernel), so the exposure is their expected co-occupation time; the form (18) is then exact to first order in  $\varepsilon_k$  and, by Jensen’s inequality, an upper bound otherwise. In M2 the index case emits infectious quanta that perform the lattice walk with coefficient  $D_{\text{air}}$  and are removed at rate  $\Lambda$ , the ventilation rate plus  $0.93 \text{ h}^{-1}$  for deposition and inactivation (the values used by Ou et al. [75] after Miller et al. [70]); a susceptible inhales in proportion to the quanta on its site. M0 is the limit  $D_{\text{air}} \rightarrow \infty$  of M2. M2 belongs to the spatially resolved models of airborne exposure of eddy-diffusion type [20, 61, 98]; it shares only the lattice propagator with M1 (Table 6).

Two generic competitors have no lattice and no physics. The *constant-ratio model* gives every susceptible in a near zone  $\varrho$  times the infection pressure of everyone else; near is the stratum of the source in the first dataset and, in the second, within two rows of a source on aircraft. In the first test it entered only after unblinding, with fixed values  $\varrho = 2$  to 3; in the second it is a pre-specified competitor with one  $\varrho$  per setting class calibrated on the same events as M2. The *exponential kernel* takes  $X_j \propto \exp(-d_j/\ell)$ , with  $d_j$  the distance to the source and one length  $\ell$  per setting class; it was added post hoc during the re-check of the second test.

### 7.3 Protocols of the two outbreak tests

*First test.* Vehicle cabins are calibrated on a cohort of contacts of 2,334 index cases on high-speed trains [47] (attack rates in 22 seat-offset cells, the adjacent-seat cell excluded because travelling companions inflate it) and seated rooms on the Guangzhou restaurant [63, 65] (interval-censored counts, because only one infection per neighbouring family is certain to have occurred at the lunch). Four single events (a bus [83], a coach [66, 75], a flight [56] and a classroom [60]) are held out and tested on the split of their cases between a near and a far stratum given the event total, and the Seoul call centre [76] on the join-count  $z$  of its case seats. The score  $S$  is the logarithm of the predictive probability of the observed split and  $\Delta = S(\text{model}) - S(\text{M0})$ . The decision rule requires the pooled  $\Delta$  to be positive with one-sided  $p < 0.05$  under M0 (A1), each event to have a two-sided predictive  $p \geq 0.05$  (A2), the case counts of two events transferred from the calibration settings (a minibus [66] and masked classrooms [43]) to lie inside their 90 % predictive intervals (level checks, A3), and no over-prediction in four groups with few or no cases (negative controls, A4); two or more failures of A2 give outcome W4, a list of what is and is not reproduced and no general statement of agreement. The protocol was frozen 87 seconds before the first model run (Supplementary Section S8).

*Second test.* It was designed around the weakness of the first, which a constant ratio passes (Section 7.4). The split does not use outcomes: the six events caused by pathogens other than SARS-CoV-2, all published before 2020, calibrate; the five SARS-CoV-2 events (four flights and the processing line) are held out. The observable is the vector of case counts over two to four distance strata, given the event total;  $\Delta_0$  and  $\Delta_C$  are the summed log-score differences of M2

**Table 7:** First test, hold-out events: near/far split (event totals fixed, one  $\varepsilon_k$  per event). RR: risk ratio near/far, observed with its 95 % interval, or the ratio of the expected stratum attack rates of a model;  $p$ : two-sided predictive  $p$ -value;  $\Delta$ : log-score difference to M0. M1 is evaluated in the exact simulator for the bus, the flight and the classroom. Sources: bus [83], coach [66, 75], flight [56], classroom [60].

| Event (near stratum)          | Near vs far   | Observed RR      | M0     | M1                     |          | M2                      |                    |
|-------------------------------|---------------|------------------|--------|------------------------|----------|-------------------------|--------------------|
|                               |               |                  | $p$    | RR; $p$                | $\Delta$ | RR; $p$                 | $\Delta$           |
| Bus (rows 5–11)               | 14/33 vs 9/34 | 1.60 (0.81–3.19) | 0.20   | 4.0; n.a. <sup>a</sup> | +0.30    | 2.4; 0.38               | +0.32              |
| Coach (rear rows)             | 3/19 vs 4/26  | 1.03 (0.26–4.06) | 1.00   | 34; 0.0002             | −7.64    | 11; 0.007               | −3.94              |
| Flight ( $\leq 2$ seats away) | 11/12 vs 1/8  | 7.33 (1.16–46.2) | 0.0008 | 1.4; 0.009             | +2.38    | 3.6; 0.61               | +6.03              |
| Classroom (front rows)        | 8/10 vs 4/14  | 2.80 (1.16–6.78) | 0.036  | 1.4; 0.31 <sup>b</sup> | +2.02    | 5.5; 0.068 <sup>c</sup> | +0.43              |
| Pooled                        |               |                  |        |                        | −2.94    |                         | +2.83 <sup>d</sup> |

<sup>a</sup> Observed total unattainable:  $\mathbb{P}(\text{total} \geq 23) = 0.0002$ ; conditional on the total,  $p = 0.46$ . <sup>b</sup> Only 2 of 24 posterior draws can reach 12 cases. <sup>c</sup> Not a genuine prediction: rejected at the posterior mode ( $p = 0.0009$ ); see text. <sup>d</sup> Exact one-sided  $p = 2.5 \times 10^{-4}$  under M0; −3.19 without the flight.

against M0 and against the constant-ratio model. The rule: B1,  $\Delta_0 > 0$  with one-sided  $p < 0.05$  under M0; B2,  $\Delta_C > 0$  with one-sided  $p < 0.05$  under the constant-ratio model (the mirror statement,  $\Delta_C < 0$  with  $p < 0.05$  under M2, means that the constant ratio predicts better); B3, every held-out event has a two-sided predictive  $p \geq 0.05$ ; two or more failures of B3 give the outcome V5, a list of the events reproduced and of those not. Computed after calibration and before any held-out stratum count was scored, the power was (with the frozen propagator; the corrected one described below gives the values in brackets): if the calibrated M2 were true, B1 and B2 would both be met with probability 0.96 (0.93; 0.82 on the four flights alone); if outbreaks followed a constant ratio, with probability 0.03 (0.05), while B1 alone would be passed with probability 0.65 (0.73), so that only B2 can count as evidence for the kernel.

*One deviation from the frozen code* affects both tests. The frozen propagator is a sum over cosine modes whose absolute error reaches about  $10^{-10}$  of the largest exposure in an event; at small mixing coefficients the true exposures of distant receivers are smaller than that, and the mode sum returns round-off noise of either sign in their place, which makes far infections look possible. This was found after the held-out events of the second test had been scored. All results below use a propagator that keeps the mode sum where it is resolved and replaces it elsewhere by a positive sum of image terms; it agrees with a third method (uniformisation) to better than  $10^{-8}$  relative in the classroom of the first test, and with the image sum to about  $10^{-4}$  in the ward of the second, where both are resolved. The correction changes no number for the flights and no outcome of either rule; the values obtained with the frozen propagator are reported in Supplementary Sections S7.3 and S10.2.

## 7.4 First test

On the train cohort both spatial models gain 26.6–26.7 units of log-likelihood over M0 for one more parameter; M2 gives  $D_{\text{air}} = 25 \text{ m}^2 \text{ h}^{-1}$  (90 % posterior interval 9.7–43; here and below the posterior is a pseudo-posterior proportional to the profile likelihood under a flat prior on  $\log_{10} D$ , Supplementary Section S7.2) and M1 gives  $D_p = 0.50 \text{ m}^2 \text{ h}^{-1}$  (0.31–0.79). Both are nevertheless rejected by the data they were calibrated on (deviance 72.6–72.9 on 20 degrees of freedom): the isotropic kernel cannot follow the anisotropy of the train table. For the restaurant  $D$  is identified only within three decades, and the fitted M2 conflicts with the tracer-gas measurement of Li et al. [63] (Supplementary Section S7.2).

Table 7 gives the hold-out results. For M2: *Flight VN54* is reproduced (11 of the 12 passengers seated at most two seats from the index case and one of the 8 farther away were infected; M0 is rejected,  $p = 0.0008$ , and M2 is not,  $p = 0.61$ ), robustly to the calibration and to the ventilation prior; this is the one robust positive result of the first test, and it is thin: 20 exposed

passengers, among them four companion couples seated side by side. The *bus* is compatible but not discriminating. The *Hunan coach* is a failure: infection was uniform along the coach (3/19 rear, 4/26 front), and M2 puts six of the seven cases in the rear half (predicted ratio 11,  $p = 0.007$ ), in 10 of the 12 pre-specified sensitivity variants. The *Marin classroom* is a pass that is not a prediction: the predictive  $p$  of 0.068 comes from the diffuse upper tail of the restaurant posterior, where the kernel is nearly flat across the room; at the posterior mode the observation is rejected ( $p = 0.0009$ ), and in 83 % of the posterior draws the emission rate needed to produce the 12 cases exceeds  $10^4$  quanta  $\text{h}^{-1}$ . With the frozen propagator this  $p$  was 0.11 (Section 7.3). The *Seoul call centre* is a failure as well. On its published seat map the 79 case seats among the 137 desks of the affected wing show no clustering at desk scale (join-count  $z = -0.64$ ; random relabelling, one-sided  $p = 0.76$ ), whereas the restaurant-calibrated M2 predicts strong clustering (median  $z \approx 11$ ); integrated over the posterior, the probability of a  $z$  as small as observed is 0.007 or 0.013, depending on how posterior draws are weighted, below the limit 0.025 of the central 95 % interval. M0 is compatible ( $P = 0.26$ ). The protocol estimated this probability from 40 posterior draws (0.031 and 0.017, on either side of the limit); the quadrature that replaces the estimate was added in the re-check after the results were known, and its value depends on the upper limit of the flat prior on  $\log_{10} D$ : 0.007 and 0.013 with the limit  $10^4 \text{ m}^2 \text{ h}^{-1}$  used throughout, 0.013 and 0.024 if the prior is extended to  $10^6 \text{ m}^2 \text{ h}^{-1}$  (post hoc).

The pooled score of M2 is  $\Delta = +2.83$  (exact one-sided  $p = 2.5 \times 10^{-4}$  under M0), so A1 is passed; it rests on the flight, without which  $\Delta = -3.19$ . Two adequacy failures give the outcome W4 for M2. And the rule can be passed without any lattice: a model with no calibration that assigns the same near/far risk ratio of 2, 2.5 or 3 to every event scores  $\Delta = +7.6$ ,  $+7.7$  and  $+7.1$ , higher than M2, with no single-event failure (a post-hoc comparison). Passing A1 shows that a proximity gradient exists in these data; it does not show that the lattice kernel or its calibration adds predictive value.

*The same-site model* M1 fails the pooled test ( $\Delta = -2.94$ ) and three single events, and its outcome is W4 as well; the failure on the bus rests on rule P4 of the post-hoc log, adopted after the results, which counts an event whose observed total the exact simulator cannot reach as a failure (Supplementary Section S8). Calibrated on the trains,  $D_p = 0.50 \text{ m}^2 \text{ h}^{-1}$ : a seated passenger changes site a few times per hour. At such mobilities the mean number of cases stays below the observed number in three of the four held-out events however large the transmission rate, because the attack rate saturates at the probability that a susceptible ever shares a site with the index case: on the bus the mean is at most about 9 cases (23 observed), and the observed totals are reached with probability 0.0002 on the bus, 0.025 on the flight and 0.081 in the classroom (Supplementary Table S21). A larger mobility would lift this ceiling but is excluded by the train data, and at walking mobility the room is well mixed and M1 loses its spatial structure.

## 7.5 Second test

*Calibration.* On the four pre-2020 flights the constant ratio is  $\varrho = 3.2$  (90 % posterior interval 1.6–5.3) and the mixing coefficient of M2 is  $D_{\text{air}} = 126 \text{ m}^2 \text{ h}^{-1}$  (52–438). In the ward the ratio is 3.5 (1.8–5.5) and the coefficient  $794 \text{ m}^2 \text{ h}^{-1}$  (302–4,390), close to the well-mixed limit, because the two ward datasets contradict each other: the students, exposed for 40 minutes, ask for a steep kernel and the inpatients, exposed for days, for a flat one.

*Against the well-mixed model: passed.* Over the five held-out events  $\Delta_0 = +14.35$  (no value as large in 200,000 draws under M0;  $+14.30$  in the re-check, with its own geometry code and propagator), and  $+10.19$  on the four flights alone; every held-out event contributes (Table 8).

*Against the constant ratio: not passed.* Pooled,  $\Delta_C = -9.15$ , a value that M2 did not produce in 200,000 draws if it were true: by the mirror criterion the constant ratio predicts better. Nine-tenths of it ( $-8.38$ ) is the processing line, where the comparison is not between equals, because the near zone of 8 m is the radius that the source singled out on these same data; with a

**Table 8:** Second test: held-out events (event totals fixed, one  $\varepsilon_k$  per event; kernel and ratio calibrated on the six pre-2020 events). Expected numbers of cases per stratum under the constant-ratio model (CR) and under M2; two-sided predictive  $p$  of the observed vector under M0, CR and M2;  $\Delta_0$  and  $\Delta_C$ : log score of M2 minus that of M0 and of CR.

| Event                      | Near vs far   | Observed by stratum (at risk) | Expected, CR       | Expected, M2       | $p$ : M0 / CR / M2                                 | $\Delta_0$ | $\Delta_C$ |
|----------------------------|---------------|-------------------------------|--------------------|--------------------|----------------------------------------------------|------------|------------|
| Flight to Naha [95]        | 7/23 vs 7/117 | 3, 4, 1, 6 (12, 11, 30, 87)   | 2.7, 2.4, 2.3, 6.6 | 3.8, 2.7, 4.5, 3.1 | 0.0050 / 0.61 / 0.047                              | +1.27      | -2.91      |
| Sydney–Perth [85]          | 8/46 vs 3/52  | 4, 4, 2, 1 (32, 14, 22, 30)   | 5.5, 2.4, 1.3, 1.8 | 5.9, 1.9, 2.0, 1.2 | 0.10 / 0.41 / 0.30                                 | +1.27      | -0.28      |
| Dubai–Auckland, EK448 [90] | 4/13 vs 0/71  | 1, 3, 0 (4, 9, 71)            | 0.4, 1.0, 2.6      | 0.8, 1.5, 1.7      | $5.2 \times 10^{-4}$ / 0.020 / 0.21                | +5.69      | +1.82      |
| Tel Aviv–Frankfurt [45]    | 2/20 vs 0/51  | 2, 0, 0, 0 (15, 5, 17, 34)    | 0.8, 0.3, 0.3, 0.6 | 1.1, 0.3, 0.4, 0.2 | 0.11 / 0.76 / 1.00                                 | +1.96      | +0.59      |
| Meat-processing line [40]  | 17/26 vs 3/52 | 5, 12, 1, 2 (9, 17, 22, 30)   | 4.0, 7.6, 3.5, 4.8 | 2.9, 4.8, 5.7, 6.5 | $1.6 \times 10^{-7}$ / 0.12 / $1.4 \times 10^{-5}$ | +4.16      | -8.38      |
| Four flights               |               |                               |                    |                    |                                                    | +10.19     | -0.77      |
| All five events            |               |                               |                    |                    |                                                    | +14.35     | -9.15      |

zone of 4 m M2 is ahead on the line by 2.2, and with 12 m it trails by 0.6 (post hoc). On the four held-out flights the two models are tied:  $\Delta_C = -0.77$ , with  $p = 0.16$  for ‘M2 better’ and  $p = 0.065$  for ‘constant ratio better’, in a test that had a power of 0.82 to show M2 better if it were true. Over the 15 pre-specified sensitivity variants  $\Delta_C$  on the flights runs from  $-3.42$  to  $+1.21$ : the sign depends on definitions of outcome, strata and near zone that the data cannot fix.

*Adequacy: two failures.* Two of the five held-out events lie outside the predictive distribution of M2: the flight to Naha ( $p = 0.047$ , marginal and fragile;  $p = 0.25$  if households are counted once) and the processing line ( $p = 1.4 \times 10^{-5}$ ), where the attack rate is flat out to 8 m and then drops, whereas the ward-calibrated kernel is almost well mixed. The outcome is V5. The competitors fail as well: the constant ratio on one flight ( $p = 0.020$ ) and the well-mixed model on three events.

*The same-site model* was scored in the second test in a closed form only, and its protocol fixed the reading in advance: the first test had shown with the exact simulator that the closed form overstates the reach of M1 (pooled  $-11.11$  against  $-2.94$  for the simulated model), so a pass of the closed form would not bear on M1 and a failure would confirm the first test. The closed form passes against the well-mixed model ( $\Delta_0 = +15.45$ , largely through the processing line), is worse than the constant ratio pooled ( $\Delta_C = -8.06$ ) and on the flights ( $-4.63$ ), and fails on three of the five held-out events.

*A physics-free kernel does as well.* In a post-hoc comparison the exponential kernel, calibrated in the same way (decay length on aircraft 2.5 m), scores  $\Delta_0 = +18.0$ ; M2 is ahead of it by 0.31 on the flights, which is not significant, and behind it by 3.7 over the five events. Nothing in this test distinguishes the lattice propagator, or the removal of aerosol by ventilation, from a generic smooth decay with distance. Further pre-specified secondary analyses, among them the transfer of the train kernel of the first test to the eight flights (it beats the well-mixed model and is rejected on four flights), are in Supplementary Section S9.1.

## 7.6 Mixing coefficients and levels

Fitted alone, the vehicle events of the first test ask for mixing coefficients that differ by orders of magnitude:  $0.06$ – $0.1 \text{ m}^2 \text{ h}^{-1}$  for the flight, 25 for the trains, 63 for the bus and at least  $6,300 \text{ m}^2 \text{ h}^{-1}$  for the coach; a common value is rejected (likelihood ratio 9.52 on 3 degrees of freedom, approximate  $p = 0.023$ , a post-hoc test). In the second test a common coefficient is

rejected for the three room events (likelihood ratio 17.4 on 2 degrees of freedom, approximate  $p = 1.6 \times 10^{-4}$ ) and, more weakly, for the eight flights (15.2 on 7, approximate  $p = 0.034$ ); these approximate  $p$ -values use a chi-square reference whose conditions are not met when maxima lie on the edge of the grid. No setting class has a single coefficient, and none transfers between classes. One infectiousness parameter is fitted per event, in both tests, so no attack rate in this section is a prediction; the emission rates implied by five single-index events and one cluster with several instructors of the first dataset span a factor of 76 (53 if only the confirmed choir cases are counted; Supplementary Table S22). Published outbreaks are selected for being large, and their level cannot be predicted from a typical index case.

## 7.7 Recorded contacts

Does the movement mechanism of the model, a random walk with contact on a shared site, describe who meets whom indoors? Six open SocioPatterns datasets of face-to-face proximity recorded with wearable RFID sensors at a resolution of 20 seconds [18] were used: an office building in 2013 and in 2015 [33, 34], a primary school [88], a hospital ward [97], a high school [68] and a scientific conference [33, 87], with 75 to 403 people over 2 to 10 days. Each model is calibrated on the first half of the days of a dataset and predicts the second half. The walk of Section 2, in discrete time with one site per contact range, has two parameters (the number of sites and the hop probability), fixed by the contact time per jointly present pair and by the mean contact duration. Its competitors are a well-mixed model and a model with a constant ratio of contact within and between the recorded groups. Seven families of contact statistics are scored, and an SIR epidemic is run on the recorded and on the model-generated contacts at the same total contact time in four scenarios; the movement component is validated on a dataset if it meets the epidemic criterion and at least six of the seven families (Supplementary Section S10.3).

The walk fails on all six datasets (Supplementary Table S31). It passes one or two of the seven families on each, namely those it was calibrated on, and does not reproduce who meets whom: people in the walk have too many partners on five of the six datasets (6.0 per person and day against 4.6 in the office of 2013, 90 against 47 in the primary school), and contact within the recorded groups is 10 to 33 % of contact time in the walk and 57 to 94 % in the data. Because the walk spreads the same contact time over too many, too equal partners, the index case infects more people on the walk’s contacts than on the recorded ones in all 24 scenarios, by 25 to 72 % on average per setting. The two models without space do neither better nor worse, and the outcome is the lowest of the protocol’s ladder of outcomes (A0 in its labels), with the sentence that the protocol reserved for this case: the lattice walk adds nothing over a non-spatial contact model. A static network in which each pair keeps the contact rate it had on the training days does better (post hoc). In the walk the number of contacts grows as the square of the number of people present; in four of the five settings where this can be estimated it grows at most linearly (exponents 0.45 to 0.95), and in the hospital ward faster than the square (2.8). Walks anchored to home sites, developed afterwards on three datasets and frozen before three confirmatory ones, reduce the error of simulated attack rates from 0.17 to 0.04–0.05 in a description after the fact, but their confirmatory test is uninformative, because the frozen fitting procedure does not recover its own synthetic truth; anchoring is a direction, not a result. A walk with two zones of different mobility, the only variant that touches heterogeneous mobility, passes two of the seven families on each dataset and does not improve on the homogeneous walk (Supplementary Section S9.2). The calibrated hop probabilities correspond to about 220 to 450 sites<sup>2</sup> per day, between  $D_0 = 1$ , which was not tested against contacts, and walking mobility.

## 7.8 The airborne kernel fixed by tracer measurements

If the non-local term of M2 describes the transport of aerosol, its mixing coefficient can be measured without any outbreak. Published tracer measurements exist for four of the settings:

ethane at eight seats of the Hunan coach [75], particles released from a breathing mannequin in a wide-body aircraft cabin [57], salt aerosol in a rail carriage [105], and tracer gas at seven tables of the Guangzhou restaurant [63]; for the classroom a regression of the turbulent diffusion coefficient on the air change rate measured in two houses [20] is extrapolated. In the three vehicles the measured fields, summarised by a diffusion coefficient, are of order 100 to 1,000 m<sup>2</sup> h<sup>-1</sup>: about 1,300 in the coach (bounded from below only), 700 and 110 for two releases in the cabin, and 900 and 350 in the carriage. These coefficients are points of a grid spaced by a factor of 1.12 (those fitted to outbreaks in both tests, of a grid spaced by 1.26), so that only their first two digits carry information. They are 4.5 to about 50 times the vehicle value of the first test (25 m<sup>2</sup> h<sup>-1</sup>), and the extrapolated room values (169 to 185 m<sup>2</sup> h<sup>-1</sup>) a few hundred times its room value (0.5 m<sup>2</sup> h<sup>-1</sup>). The aircraft coefficient of the second test (126 m<sup>2</sup> h<sup>-1</sup>) lies inside the range measured in cabins; that does not make an outbreak-fitted coefficient a measurement of air mixing, since within the same test no common coefficient holds. The coefficient that outbreaks ask for absorbs whatever makes risk fall steeply with distance: exposure at close range, companions seated together, behaviour.

With the kernel fixed in this way, the events of the first dataset were predicted with one free parameter each (Supplementary Table S32). The coach and the call centre are no longer rejected, but only because the measured kernel is close to well mixed, and the well-mixed model fits as well or better. The classroom is the one gain over the well-mixed model ( $p = 0.42$  against 0.036), matched by constant ratios of 2 and 3. Flight VN54 is not reproduced: the predicted ratio is 1.66 against 7.3 observed, and the formal non-rejection ( $p = 0.14$ ) depends on zero sensor readings (with the zeros treated as missing, a variant listed in the protocol,  $p = 0.003$ ). The restaurant is not reproduced at the reported count, and the train table is a failure for every model. Over the five stratum events the tracer kernel scores +7.87 against the well-mixed model and -0.80 and -0.78 against the constant ratios 2 and 3; the outcome of the tracer protocol is the one for one adequacy failure and no gain over a generic gradient (S4 in its labels), and the one for two adequacy failures (S5) if the flight, whose formal pass rests on zero sensor readings, is counted as a failure. The tracer kernel is not validated as a spatial predictor. In an exploratory extension to the four held-out flights of the second test, written after its results were known, the two cabin coefficients beat the well-mixed model and tie with a constant ratio of 3 (-0.16; Supplementary Section S9.3).

## 7.9 Zone-level structure

At the scale of zones the model says that transmission between two groups is proportional to the time their members spend in the same place, with the frequency-dependent normalisation of Section 2: if a person of group  $i$  spends a fraction  $f_{iz}$  of the time in zone  $z$ , the force of infection on group  $i$  is

$$\lambda_i = \beta \sum_z q_z f_{iz} \frac{\sum_j f_{jz} I_j}{\sum_j f_{jz} N_j}, \quad (22)$$

a residence-time formulation in the sense of Bichara et al. [12]. The model fitted puts measured shares of contact duration on three exclusive levels (the own class, the other classes of the grade, the rest of the school), which identifies contact time measured by RFID with time spent together. The epidemic is seasonal influenza in the 29 public primary schools of Matsumoto, Japan, in 2014/15, with the day of onset of 2,548 infected among 10,923 pupils in 455 classes [26]; the mixing is the share of a pupil's contact time measured in a primary school in Lyon [32, 88]: 0.762 with the own class, 0.104 with the other classes of the grade and 0.134 with the rest of the school. A daily chain-binomial model with the serial-interval kernel of the source is fitted under two-fold cross-validation by school; the rule has three tiers (Supplementary Section S10.5).

The zone model predicts held-out schools far better than a well-mixed school (ahead in 27 of 29 schools) and than independent classes (Supplementary Table S33). The share between classes

of the same grade agrees to two decimals; exact agreement of the three shares is rejected, the fitted school-level share being almost twice the measured one. This comparison is confounded: the model has no household term, whereas the source attributes 41 % of the pupils’ infections to households, and siblings attend the same school in different grades. The fitted shares lie within the tolerance, but not securely (11 % of bootstrap resamples exceed it). The pre-specified generic two-level model with one fitted ratio is not beaten, and a guess of 0.7, 0.1 and 0.2, chosen after the fitted shares were known, does better. The verdict of the rule, ‘right structure within tolerance, not exact’, is reported as a weak pass: the measurement is not shown to matter beyond the statement that most transmission is within the class. The test rests on the residence-time reduction (22), which is an assumption; the lattice walk, the gradient within a room, the airborne term and attack-rate levels are not involved. A zone model of the Seoul call-centre building by floor excludes a well-mixed building, which needs no model, and nothing more: the time spent in the shared lifts and lobby is not measured, and with a longer assumed time the same model is inconsistent with the data (Supplementary Section S9.4).

## 7.10 What the evidence supports

The evidence supports the following statement and no stronger one. Tested out of sample against documented outbreaks, recorded contacts, tracer measurements and a school epidemic, the model does not show general agreement, and what it gets right is not specific to it (Table 9, Supplementary Fig. S21). The model with transmission only between people on the same lattice site is not supported. The extension with a diffusing-aerosol term predicts that risk falls with distance from the source better than a well-mixed model, with the kernel calibrated on other outbreaks and one infectiousness parameter fitted per event, but it is not shown to predict the gradient better than a constant ratio of near to far risk: pooled over the five held-out events of the second test the pre-specified comparison favours the constant ratio, nine-tenths of it on an event whose near zone was chosen by its source on the same data, and on four held-out flights the two are statistically indistinguishable. No single mixing coefficient holds across rooms, across vehicles or from trains to aircraft. Two structural statements are supported, and the model shares them with simpler ones: risk falls smoothly with distance from the source (pooled near/far ratio with random effects about 4 in the second dataset, 3.3 without the processing line, post hoc; 2.05 over the four held-out events of the first test with distance strata), and transmission between zones is dominated by the group in which people spend their time. The contact mechanism, the shape and calibration of the lattice kernel, the reading of fitted coefficients as a property of the air and their transfer between settings are not supported; attack-rate levels were fitted, not predicted.

Above all, none of these tests bears on the heterogeneity of mobility, which is what distinguishes the model of Sections 2–6 from simpler ones. In every outbreak that publishes positions, and hence in every event tested, people are seated or at fixed stations, so the heterogeneity of mobility between zones never enters, and the model that has a positive result is a standard aerosol model; the contact records carry no positions; and no metro-carriage or supermarket outbreak with a denominator was found. Nor do the pre-specified tests bear on the illustrative values  $\beta = 0.5 \text{ day}^{-1}$  and  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$  of the earlier sections: in every test the level of transmission was fitted (one infectiousness parameter per outbreak, one rate for the school epidemic, a matched contact time for the contact records), and mobilities and mixing coefficients were calibrated on data. The default values are not meant for single events, and they cannot produce the large ones. Under the frequency-dependent hazard (5) an infective at  $x$  infects at expected rate  $\beta q_x$  (Section 2.3). Summing, over chains of transmission between distinct people, the expected number of ordered contact events along each chain (people move independently and remain at stationarity), the expected number of infections caused by one index case in an event of duration  $T$  is at most  $e^{\beta q_{\max} T} - 1$ . With  $\beta = 0.5 \text{ day}^{-1}$  and the largest  $q$  of the scenes (2.8) this is 0.10 for the 1.7-hour Zhejiang bus trip [83], on which 23 of 67 passengers were infected,

**Table 9:** The model and its components after both outbreak tests and the analyses of contacts, tracers and zones. Every entry is the outcome after the re-check; Supplementary Table S39 registers all tests, with what was pre-specified and what was added post hoc.

| Component                                                                                 | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Position                                                                                   |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| M1: people walk on the lattice, transmission on the same site (the model of Sections 2–6) | expected attack rate below the observed one in three of four seated events of the first test (bus total reached with probability 0.0002); in the second test its closed form, which overstates its reach and whose pass by the protocol does not bear on the model, beats well mixed (+15.45) but fails on three of five new events and is worse than a constant ratio; its contacts fail against six of six contact datasets and add nothing over a model without space | not supported                                                                              |
| M2: non-local airborne term, coefficient fitted to outbreaks                              | better than well mixed in both tests; the pre-specified pooled comparison favours a constant near/far ratio (−9.15, nine-tenths from the processing line); tied with a constant two-row ratio on four flights; an exponential decay does as well (post hoc); failures on a coach, a call centre, a processing line, one flight; no common coefficient; no transfer from trains to aircraft                                                                               | supported only as: a smooth one-parameter gradient beats a well-mixed model                |
| M2 with the coefficient measured by tracers                                               | one gain over well mixed (classroom), matched by a constant ratio; flight VN54, restaurant and trains not reproduced; exploratory extension to four flights ties with a ratio of 3                                                                                                                                                                                                                                                                                       | not validated as a spatial predictor                                                       |
| Zone-level reduction, eq. (22)                                                            | weak pass on influenza in 29 schools; a fitted ratio does as well, and a guess of 0.7, 0.1, 0.2 (post hoc) does better; building by floor inconclusive                                                                                                                                                                                                                                                                                                                   | weakly supported as structure (class-dominated, three levels); the quantitative law is not |
| Walkers anchored to home sites                                                            | error of simulated epidemics smaller in a description after the fact; confirmatory test uninformative                                                                                                                                                                                                                                                                                                                                                                    | neither supported nor refuted                                                              |
| Heterogeneous mobility $D_x$                                                              | no dataset with a measured mobility field; a two-zone walk does not improve on the homogeneous one                                                                                                                                                                                                                                                                                                                                                                       | untested                                                                                   |

and 0.16 for the 2.5-hour Skagit Valley choir rehearsal [41], at which 52 of 60 attendees (32 of them confirmed) were; by Markov’s inequality the model gives such counts a probability below 0.005, and the bound reaches them only for  $\beta$  of about  $16 \text{ day}^{-1}$  on the bus and  $14 \text{ day}^{-1}$  for the choir ( $12 \text{ day}^{-1}$  with the 32 confirmed cases). This comparison was added after the tests, for the two events of the first dataset with the highest attack rates after a short seated exposure (Supplementary Table S39).

## 8 Discussion and conclusions

### 8.1 What stands

*The mean-field theory.* For the reaction–diffusion model,  $R_0 = \rho(\mathbf{K})$  (Theorem 3.1). With the same-cell kernel it lies between  $\beta\langle q \rangle_\pi/\gamma$  and  $\beta q_{\max}/\gamma$  (Theorem 3.2) and decreases with the mobility scale (Theorem 3.3); for other contact kernels the lower bound and the monotonicity can fail (Proposition 3.7). In a closed room with uniform  $q$  it equals  $\beta q/\gamma$  whatever the size,

shape, barriers or mobility (Theorem 4.1), and a leak at rate  $\mu_{\text{leak}}$  replaces  $\gamma$  by  $\gamma + \mu_{\text{leak}}$  under the conditions of Theorem 4.2. These statements are known for epidemic patch models [1, 3, 19, 23, 29–31, 96], and loss through a leak is the classical critical-patch-size mechanism [16, 84]. What is added here is their statement for the indoor lattice model, with the condition under which each holds and counter-examples where it does not.

*The pair-level description.* It applies the two-walker theory of transmission [38, 51] with an occupancy-normalised hazard.  $R_1(x)$  is exact for the first generation of a lone index case. It accounts for the gap between the counted number of secondary cases in the office scene,  $\hat{R} = 2.265 \pm 0.004$ , and  $R_0 = 5.11$  ( $\bar{R}_1 = 2.27$ ;  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ , 100 people), and for the rise of the counted number with mobility while  $R_0$  falls. The closed form is proved for periodic rooms and the infinite floor; for rooms with reflecting walls it is an approximation, within 1 % in the cases tested.

*The simulator* has been checked against exact results and against a separately written implementation; the scene and intervention results are obtained from it. Barriers, occupancy and schedules are outputs of these simulations; the reductions of infection efficiency called ventilation and masks are inputs.

*The data and the tests.* The two datasets hold 38 records of documented outbreaks and exposure cohorts (35 distinct outbreaks), among them eleven records of ten outbreaks with transcribed seat or position data. Two pre-specified out-of-sample tests compare two spatial models with a well-mixed model on 18 of the records (22 with the level checks and negative controls of the first test), and three further pre-specified analyses confront the model with recorded contacts in six indoor settings, tracer measurements and an influenza epidemic in 29 schools. Every primary analysis was re-checked with separately written code within the same project (parts of the re-check reused data loaders, event geometries and seat maps of the analyses) and is reported as it stands after that re-check; the exceptions (analyses not re-run, and comparisons added post hoc) are named where they occur and in the register of tests. The protocols were written with the data known, were frozen by file hashes on the author’s machine and were not lodged with a registry.

## 8.2 Limits

**Physical regime.** Every spatial effect in a closed room reported in Sections 3–6 is large only near the default  $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ , which is three to almost six orders of magnitude below estimates of walking mobility, depending on the scene. In the office scene  $R_0$  exceeds the well-mixed value  $\beta\langle q \rangle/\gamma$  by 10.0 % at  $D_0 = 1$ , by 0.72 % at  $D_0 = 10$ , by 0.06 % at  $D_0 = 100$  and by less than 0.003 % at  $D_0 = 2.4 \times 10^3 \text{ cell}^2 \text{ day}^{-1}$ : at realistic mobility the mean-field model of a closed room is a well-mixed model. Leaks through a localised door are the exception; their effect grows with mobility, and they should be modelled by an input dwell time.

**Movement model.** A random walk without home positions is not how people at desks behave: at  $D_0 = 1$  the residence time in the meeting room of the office scene is 114 days. Calibrated to the amount and the mean duration of contact in six indoor settings, the walk gives each person too many partners (in five of the six settings), too equal a share of contact and no group structure, and an epidemic simulated on its contacts over-predicts the secondary cases of the index case in all 24 scenarios, by 25 to 72 % on average per setting, as a well-mixed model does. That a memoryless walk misses the bursty statistics of face-to-face contact was known [86]; what is added here is the epidemic consequence and the comparison with models without space. Walkers anchored to home cells are the natural next model; the corresponding two-walker problem has been treated by Kenkre and Sugaya [51] and Sugaya and Kenkre [89], and the consequences of anchoring for the reproduction numbers of Sections 3–5 have not been computed.

**No threshold theory for discrete individuals.**  $R_1$  and  $\rho(\mathbf{K}_1)$  are not threshold quantities, and the mean-field branching probability of a major outbreak in the office scene is 0.78, against 0.50 simulated. For walkers on infinite or large homogeneous lattices the existence of a threshold is established [17, 54, 67]; what is missing is a computable threshold, and a final size, for small heterogeneous rooms. A moment closure beyond pairs [48, 58] is one possible next step.

**Modelling choices that change conclusions.** The first is the contact kernel: at  $D_0 = 1$  the simulated probability of a major outbreak in the metro scene is 0.88 with the 1 m kernel and zero with the same-cell kernel, and the lower bound  $\beta\langle q \rangle_\pi/\gamma \leq R_0$  and the monotonicity in mobility require the same-cell kernel. The second is the incidence law, frequency-dependent or density-dependent [4, 49]: halving the occupancy of the office leaves the mean-field  $R_0$  at 5.11 under the first and lowers it to 2.53 under the second.

**Airborne transport.** A non-local term (here an airborne one) or exposure outside the modelled event is needed to reach the observed attack rates of seated events, and the one added here (M2) is a standard diffusing-aerosol model [61, 98]. Its isotropic kernel fails where airflow is directed or recirculating, as on the Hunan coach and in the Guangzhou restaurant; coefficients fitted to outbreaks are not a property of the air; and with a measured coefficient the kernel cannot produce the steep gradients that close-range exposure, companions seated together or directed airflow produce.

**Parameters, data and protocols.** The values  $\beta = 0.5 \text{ day}^{-1}$  and  $\gamma = 0.14 \text{ day}^{-1}$  and the zone values of  $q$  and  $D$  are illustrative and not calibrated; at these values the model cannot produce the attack rates of short seated outbreaks (Section 7.10). All scenes are closed rooms with an SIR course, an exponential infectious period, no latent period and one index case. In every test the level was fitted, not predicted. Most of the 38 records are investigated outbreaks, which over-represent highly infectious index cases [25, 64]; strata are small (the positive result of the first test rests on 20 passengers, and the five held-out events of the second have 2 to 20 cases each); seat maps were transcribed by eye from published figures; six sources of the first dataset were read as abstracts only; the tracer coefficients of the cabin, the carriage and the rooms are transferred from another airframe, other rolling stock and two houses; and the zone-level test has one pathogen, one type of setting, mixing measured over two days in a school of another country, and no household term. All protocols were written with the data already known, and their timing is attested by file hashes and file times on the author’s machine only. As Section 7.10 sets out, heterogeneous mobility is untested by data.

### 8.3 What would change the verdict

Four kinds of data would, and none of them was available. The first is a dataset with a *measured mobility field*: tracked positions, or dwell times by zone, together with infection or contact outcomes; without it the heterogeneity of mobility cannot be tested, and neither  $D_0$  nor the anchoring of walkers can be calibrated. The second, for the airborne term against generic gradients, is a set of outbreaks with exposure known seat by seat and three or more distance strata chosen before the outcomes are seen, large enough to separate a diffusive kernel from a step function; the test on four flights had a power of 0.82 and returned a tie. The third, for anchored walkers, is a fitting procedure that recovers its own synthetic truth, followed by a fresh confirmatory set of contact records. The fourth, for zones, is an outbreak with denominators by zone and mixing measured independently in the same population. Any such test should be registered with a third party before the data are examined, and must beat the generic benchmarks of this article, which were not beaten here: a constant near/far risk ratio of 2 to 4 for distance, a static network of pair contact rates for contacts, and rough shares of 0.7, 0.1 and 0.2 for the group, the next level and the rest for zones (the last two post hoc).

## 8.4 A practical reading, with reservations

Nothing below is validated against data; these are properties of the model, not recommendations. Uniform reductions of  $q$  or  $\beta$  scale  $R_0$  in proportion, and the factors are inputs, not predictions. With symmetric hop rates and a fixed set of cells, slowing or closing connections cannot lower the mean-field  $R_0$  (Theorem 3.6); removing floor cells changes  $\langle q \rangle$  and, in the office scene, shifts  $R_0$  by +0.07 % at  $D_0 = 1$  and by -1.1 % and -1.5 % at  $D_0 = 10$  and 100. Barriers, separation of zones and capacity limits reduce simulated outbreaks appreciably only at low mobility and occupancy, through the discreteness of individuals, which is absent from the mean field and fades as occupancy grows; because  $R_1$  is not a threshold, ' $R_1 < 1$ ' is not a control criterion. Residence time dominates: an 8-hour day divides the office  $R_0$  by almost three, and the expected number of infections caused during one visit is 0.21 for an office day, 0.027 for a one-hour lecture, 0.018 for a 20-minute metro ride and 0.0086 for a 27-minute supermarket visit (mean field).

## 8.5 Conclusions

Three consequences follow. First, a lattice random-walk model of a closed room has, for uniform efficiency, no size effect in its mean-field reproduction number and, at the mobility of people who walk, hardly any spatial effect at all: what the floor plan contributes is the area-weighted mean of  $q$ , and what matters beyond it is how long people stay. Second, where spatial effects are large, at low mobility and low occupancy, the mean-field number is the wrong guide for discrete people. The pair-level number gives the first generation exactly, but neither it nor any other quantity computed here is a threshold, so outbreak probabilities and sizes have to be simulated; and the sign of an intervention can differ between the two levels (separation of zones raises  $R_0$  and lowers simulated outbreaks). Third, confronted with 22 of 38 records of documented outbreaks, recorded contacts in six settings, tracer measurements and an epidemic in 29 schools, the model does not show general agreement. The same-site model is not supported. The extension with a diffusing aerosol predicts the decline of risk with distance better than a well-mixed model, as does the closed form of the same-site model (a pass that, by the protocol, does not bear on the same-site model; and a criterion that outbreaks with a constant near/far ratio would meet with probability 0.65 to 0.73), but the pre-specified pooled comparison favours a constant ratio of near to far risk, mostly through one event whose near zone was chosen on the same data, and on four held-out flights the two are tied. What these tests give in the end is a set of benchmarks that a spatial model of indoor transmission has to beat, and a list of the data that would be needed to test the heterogeneity of mobility at all.

## Data and code availability

All code, scene definitions, simulation outputs, the two datasets of outbreak records with their sources, the tracer and school data that may be redistributed, a script that downloads the contact records, the protocol documents, the code of the re-checks and the scripts that generate every figure and table of this article and of its Supplementary Material form one repository (version 1.2.0), together with `reproduce.md`, which gives the commands to re-run each result, and `notes/VERIFICATION.md`, which describes how the results were checked. The protocol records in the repository are redacted copies, with every changed line listed in `code/REDACTIONS.md`; the hashes of the freeze records refer to the unredacted originals, which the author can provide to the editor and referees, so the timing of the protocols cannot be checked from the repository alone. The repository is public at <https://github.com/zhouyi-xiaoxiao/indoor-epidemic-random-walk>. This article accompanies version 1.2.0 of the repository, identified by the file `VERSION`.

## Ethics and data

Only published outbreak reports and published or publicly released, anonymised contact, tracer and school data were analysed; no data on human participants were collected for this study.

## Acknowledgements

This article develops the author’s undergraduate thesis (School of Mathematics, Tianjin University, 2025).

## References

- [1] Linda J. S. Allen, Benjamin M. Bolker, Yuan Lou, and Andrew L. Nevai. Asymptotic profiles of the steady states for an SIS epidemic patch model. *SIAM Journal on Applied Mathematics*, 67(5): 1283–1309, 2007. doi: 10.1137/060672522.
- [2] Linda J. S. Allen, Benjamin M. Bolker, Yuan Lou, and Andrew L. Nevai. Asymptotic profiles of the steady states for an SIS epidemic reaction-diffusion model. *Discrete and Continuous Dynamical Systems. Series A*, 21(1):1–20, 2008. doi: 10.3934/dcds.2008.21.1.
- [3] Lee Altenberg. Resolvent positive linear operators exhibit the reduction phenomenon. *Proceedings of the National Academy of Sciences*, 109(10):3705–3710, 2012. doi: 10.1073/pnas.1113833109.
- [4] Roy M. Anderson and Robert M. May. *Infectious Diseases of Humans: Dynamics and Control*. Oxford University Press, Oxford, 1991. doi: 10.1093/oso/9780198545996.001.0001.
- [5] Håkan Andersson and Tom Britton. *Stochastic Epidemic Models and Their Statistical Analysis*. Springer, New York, 2000. doi: 10.1007/978-1-4612-1158-7.
- [6] M. G. Baker, C. N. Thornley, C. Mills, S. Roberts, S. Perera, J. Peters, A. Kelso, I. Barr, and N. Wilson. Transmission of pandemic A/H1N1 2009 influenza on passenger aircraft: retrospective cohort study. *BMJ*, 340:c2424, 2010. doi: 10.1136/bmj.c2424.
- [7] Frank Ball, Denis Mollison, and Gianpaolo Scalia-Tomba. Epidemics with two levels of mixing. *The Annals of Applied Probability*, 7(1):46–89, 1997. doi: 10.1214/aoap/1034625252.
- [8] Martin Z. Bazant and John W. M. Bush. A guideline to limit indoor airborne transmission of COVID-19. *Proceedings of the National Academy of Sciences*, 118(17):e2018995118, 2021. doi: 10.1073/pnas.2018995118.
- [9] M. Begon, M. Bennett, R. G. Bowers, N. P. French, S. M. Hazel, and J. Turner. A clarification of transmission terms in host-microparasite models: numbers, densities and areas. *Epidemiology and Infection*, 129(1):147–153, 2002. doi: 10.1017/S0950268802007148.
- [10] Vitaly Belik, Theo Geisel, and Dirk Brockmann. Natural human mobility patterns and spatial spread of infectious diseases. *Physical Review X*, 1(1):011001, 2011. doi: 10.1103/PhysRevX.1.011001.
- [11] Abraham Berman and Robert J. Plemmons. *Nonnegative Matrices in the Mathematical Sciences*. Society for Industrial and Applied Mathematics, Philadelphia, PA, 1994. doi: 10.1137/1.9781611971262.
- [12] Derdei Bichara, Yun Kang, Carlos Castillo-Chavez, Richard Horan, and Charles Perrings. SIS and SIR epidemic models under virtual dispersal. *Bulletin of Mathematical Biology*, 77(11):2004–2034, 2015. doi: 10.1007/s11538-015-0113-5.
- [13] David Blackman and Sebastiano Vigna. Scrambled linear pseudorandom number generators. *ACM Transactions on Mathematical Software*, 47(4):1–32, 2021. doi: 10.1145/3460772.
- [14] N. Boccara and K. Cheong. Automata network SIR models for the spread of infectious diseases in populations of moving individuals. *Journal of Physics A: Mathematical and General*, 25(9): 2447–2461, 1992. doi: 10.1088/0305-4470/25/9/018.
- [15] A. Buscarino, L. Fortuna, M. Frasca, and V. Latora. Disease spreading in populations of moving agents. *EPL (Europhysics Letters)*, 82(3):38002, 2008. doi: 10.1209/0295-5075/82/38002.
- [16] Robert Stephen Cantrell and Chris Cosner. *Spatial Ecology via Reaction-Diffusion Equations*. John Wiley & Sons, Chichester, 2003. doi: 10.1002/0470871296.
- [17] V. R. Carvalho, T. F. A. Alves, G. A. Alves, D. S. M. Alencar, F. W. S. Lima, A. Macedo-Filho, and R. S. Ferreira. Generalized diffusive epidemic process with permanent immunity in two dimensions. *Journal of Statistical Mechanics: Theory and Experiment*, 2025(1):013102, 2025. doi: 10.1088/1742-5468/ad9f4d.

- [18] Ciro Cattuto, Wouter Van den Broeck, Alain Barrat, Vittoria Colizza, Jean-François Pinton, and Alessandro Vespignani. Dynamics of person-to-person interactions from distributed RFID sensor networks. *PLoS ONE*, 5(7):e11596, 2010. doi: 10.1371/journal.pone.0011596.
- [19] Shanshan Chen, Junping Shi, Zhisheng Shuai, and Yixiang Wu. Asymptotic profiles of the steady states for an SIS epidemic patch model with asymmetric connectivity matrix. *Journal of Mathematical Biology*, 80(7):2327–2361, 2020. doi: 10.1007/s00285-020-01497-8.
- [20] Kai-Chung Cheng, Viviana Acevedo-Bolton, Ruo-Ting Jiang, Neil E. Klepeis, Wayne R. Ott, Oliver B. Fringer, and Lynn M. Hildemann. Modeling exposure close to air pollution sources in naturally ventilated residences: association of turbulent diffusion coefficient with air change rate. *Environmental Science & Technology*, 45(9):4016–4022, 2011. doi: 10.1021/es103080p.
- [21] Vittoria Colizza, Romualdo Pastor-Satorras, and Alessandro Vespignani. Reaction–diffusion processes and metapopulation models in heterogeneous networks. *Nature Physics*, 3(4):276–282, 2007. doi: 10.1038/nphys560.
- [22] J. N. Darroch and E. Seneta. On quasi-stationary distributions in absorbing continuous-time finite Markov chains. *Journal of Applied Probability*, 4(1):192–196, 1967. doi: 10.2307/3212311.
- [23] O. Diekmann, J. A. P. Heesterbeek, and J. A. J. Metz. On the definition and the computation of the basic reproduction ratio  $R_0$  in models for infectious diseases in heterogeneous populations. *Journal of Mathematical Biology*, 28(4):365–382, 1990. doi: 10.1007/BF00178324.
- [24] Richard Durrett and Simon Levin. The importance of being discrete (and spatial). *Theoretical Population Biology*, 46(3):363–394, 1994. doi: 10.1006/tpbi.1994.1032.
- [25] Akira Endo, Centre for the Mathematical Modelling of Infectious Diseases COVID-19 Working Group, Sam Abbott, Adam J. Kucharski, and Sebastian Funk. Estimating the overdispersion in COVID-19 transmission using outbreak sizes outside China. *Wellcome Open Research*, 5:67, 2020. doi: 10.12688/wellcomeopenres.15842.3.
- [26] Akira Endo, Mitsuo Uchida, Naoki Hayashi, Yang Liu, Katherine E. Atkins, Adam J. Kucharski, and Sebastian Funk. Within and between classroom transmission patterns of seasonal influenza among primary school students in Matsumoto city, Japan. *Proceedings of the National Academy of Sciences*, 118(46):e2112605118, 2021. doi: 10.1073/pnas.2112605118.
- [27] Mattia Frasca, Arturo Buscarino, Alessandro Rizzo, Luigi Fortuna, and Stefano Boccaletti. Dynamical network model of infective mobile agents. *Physical Review E*, 74(3):036110, 2006. doi: 10.1103/PhysRevE.74.036110.
- [28] Laura Gammaitoni and Maria Clara Nucci. Using a mathematical model to evaluate the efficacy of TB control measures. *Emerging Infectious Diseases*, 3(3):335–342, 1997. doi: 10.3201/eid0303.970310.
- [29] Daozhou Gao. Travel frequency and infectious diseases. *SIAM Journal on Applied Mathematics*, 79(4):1581–1606, 2019. doi: 10.1137/18M1211957.
- [30] Daozhou Gao and Chao-Ping Dong. Fast diffusion inhibits disease outbreaks. *Proceedings of the American Mathematical Society*, 148(4):1709–1722, 2020. doi: 10.1090/proc/14868.
- [31] Daozhou Gao and Shigui Ruan. An SIS patch model with variable transmission coefficients. *Mathematical Biosciences*, 232(2):110–115, 2011. doi: 10.1016/j.mbs.2011.05.001.
- [32] Valerio Gemmetto, Alain Barrat, and Ciro Cattuto. Mitigation of infectious disease at school: targeted class closure vs school closure. *BMC Infectious Diseases*, 14(1):695, 2014. doi: 10.1186/s12879-014-0695-9.
- [33] Mathieu Génois and Alain Barrat. Can co-location be used as a proxy for face-to-face contacts? *EPJ Data Science*, 7(1):11, 2018. doi: 10.1140/epjds/s13688-018-0140-1.
- [34] Mathieu Génois, Christian L. Vestergaard, Julie Fournet, André Panisson, Isabelle Bonmarin, and Alain Barrat. Data on face-to-face contacts in an office building suggest a low-cost vaccination strategy based on community linkers. *Network Science*, 3(3):326–347, 2015. doi: 10.1017/nws.2015.10.
- [35] Daniel T. Gillespie. A general method for numerically simulating the stochastic time evolution of coupled chemical reactions. *Journal of Computational Physics*, 22(4):403–434, 1976. doi: 10.1016/0021-9991(76)90041-3.
- [36] Daniel T. Gillespie. Exact stochastic simulation of coupled chemical reactions. *The Journal of Physical Chemistry*, 81(25):2340–2361, 1977. doi: 10.1021/j100540a008.
- [37] Luca Giuggioli. Exact spatiotemporal dynamics of confined lattice random walks in arbitrary dimensions: a century after Smoluchowski and Pólya. *Physical Review X*, 10(2):021045, 2020. doi: 10.1103/PhysRevX.10.021045.
- [38] Luca Giuggioli and Seeralan Sarvaharman. Spatio-temporal dynamics of random transmission

- events: from information sharing to epidemic spread. *Journal of Physics A: Mathematical and Theoretical*, 55(37):375005, 2022. doi: 10.1088/1751-8121/ac8587.
- [39] Marta C. González and Hans J. Herrmann. Scaling of the propagation of epidemics in a system of mobile agents. *Physica A: Statistical Mechanics and its Applications*, 340(4):741–748, 2004. doi: 10.1016/j.physa.2004.05.017.
  - [40] Thomas Günther, Manja Czech-Sioli, Daniela Indenbirken, Alexis Robitaille, Peter Tenhaken, Martin Exner, Matthias Ottinger, Nicole Fischer, Adam Grundhoff, and Melanie M. Brinkmann. SARS-CoV-2 outbreak investigation in a German meat processing plant. *EMBO Molecular Medicine*, 12(12):EMMM202013296, 2020. doi: 10.15252/emmm.202013296.
  - [41] Lea Hamner, Polly Dubbel, Ian Capron, Andy Ross, Amber Jordan, Jaxon Lee, Joanne Lynn, Amelia Ball, Simranjit Narwal, Sam Russell, Dale Patrick, and Howard Leibrand. High SARS-CoV-2 attack rate following exposure at a choir practice — Skagit County, Washington, March 2020. *MMWR. Morbidity and Mortality Weekly Report*, 69(19):606–610, 2020. doi: 10.15585/mmwr.mm6919e6.
  - [42] Günther Hasibeder and Christopher Dye. Population dynamics of mosquito-borne disease: persistence in a completely heterogeneous environment. *Theoretical Population Biology*, 33(1):31–53, 1988. doi: 10.1016/0040-5809(88)90003-2.
  - [43] Rebecca B. Hershov, Karen Wu, Nathaniel M. Lewis, Alison T. Milne, Dustin Currie, Amanda R. Smith, Spencer Lloyd, Brian Orleans, Erin L. Young, Brandi Freeman, et al. Low SARS-CoV-2 transmission in elementary schools — Salt Lake County, Utah, December 3, 2020–January 31, 2021. *MMWR. Morbidity and Mortality Weekly Report*, 70(12):442–448, 2021. doi: 10.15585/mmwr.mm7012e3.
  - [44] Vicki Stover Hertzberg and Howard Weiss. On the 2-row rule for infectious disease transmission on aircraft. *Annals of Global Health*, 82(5):819–823, 2016. doi: 10.1016/j.aogh.2016.06.003.
  - [45] Sebastian Hoehl, Onur Karaca, Niko Kohmer, Sandra Westhaus, Jürgen Graf, Udo Goetsch, and Sandra Ciesek. Assessment of SARS-CoV-2 transmission on an international flight and among a tourist group. *JAMA Network Open*, 3(8):e2018044, 2020. doi: 10.1001/jamanetworkopen.2020.18044.
  - [46] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, Cambridge, 2nd edition, 2013. doi: 10.1017/CBO9781139020411.
  - [47] Maogui Hu, Hui Lin, Jinfeng Wang, Chengdong Xu, Andrew J. Tatem, Bin Meng, Xin Zhang, Yifeng Liu, Pengda Wang, Guizhen Wu, Haiyong Xie, and Shengjie Lai. Risk of coronavirus disease 2019 transmission in train passengers: an epidemiological and modeling study. *Clinical Infectious Diseases*, 72(4):604–610, 2021. doi: 10.1093/cid/ciaa1057.
  - [48] Matt J. Keeling. The effects of local spatial structure on epidemiological invasions. *Proceedings of the Royal Society of London. Series B: Biological Sciences*, 266(1421):859–867, 1999. doi: 10.1098/rspb.1999.0716.
  - [49] Matt J. Keeling and Pejman Rohani. *Modeling Infectious Diseases in Humans and Animals*. Princeton University Press, Princeton, NJ, 2008. doi: 10.1515/9781400841035.
  - [50] V. M. Kenkre and Luca Giuggioli. *Theory of the Spread of Epidemics and Movement Ecology of Animals*. Cambridge University Press, 2021. doi: 10.1017/9781108882279.
  - [51] V. M. Kenkre and S. Sugaya. Theory of the transmission of infection in the spread of epidemics: interacting random walkers with and without confinement. *Bulletin of Mathematical Biology*, 76(12):3016–3027, 2014. doi: 10.1007/s11538-014-0042-8.
  - [52] Thomas A. Kenyon, Sarah E. Valway, Walter W. Ihle, Ida M. Onorato, and Kenneth G. Castro. Transmission of multidrug-resistant *Mycobacterium tuberculosis* during a long airplane flight. *New England Journal of Medicine*, 334(15):933–938, 1996. doi: 10.1056/NEJM199604113341501.
  - [53] Harry Kesten and Vladas Sidoravicius. The spread of a rumor or infection in a moving population. *The Annals of Probability*, 33(6):2402–2462, 2005. doi: 10.1214/009117905000000413.
  - [54] Harry Kesten and Vladas Sidoravicius. A phase transition in a model for the spread of an infection. *Illinois Journal of Mathematics*, 50(1–4):547–634, 2006. doi: 10.1215/ijm/1258059486.
  - [55] Harry Kesten and Vladas Sidoravicius. A shape theorem for the spread of an infection. *Annals of Mathematics*, 167(3):701–766, 2008. doi: 10.4007/annals.2008.167.701.
  - [56] Nguyen Cong Khanh, Pham Quang Thai, Ha-Linh Quach, Ngoc-Anh Hoang Thi, Phung Cong Dinh, Tran Nhu Duong, Le Thi Quynh Mai, Ngu Duy Nghia, Tran Anh Tu, La Ngoc Quang, et al. Transmission of SARS-CoV 2 during long-haul flight. *Emerging Infectious Diseases*, 26(11):2617–2624, 2020. doi: 10.3201/eid2611.203299.
  - [57] Sean M. Kinahan, David B. Silcott, Blake E. Silcott, Ryan M. Silcott, Peter J. Silcott, Braden J.

- Silcott, Steven L. Distelhorst, Vicki L. Herrera, Danielle N. Rivera, Kevin K. Crown, et al. Aerosol tracer testing in Boeing 767 and 777 aircraft to simulate exposure potential of infectious aerosol such as SARS-CoV-2. *PLoS ONE*, 16(12):e0246916, 2021. doi: 10.1371/journal.pone.0246916.
- [58] István Z. Kiss, Joel C. Miller, and Péter L. Simon. *Mathematics of Epidemics on Networks: From Exact to Approximate Models*. Springer, Cham, 2017. doi: 10.1007/978-3-319-50806-1.
- [59] Ryogo Kubo. The fluctuation-dissipation theorem. *Reports on Progress in Physics*, 29(1):255–284, 1966. doi: 10.1088/0034-4885/29/1/306.
- [60] Tracy Lam-Hine, Stephen A. McCurdy, Lisa Santora, Lael Duncan, Russell Corbett-Detig, Beatrix Kapusinszky, and Matthew Willis. Outbreak associated with SARS-CoV-2 B.1.617.2 (Delta) variant in an elementary school — Marin County, California, May–June 2021. *MMWR. Morbidity and Mortality Weekly Report*, 70(35):1214–1219, 2021. doi: 10.15585/mmwr.mm7035e2.
- [61] Zechariah Lau, Ian M. Griffiths, Aaron English, and Katerina Kaouri. Predicting the spatio-temporal infection risk in indoor spaces using an efficient airborne transmission model. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 478(2259):20210383, 2022. doi: 10.1098/rspa.2021.0383.
- [62] Quentin J. Leclerc, Naomi M. Fuller, Lisa E. Knight, Centre for the Mathematical Modelling of Infectious Diseases COVID-19 Working Group, Sebastian Funk, and Gwenan M. Knight. What settings have been linked to SARS-CoV-2 transmission clusters? *Wellcome Open Research*, 5:83, 2020. doi: 10.12688/wellcomeopenres.15889.2.
- [63] Yuguo Li, Hua Qian, Jian Hang, Xuguang Chen, Pan Cheng, Hong Ling, Shengqi Wang, Peng Liang, Jiansen Li, Shenglan Xiao, et al. Probable airborne transmission of SARS-CoV-2 in a poorly ventilated restaurant. *Building and Environment*, 196:107788, 2021. doi: 10.1016/j.buildenv.2021.107788.
- [64] James O. Lloyd-Smith, Sebastian J. Schreiber, P. Ekkehard Kopp, and Wayne M. Getz. Super-spreading and the effect of individual variation on disease emergence. *Nature*, 438(7066):355–359, 2005. doi: 10.1038/nature04153.
- [65] Jianyun Lu, Jieni Gu, Kuibiao Li, Conghui Xu, Wenzhe Su, Zhisheng Lai, Deqian Zhou, Chao Yu, Bin Xu, and Zhicong Yang. COVID-19 outbreak associated with air conditioning in restaurant, Guangzhou, China, 2020. *Emerging Infectious Diseases*, 26(7):1628–1631, 2020. doi: 10.3201/eid2607.200764.
- [66] Kaiwei Luo, Zhao Lei, Zheng Hai, Shanliang Xiao, Jia Rui, Hao Yang, Xinpeng Jing, Hui Wang, Zhengshen Xie, Ping Luo, et al. Transmission of SARS-CoV-2 in public transportation vehicles: a case study in Hunan Province, China. *Open Forum Infectious Diseases*, 7(10):ofaa430, 2020. doi: 10.1093/ofid/ofaa430.
- [67] Daniel Souza Maia and Ronald Dickman. Diffusive epidemic process: theory and simulation. *Journal of Physics: Condensed Matter*, 19(6):065143, 2007. doi: 10.1088/0953-8984/19/6/065143.
- [68] Rossana Mastrandrea, Julie Fournet, and Alain Barrat. Contact patterns in a high school: a comparison between data collected using wearable sensors, contact diaries and friendship surveys. *PLoS ONE*, 10(9):e0136497, 2015. doi: 10.1371/journal.pone.0136497.
- [69] H. McCallum, N. Barlow, and J. Hone. How should pathogen transmission be modelled? *Trends in Ecology & Evolution*, 16(6):295–300, 2001. doi: 10.1016/S0169-5347(01)02144-9.
- [70] Shelly L. Miller, William W. Nazaroff, Jose L. Jimenez, Atze Boerstra, Giorgio Buonanno, Stephanie J. Dancer, Jarek Kurnitski, Linsey C. Marr, Lidia Morawska, and Catherine Noakes. Transmission of SARS-CoV-2 by inhalation of respiratory aerosol in the Skagit Valley Chorale superspreading event. *Indoor Air*, 31(2):314–323, 2021. doi: 10.1111/ina.12751.
- [71] Sirish Namilae, Pierrot Derjany, Anuj Mubayi, Mathew Scotch, and Ashok Srinivasan. Multiscale model for pedestrian and infection dynamics during air travel. *Physical Review E*, 95(5):052320, 2017. doi: 10.1103/PhysRevE.95.052320.
- [72] C. J. Noakes, C. B. Beggs, P. A. Sleight, and K. G. Kerr. Modelling the transmission of airborne infections in enclosed spaces. *Epidemiology and Infection*, 134(5):1082–1091, 2006. doi: 10.1017/S0950268806005875.
- [73] Catherine J. Noakes and P. Andrew Sleight. Mathematical models for assessing the role of airflow on the risk of airborne infection in hospital wards. *Journal of the Royal Society Interface*, 6(Suppl. 6):S791–S800, 2009. doi: 10.1098/rsif.2009.0305.focus.
- [74] Sonja J. Olsen, Hsiao-Ling Chang, Terence Yung-Yan Cheung, Antony Fai-Yu Tang, Tamara L. Fisk, Steven Peng-Lim Ooi, Hung-Wei Kuo, Donald Dah-Shyong Jiang, Kow-Tong Chen, Jim Lando,

- et al. Transmission of the severe acute respiratory syndrome on aircraft. *New England Journal of Medicine*, 349(25):2416–2422, 2003. doi: 10.1056/NEJMoa031349.
- [75] Cuiyun Ou, Shixiong Hu, Kaiwei Luo, Hongyu Yang, Jian Hang, Pan Cheng, Zheng Hai, Shanliang Xiao, Hua Qian, Shenglan Xiao, et al. Insufficient ventilation led to a probable long-range airborne transmission of SARS-CoV-2 on two buses. *Building and Environment*, 207:108414, 2022. doi: 10.1016/j.buildenv.2021.108414.
  - [76] Shin Young Park, Young-Man Kim, Seonju Yi, Sangeun Lee, Baeg-Ju Na, Chang Bo Kim, Jung-il Kim, Hea Sook Kim, Young Bok Kim, Yoojin Park, et al. Coronavirus disease outbreak in call center, South Korea. *Emerging Infectious Diseases*, 26(8):1666–1670, 2020. doi: 10.3201/eid2608.201274.
  - [77] Suhas V. Patankar. *Numerical Heat Transfer and Fluid Flow*. Hemisphere Publishing Corporation, Washington, DC, 1980. doi: 10.1201/9781482234213.
  - [78] Z. Peng, A. L. Pineda Rojas, E. Kropff, W. Bahnfleth, G. Buonanno, S. J. Dancer, J. Kurnitski, Y. Li, M. G. L. C. Loomans, L. C. Marr, et al. Practical indicators for risk of airborne transmission in shared indoor environments and their application to COVID-19 outbreaks. *Environmental Science & Technology*, 56(2):1125–1137, 2022. doi: 10.1021/acs.est.1c06531.
  - [79] Hua Qian, Te Miao, Li Liu, Xiaohong Zheng, Danting Luo, and Yuguo Li. Indoor transmission of SARS-CoV-2. *Indoor Air*, 31(3):639–645, 2021. doi: 10.1111/ina.12766.
  - [80] C. J. Rhodes and R. M. Anderson. Persistence and dynamics in lattice models of epidemic spread. *Journal of Theoretical Biology*, 180(2):125–133, 1996. doi: 10.1006/jtbi.1996.0088.
  - [81] E. C. Riley, G. Murphy, and R. L. Riley. Airborne spread of measles in a suburban elementary school. *American Journal of Epidemiology*, 107(5):421–432, 1978. doi: 10.1093/oxfordjournals.aje.a112560.
  - [82] Seeralan Sarvaharman and Luca Giuggioli. Particle-environment interactions in arbitrary dimensions: a unifying analytic framework to model diffusion with inert spatial heterogeneities. *Physical Review Research*, 5(4):043281, 2023. doi: 10.1103/PhysRevResearch.5.043281. A correction to this article was recorded by the publisher on 9 April 2025.
  - [83] Ye Shen, Changwei Li, Hongjun Dong, Zhen Wang, Leonardo Martinez, Zhou Sun, Andreas Handel, Zhiping Chen, Enfu Chen, Mark H. Ebell, et al. Community outbreak investigation of SARS-CoV-2 transmission among bus riders in eastern China. *JAMA Internal Medicine*, 180(12):1665–1671, 2020. doi: 10.1001/jamainternmed.2020.5225.
  - [84] J. G. Skellam. Random dispersal in theoretical populations. *Biometrika*, 38(1-2):196–218, 1951. doi: 10.1093/biomet/38.1-2.196.
  - [85] Hollie Speake, Anastasia Phillips, Tracie Chong, Chisha Sikazwe, Avram Levy, Jurissa Lang, Benjamin Scalley, David J. Speers, David W. Smith, Paul Effler, et al. Flight-associated transmission of severe acute respiratory syndrome coronavirus 2 corroborated by whole-genome sequencing. *Emerging Infectious Diseases*, 26(12):2872–2880, 2020. doi: 10.3201/eid2612.203910.
  - [86] Michele Starnini, Andrea Baronchelli, and Romualdo Pastor-Satorras. Modeling human dynamics of face-to-face interaction networks. *Physical Review Letters*, 110(16):168701, 2013. doi: 10.1103/PhysRevLett.110.168701.
  - [87] Juliette Stehlé, Nicolas Voirin, Alain Barrat, Ciro Cattuto, Vittoria Colizza, Lorenzo Isella, Corinne Régis, Jean-François Pinton, Nagham Khanafer, Wouter Van den Broeck, et al. Simulation of an SEIR infectious disease model on the dynamic contact network of conference attendees. *BMC Medicine*, 9(1):87, 2011. doi: 10.1186/1741-7015-9-87.
  - [88] Juliette Stehlé, Nicolas Voirin, Alain Barrat, Ciro Cattuto, Lorenzo Isella, Jean-François Pinton, Marco Quagiotto, Wouter Van den Broeck, Corinne Régis, Bruno Lina, et al. High-resolution measurements of face-to-face contact patterns in a primary school. *PLoS ONE*, 6(8):e23176, 2011. doi: 10.1371/journal.pone.0023176.
  - [89] S. Sugaya and V. M. Kenkre. Analysis of transmission of infection in epidemics: confined random walkers in dimensions higher than one. *Bulletin of Mathematical Biology*, 80(12):3106–3126, 2018. doi: 10.1007/s11538-018-0507-2.
  - [90] Tara Swadi, Jemma L. Geoghegan, Tom Devine, Caroline McElnay, Jillian Sherwood, Phil Shoemack, Xiaoyun Ren, Matt Storey, Sarah Jefferies, Erasmus Smit, et al. Genomic evidence of in-flight transmission of SARS-CoV-2 despite predeparture testing. *Emerging Infectious Diseases*, 27(3):687–693, 2021. doi: 10.3201/eid2703.204714.
  - [91] Attila Szabo, Gene Lamm, and George H. Weiss. Localized partial traps in diffusion processes and random walks. *Journal of Statistical Physics*, 34(1-2):225–238, 1984. doi: 10.1007/BF01770356.
  - [92] G. N. Sze To and C. Y. H. Chao. Review and comparison between the Wells–Riley and dose-response

- approaches to risk assessment of infectious respiratory diseases. *Indoor Air*, 20(1):2–16, 2010. doi: 10.1111/j.1600-0668.2009.00621.x.
- [93] Geoffrey Ingram Taylor. Diffusion by continuous movements. *Proceedings of the London Mathematical Society*, s2-20(1):196–212, 1922. doi: 10.1112/plms/s2-20.1.196.
  - [94] Joseph H. Tien, Zhisheng Shuai, Marisa C. Eisenberg, and P. van den Driessche. Disease invasion on community networks with environmental pathogen movement. *Journal of Mathematical Biology*, 70(5):1065–1092, 2015. doi: 10.1007/s00285-014-0791-x.
  - [95] Takao Toyokawa, Tomoe Shimada, Takahiro Hayamizu, Tsuyoshi Sekizuka, Yuji Zukeyama, Miyako Yasuda, Yuko Nakamura, Sho Okano, Jun Kudaka, Tetsuya Kakita, et al. Transmission of SARS-CoV-2 during a 2-h domestic flight to Okinawa, Japan, March 2020. *Influenza and Other Respiratory Viruses*, 16(1):63–71, 2022. doi: 10.1111/irv.12913.
  - [96] P. van den Driessche and James Watmough. Reproduction numbers and sub-threshold endemic equilibria for compartmental models of disease transmission. *Mathematical Biosciences*, 180(1–2): 29–48, 2002. doi: 10.1016/S0025-5564(02)00108-6.
  - [97] Philippe Vanhems, Alain Barrat, Ciro Cattuto, Jean-François Pinton, Nagham Khanafer, Corinne Régis, Byeul-a Kim, Brigitte Comte, and Nicolas Voirin. Estimating potential infection transmission routes in hospital wards using wearable proximity sensors. *PLoS ONE*, 8(9):e73970, 2013. doi: 10.1371/journal.pone.0073970.
  - [98] Akula Venkatram and Jeffrey Weil. Modeling turbulent transport of aerosols inside rooms using eddy diffusivity. *Indoor Air*, 31(6):1886–1895, 2021. doi: 10.1111/ina.12901.
  - [99] Ville Vuorinen, Mia Aarnio, Mikko Alava, Ville Alopaeus, Nina Atanasova, Mikko Auvinen, Nallannan Balasubramanian, Hadi Bordbar, Panu Erästö, Rafael Grande, et al. Modelling aerosol transport and virus exposure with numerical simulations in relation to SARS-CoV-2 transmission by inhalation indoors. *Safety Science*, 130:104866, 2020. doi: 10.1016/j.ssci.2020.104866.
  - [100] Lin Wang and Xiang Li. Spatial epidemiology of networked metapopulation: an overview. *Chinese Science Bulletin*, 59(28):3511–3522, 2014. doi: 10.1007/s11434-014-0499-8.
  - [101] Wendi Wang and Xiao-Qiang Zhao. Basic reproduction numbers for reaction-diffusion epidemic models. *SIAM Journal on Applied Dynamical Systems*, 11(4):1652–1673, 2012. doi: 10.1137/120872942.
  - [102] Ulrich Weidmann. Transporttechnik der Fussgänger: Transporttechnische Eigenschaften des Fussgängerverkehrs, Literatúrauswertung. Technical Report Schriftenreihe des IVT Nr. 90, 2. ergänzte Auflage, Institut für Verkehrsplanung, Transporttechnik, Strassen- und Eisenbahnbau (IVT), ETH Zürich, Zürich, 1993. URL <https://doi.org/10.3929/ethz-b-000242008>.
  - [103] William Firth Wells. *Airborne Contagion and Air Hygiene: An Ecological Study of Droplet Infections*. Harvard University Press, Cambridge, MA, 1955. URL <https://lccn.loc.gov/55007856>.
  - [104] Tze-wai Wong, Chin-kei Lee, Wilson Tam, Joseph Tak-fai Lau, Tak-sun Yu, Siu-fai Lui, Paul K. S. Chan, Yuguo Li, Joseph S. Bresee, Joseph J. Y. Sung, et al. Cluster of SARS among medical students exposed to single patient, Hong Kong. *Emerging Infectious Diseases*, 10(2):269–276, 2004. doi: 10.3201/eid1002.030452.
  - [105] Huw Woodward, Rick J. B. de Kreijl, Emily S. Kruger, Shiwei Fan, Arvind Tiwari, Sarkawt Hama, Simon Noel, Megan S. Davies Wykes, Prashant Kumar, and Paul F. Linden. An evaluation of the risk of airborne transmission of COVID-19 on an inter-city train carriage. *Indoor Air*, 32(10):e13121, 2022. doi: 10.1111/ina.13121.
  - [106] Fabian Ying and Neave O’Clery. Modelling COVID-19 transmission in supermarkets using an agent-based model. *PLoS ONE*, 16(4):e0249821, 2021. doi: 10.1371/journal.pone.0249821.
  - [107] Nicholas Young, Richard Pebody, Gillian Smith, Babatunde Olowokure, Giri Shankar, Katja Hoschler, Monica Galiano, Helen Green, Anders Wallensten, Angela Hogan, et al. International flight-related transmission of pandemic influenza A(H1N1)pdm09: an historical cohort study of the first identified cases in the United Kingdom. *Influenza and Other Respiratory Viruses*, 8(1):66–73, 2014. doi: 10.1111/irv.12181.
  - [108] I. T. S. Yu, T. W. Wong, Y. L. Chiu, N. Lee, and Y. Li. Temporal-spatial analysis of severe acute respiratory syndrome among hospital inpatients. *Clinical Infectious Diseases*, 40(9):1237–1243, 2005. doi: 10.1086/428735.